General Relativity I

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These notes are a work in progress. The few subsections marked with an asterisk, ⊛, discuss things not lectured in class, while those marked with a black dot, ⦿, summarize material from other references and can be skipped entirely.

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0 Recommended texts

The course does not have an official textbook, but here are a few books that you can refer to.

1. Wald, *General Relativity* (1984). This is the instructor's default perspective, but some may find it a bit terse and unmotivated. Described as a less chatty version of MTW.

2. Ferrari, Gualtieri, and Pani, *General Relativity and its Applications* (2021). A great book. Unfortunately, has no exercises, or material on cosmology.

3. d'Inverno and Vickers, *Introducing Einstein's Relativity: A Deeper Understanding* (2022). Something the instructor used early on as a student; the first edition was very good. When this lecture was given, the second edition hadn't actually come out yet!

4. Padmanabhan, *Gravitation: Foundations and Frontiers* (2010). Takes a very particle-physics approach to things; big and comprehensive, and covered most ideally over three semesters.

5. Zee, *Einstein Gravity in a Nutshell* (2013). A fun book that bears Zee's stamp, with nice discussion on AdS/CFT. Recommended for the unique perspective he offers on the subject. There are times when Zee can be rather antagonistic towards the math, which is not really necessary, in the instructor's opinion. (For example, you should not have to feel bad about using the term "diffeomorphism.")

6. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (2003). Much friendlier than Wald, and in some ways the canonical text for a graduate-level introduction.

7. Choquet-Bruhat, *Introduction to General Relativity, Black Holes, and Cosmology* (2015). Purportedly for undergraduates, but is quite intense. Choquet-Bruhat's perspective differs at times from the instructor's, at least for a first course, since it is written with a view toward mathematical relativity.

8. Hartle, *Gravity: An Introduction to Einstein's General Relativity* (2003). You can turn to these next two books when you have trouble following the lectures; they are typically used in undergraduate courses.

9. Schutz, *A First Course in General Relativity* (2009).

10. Hobson, Efstathiou, and Lasenby, *General Relativity: An Introduction for Physicists* (2006). Especially good for cosmology.

This is not a differential geometry class, but we'll spend quite a bit of time on the subject, because it is the language of general relativity. The discussion will be modern, so that you can talk to mathematicians!

1. O'Neill, *Semi-Riemannian Geometry* (1983). The instructor has colleagues who have attempted to teach introductory GR out of O'Neill (!!). Leads into Hawking and Ellis.

2. Lovett, *Differential Geometry of Manifolds* (2020). A differential geometry book at its core, but has its eye always on the physics.

1 A review of Newtonian gravity

In this lecture, we discuss what kind of theory general relativity (GR) is, we make a few important observations about Newtonian gravity, and then we talk briefly about what a theory of gravity ought to possess.

General relativity is more than just a theory about a particular interaction – for examples of these, think Maxwell's equations and electromagnetism, or the gravitational force law and Newtonian gravity. This is because GR is a theory of both gravity and spacetime. We'll talk about the distinction between the two in a bit, but first, let's review what we know about NG.

(1) *Newton's law of gravitation* says that

$$\begin{aligned}\vec{F} &= -\frac{GMm_g}{r^2}\,\hat{r} \\ &= m_g \underbrace{\left(-\frac{GM}{r^2}\,\hat{r}\right)}_{\text{grav field}}.\end{aligned}$$

We use m_g to denote *gravitational mass.* You can think of this as a "charge" that determines the strength of the interaction. More generally, we may write

$$\boxed{\vec{F} = m_g(-\nabla\Phi).}$$

In the special case above, we had $\Phi = -GM/r$.

(2) *Newtonian gravity is linear, and it obeys the principle of superposition.* In other words,

$$\Phi(\vec{x}) = \sum_i \Phi_i(\vec{x}) = -G\sum_i \frac{m_i}{|\vec{x}-\vec{r}_i|}.$$

For continuous distributions, we replace the sum with an integral,

$$\Phi(\vec{x}) = -G\int d^3x'\,\frac{\rho(t,\vec{x}')}{|\vec{x}-\vec{x}'|}.$$

Recalling the identity

$$\nabla^2\frac{1}{|\vec{x}-\vec{x}'|} = -4\pi\delta(\vec{x}-\vec{x}'),$$

we obtain

$$\begin{aligned}\nabla^2\Phi &= -G\int d^3x'\,\rho(t,\vec{x}')\,\nabla^2\frac{1}{|\vec{x}-\vec{x}'|} \\ &= 4\pi G\int d^3x'\,\rho(t,\vec{x}')\delta(\vec{x}-\vec{x}')\end{aligned}$$

and subsequently

$$\boxed{\nabla^2\Phi = 4\pi G\,\rho(t,\vec{x}).}$$

This is the Newtonian field equation. There is an issue with this – the equation does not come with retardation (time delay). Changes propagate instantaneously, which we know to be inconsistent with SR, so GR must replace this with something else.

(3) The *Newtonian EOM* in a gravitational field is

$$\vec{a} = \frac{1}{m_i} \sum \vec{F}.$$

One thing we notice is that this is a second-order ODE. Due to classical mechanics, we are familiar with solving these. (We mention this because the analogous equations in GR are really, really difficult in comparison!)

Note also that the subscript i is not an index. Instead, it denotes *inertial mass.* You may wonder why we make a distinction, since until now we have taken the equivalence of m_i and m_g for granted. In an elementary class, we simply say that they are the same. This is true, as far as our experiments have been able to show, but from the Newtonian point of view, this equivalence comes as a surprise. To understand why, recall that inertial and gravitational mass were defined in operationally different ways. We saw that

m_i — measure of resistance to changes in motion

m_g — charge that determines strength of interaction

In principle, these have nothing to do with each other. Here's an example from electromagnetism – the motion of a particle in an electric field is given by

$$\vec{a} = \frac{q\vec{E}}{m},$$

so the analogous concept here would be finding that, in every experiment we conduct, we observe that $q/m = 1$.

(4) *Inertial mass m_i and gravitational mass m_g are equivalent.* As we mentioned earlier, this is rather surprising, and it is natural to ask why this is the case. While this is merely an experimental fact in NG, we will see that in GR, this equivalence is a consequence of the theory. What does this imply for Newtonian gravity? Going back to our previous equations, we see that for objects in freefall (moving solely under the influence of gravity), we have

$$\boxed{\vec{a} = -\nabla\Phi.}$$

This says that the motion of the body has nothing to do with the properties of the body itself! To better see this, we rewrite as

$$\frac{d^2\vec{x}}{dt^2} = -\nabla\Phi(\vec{x}(t), t).$$

There is no information about the body here. Compare to the analogous equation in electromagnetism, where the motion would still be dependent upon the ratio q/m, which is a property of the body and is not fixed in general. So if you were to give the same initial velocity to a rock, a feather, or a small animal,[1] letting them all go from the same point, you would find that all would trace out the same path. We can think of this path as a sort of universal trajectory.

Clearly, there must be something that determines the trajectory the body follows. We'll say that this influence is "what's left" when the body is removed. This "what's left" is what is known as *spacetime.* Thus, we may say that universal trajectories are determined by spacetime.

[1] I believe this only applies in the test-particle limit, so we would need to pick a small animal with a negligible effect on the spacetime geometry.

Now, it turns out that physicists have differing opinions as to what spacetime really is. Ask a relativist, and they will tell you that the structure of spacetime is carried by the *metric* $g_{\mu\nu}$. Among other things, this intepretation provides an elegant explanation for the equivalence of inertial and gravitational mass. A particle physicist, however, would say that $g_{\mu\nu}$ is merely something that couples universally to everything. Some feel that this is an inelegant (Dr. Vega would say emaciated) way of viewing things.

But everyone can agree that GR is a theory of gravity. As such, it must explain gravitational phenomena, such as relativistic celestial mechanics, gravitational waves, stellar structure, black holes, and the expansion of the universe.

We will wrap things up by asserting that a theory of gravity must…

(0) …have a dynamical variable. (Φ)

(1) …tell us how this variable affects objects. ($\vec{a} = -\nabla\Phi$)

(2) …tell us how this variable is produced by other objects. ($\nabla^2\Phi = 4\pi G\rho$)

These requirements are not so restrictive; in fact, any classical field theory satisfies these. In particular, it's clear that Newtonian gravity satisfies each of these requirements (see expressions in parentheses), and eventually we will establish the following correspondence:

$$\begin{aligned} \Phi \quad &\longleftrightarrow \quad g_{\mu\nu} \\ \vec{a} = -\nabla\Phi \quad &\longleftrightarrow \quad \frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0 \\ \nabla^2\Phi = 4\pi G\rho \quad &\longleftrightarrow \quad R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R = 8\pi T_{\alpha\beta} \end{aligned}$$

The symbols on the right-hand side of this figure will remain mysterious for now, but in time, we will understand how this works out.

2 The equivalence principle; Lorentz transformations

In this lecture, we discuss the equivalence principle and a bit of the history leading up to it, and then we derive the Lorentz transformations without making any reference to photons, thereby showing that relativity has nothing to do with light.

The equivalence principle

(please revisit this section)

A bit of history

Relativity is not an invention of Einstein – in fact, it is quite old. One can characterize relativity as asserting the equivalence of the laws of physics for all distinguished observers. This is encoded, for example, in the *Galilean transformations*,

$$\begin{aligned} t' &= t, \\ \vec{x}\,' &= \vec{x} - \vec{v}_0\, t. \end{aligned}$$

These imply simple velocity addition, and it is easy to verify that they leave $\vec{F} = m\vec{a}$ invariant, assuming that your forces are real ones.

Now, when Maxwell's equations were introduced, they predicted a speed of propagation that was not invariant with respect to Galilean transformations. Initially, some took this to mean that these equations were not fundamental – not as fundamental, at least, as the laws of mechanics. In particular, they were believed only to hold in a certain preferred frame. This was the 'ether' theory, and it was actually not as contrived as it sounds.[2] For example, wave mechanics are not invariant under Galilean transformations, but this is OK because the fluid comes with its own preferred rest frame.

The ether theory came with an important prediction. If it were true, then it would be possible to detect the preferred frame. As we know, the Michelson-Morley experiment, which was designed to do this, gave a null result. Despite this, a number of theories were proposed to save the ether idea. For example, one proposal attributed the null result to an 'ether-dragging' effect; the Earth, as it moved through the ether, dragged along a thin layer of ether that was essentially at rest.

Anyway, the dominant viewpoint was that Galilean relativity, which came with notions of absolute time and simultaneity, provided the correct way of viewing things, and that Maxwell's equations were not fundamental. Instead, the laws of mechanics were taken to be fundamental. In order for this to align with observation, scientists needed to propose the ether theory and a number of other band-aid explanations. In the end, though, this meant that relativity was something that conformed closely to our intuition.

Then Einstein came along and said – no, special relativity is the way, Maxwell's equations are the fundamental ones, and there is no ether. We'll discuss this in a moment, because now that we've caught up to Einstein, we can provide the following description:

> *Relativity*: The existence of a privileged class of observers (inertial) with respect to whom the laws of physics look the same.

[2] In hindsight, we can see that the theory was wrong, but chances are that if you had been a practicing scientist at the time, you would have been an 'ether person', too.

This is how we can think of relativity now, and this is how Einstein thought as well. Note that he did not have to change the class of privileged observers – these are still inertial, which we recall to be the frames characterized by the absence of fictitious forces.

Now, part of what made special relativity so radical was the fact that *Lorentz transformations* (the SR analogue of Galilean transformations) necessarily mixed time and space:

$$\begin{aligned} ct' &= \gamma(ct - \beta x) \\ x' &= \gamma(x - \beta ct) \end{aligned}$$

where we have

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} \qquad \text{and} \qquad \beta = \frac{v}{c}.$$

The quantity v is understood to be the relative velocity between primed and unprimed observers. We may write this transformation in matrix form as

$$x^{\mu\prime} = \Lambda^{\mu}{}_{\nu} x^{\nu},$$

where x^μ is an object called a 4-vector,

$$x^\mu = (x^0, x^1, x^2, x^3), \quad \text{with } x^0 = ct,$$

and

$$\Lambda^{\mu}{}_{\nu} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

but this is getting a bit ahead of things, and we'll discuss these in more detail next lecture. The reason we mention these is that, as discussed earlier, we want to write Newton's laws so that they are covariant. Formally, we express this as going from

$$\frac{d\vec{p}}{dt} = \vec{F} \quad \longrightarrow \quad \frac{dp^\mu}{d\tau} = F^\mu,$$

where the things on the right side are invariant under LTs, but the things on the left side aren't. We were able to do this for all forces, with the exception of gravity. That is, we knew that

$$\nabla^2\Phi = 4\pi G\,\rho \iff \vec{F} = -\frac{GMm}{r^2}\,\hat{r};$$

but we weren't quite sure how to generalize this so that it would respect SR. One attempt to do this (by Einstein-Fokker-Nordström) involved promoting ∇^2 to the d'Alembertian $\Box$,

$$\Box = -\partial_t^2 + \partial_i\partial^i, \qquad \text{where } \partial_i\partial^i = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}.$$

This resulted in a special relativistic theory of gravity, but damningly, it did not predict light bending, and $m_i = m_g$ was still an accident. Our little history of relativity ends here; presumably, this is where GR came in to explain everything.

Deriving the Lorentz transformations without light

In a small enough neighborhood of your spacetime, everything reduces to SR. This means that Lorentz transformations are important. But how do we derive these in the first place? The typical derivation starts by asserting that photons move at the same speed in every frame, and then considers thought experiments involving trains and mirrors and light-clocks. This line of reasoning eventually arrives at the Lorentz transformations.

There's nothing incorrect about this, but this argument seems to privilege light. The reality, however, is that relativity is not subservient to Maxwell's equations. It is possible to derive the Lorentz transformations without making reference to photons, and we will do this now.

Consider two frames $\mathcal{O}$ and $\overline{\mathcal{O}}$, with time and space coordinates (t, x) and $(\bar{t}, \overline{x})$, respectively. Suppose that $\overline{\mathcal{O}}$ is moving with uniform speed v in the x-direction of $\mathcal{O}$. Now, the most general mapping from $\mathcal{O} \to \overline{\mathcal{O}}$ is of the form

$$\begin{aligned}\bar{t} &= \overline{T}(t, x; v)\\ \overline{x} &= \overline{X}(t, x; v)\end{aligned}$$

We distinguish v from the other two arguments because it is fixed, but we expect $\overline{T}$ and $\overline{X}$ to depend on it in some way. We want to show that these are the Lorentz transformations. To do this, we will need to make three assumptions about $\overline{T}$ and $\overline{X}$.

(1) *Reciprocity.*

$$\begin{aligned}t &= \overline{T}(\bar{t}, \overline{x}; -v) = \overline{T}\Big(\overline{T}(t, x; v),\, \overline{X}(t, x; v); -v\Big)\\ x &= \overline{X}(\bar{t}, \overline{x}; -v) = \overline{X}\Big(\overline{T}(t, x; v),\, \overline{X}(t, x; v); -v\Big)\end{aligned}$$

The content of our assumption is in the first of both equalities.

(2) *Isotropy of space.* If we were to flip our coordinate axes around, so that

$$x \to -x, \quad v \to -v, \quad \text{and} \quad t \to t,$$

this should represent the same physical scenario. In terms of our transformations, this is the assumption that

$$\begin{aligned}\overline{T}(t, -x; -v) &= \overline{T}(t, x; v)\\ \overline{X}(t, -x; -v) &= -\overline{X}(t, x; v)\end{aligned}$$

(3) *Invariance of uniform motion.* Some people call this the 'homogeneity of space and time'. This says that if in the unbarred frame,

$$\frac{dx_\mu}{d\tau} = \text{constant}, \quad \text{so that} \quad \frac{d^2 x_\mu}{d\tau^2} = 0,$$

then we must have

$$\frac{d^2 \overline{x}_\mu}{d\tau^2} = 0.$$

We are being intentionally ambiguous about what sort of parameter τ is; the point is that uniform motion gets mapped to uniform motion.

All our assumptions have been stated, so we proceed with the derivation. First, note that

$$\frac{d^2 \overline{x}_\mu}{d\tau^2} = \frac{\partial \overline{x}_\mu}{\partial x_\nu}\frac{d^2 x_\nu}{d\tau^2} + \frac{\partial^2 \overline{x}_\mu}{\partial x_\nu \partial x_\sigma}\frac{dx_\nu}{d\tau}\frac{dx_\sigma}{d\tau}, \tag{2.1}$$

where we are summing over repeated indices. Let x_ν be such that

$$\frac{dx_\nu}{d\tau} = \text{constant}.$$

By (3), this implies that the left-hand side of (2.1) is zero, which in turn implies that

$$\frac{\partial^2 \overline{x}_\mu}{\partial x_\nu \partial x_\sigma} \frac{dx_\nu}{d\tau} \frac{dx_\sigma}{d\tau} = 0.$$

But both $dx_\nu/d\tau$ and $dx_\sigma/d\tau$ are arbitrary constants, so for this to be true in general, we need

$$\frac{\partial^2 \overline{x}_\mu}{\partial x_\nu \partial x_\sigma} = 0.$$

This implies that $\overline{x}_\mu$ is linear in x^μ. (This really just implies that $\overline{x}_\mu$ is affine, but by isotropy, we may choose our origin such that it is linear.) Thus, without loss of generality,

$$\begin{aligned} \overline{X}(t, x; v) &= A_v x + B_v t \\ \overline{T}(t, x; v) &= C_v x + D_v t \end{aligned}$$

In matrix form, we have

$$\begin{pmatrix} \overline{x} \\ \overline{t} \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

Sometimes we will suppress the arguments of A, B, C, D, but you should always keep in mind that these are functions of v. This is a pretty impressive simplification – the homogeneity of space and time implies that the spacetime transformations must be linear in x^μ.

Next, we note that isotropy applied to $\overline{T}$ implies

$$C_{-v}(-x) + D_{-v} t = C_v x + D_v t.$$

Since this holds for arbitrary x and t , we have

$$\begin{aligned} C_{-v} &= -C_v \\ D_{-v} &= D_v \end{aligned}$$

That is, C is an odd function of v, while D is an even function of v. Similarly, isotropy applied to $\overline{X}$ tells us about A_v and B_v:

$$\begin{aligned} A_{-v} &= A_v \\ B_{-v} &= -B_v \end{aligned}$$

Then we use our remaining assumption – by reciprocity, we have

$$\begin{aligned} t &= C_{-v}\overline{X}(t, x; v) + D_{-v}\overline{T}(t, x; v) \\ &= -C_v[A_v x + B_v t] + D_v[C_v x + D_v t]. \end{aligned}$$

We used the linearity of the transformation in both lines, and the even- and odd-ness of C and D in the second line. Rearranging a bit, this implies that

$$t = C(D - A)x + (D^2 - CB)t.$$

But this must hold for all x and t, so

$$\begin{aligned} C(D - A) &= 0 \\ D^2 - CB &= 1 \end{aligned}$$

In a similar manner (using reciprocity for x), it follows that

$$x = (A^2 - BC)x + B(A - D)t,$$

thus

$$\begin{aligned} B(A - D) &= 0 \\ A^2 - BC &= 1 \end{aligned}$$

Now we want to determine the form of these functions. There are two cases that seem natural to examine here – either $A \neq D$ or $A = D$.

Suppose first that $A \neq D$. It's easy to see that this implies $B = C = 0$, which forces $A^2 = 1$ and $D^2 = 1$. But, referring back to our matrix equation, we see that this solution corresponds to

$$\begin{pmatrix} \overline{x} \\ \overline{t} \end{pmatrix} = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

but this is not a solution we're interested in. (For one, this transformation has no dependence on v, which is obviously incorrect.)

So we are left with the $A = D$ case. Solving the system of equations gives

$$C_v = \frac{A_v^2 - 1}{B_v}.$$

To further specify the form of this, consider the trajectory $x = vt$ in $\mathcal{O}$. Since we chose our coordinate systems to coincide at $t = 0$, this just gets mapped to $\overline{x} = 0$ in $\overline{\mathcal{O}}$. This means

$$\overline{x} = 0 = A_v(vt) + B_v t \implies B_v = -vA_v$$

Thus

$$C_v = \frac{1 - A_v^2}{vA_v}.$$

We've managed to narrow things down to a single function A, and to summarize, we've shown

$$\begin{pmatrix} \overline{x} \\ \overline{t} \end{pmatrix} = \begin{pmatrix} A & -vA \\ \frac{1-A^2}{vA} & A \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

We want to determine what A is, and to do this, we will assume that the set of our transformations is closed under compositions. So if we consider a third frame $\overline{\overline{\mathcal{O}}}$ moving uniformly with speed u with respect to $\overline{\mathcal{O}}$, we obtain

$$\begin{pmatrix} \overline{\overline{x}} \\ \overline{\overline{t}} \end{pmatrix} = \begin{pmatrix} A_u & -uA_u \\ \frac{1-A_u^2}{uA_u} & A_u \end{pmatrix} \begin{pmatrix} A_v & -vA_v \\ \frac{1-A_v^2}{vA_v} & A_v \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

It is not hard to show that the upper left entry of the resulting matrix is

$$A_u A_v - uA_u \frac{1 - A_v^2}{vA_v},$$

and that the lower right entry is

$$-vA_v \frac{1 - A_u^2}{uA_u} + A_u A_v.$$

Now, since we're assuming that this resulting matrix is itself a spacetime transformation, its diagonal entries must be equal. (We derived this when we rejected the $A \neq D$ case earlier.) A little bit of manipulating shows that this requirement is equivalent to

$$\frac{1 - A_v^2}{v^2 A_v} = \frac{1 - A_u^2}{u^2 A_u}.$$

But both u and v were arbitrary, so we see that

$$\frac{A_v^2 - 1}{v^2 A_v^2} = k = \text{constant}.$$

Since we've been working with geometric units, A is dimensionless, and the constant k has units of $1/v^2$. We also see that

$$A = \frac{1}{\sqrt{1 - kv^2}},$$

where we've chosen the positive square root.[3] Thus, our spacetime transformation must be of the form

$$\begin{pmatrix} \overline{x} \\ \overline{t} \end{pmatrix} = \frac{1}{\sqrt{1 - kv^2}} \begin{pmatrix} 1 & -v \\ -kv & 1 \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix}$$

The only thing that is free to vary is the value of the constant k, which defines an invariant velocity. (The term 'invariant' can be justified by considering the velocity addition formula, which will be derived later on.) More precisely, we have

$$k = \frac{1}{v_0^2}.$$

This is the only thing we need to 'put in'. So what relativity tells us is that there is some fundamental constant v_0 which plays a special role in spacetime transformations.

To rewrite in a more familiar form, we see that the transformation laws that must follow from our three assumptions are

$$\boxed{\begin{aligned} \overline{t} &= \gamma(t - kvx) \\ \overline{x} &= \gamma(x - vt) \end{aligned} \qquad \text{with } \gamma = \frac{1}{\sqrt{1 - kv^2}}.}$$

These are a sort of generalized Lorentz transformation. Note that when $v_0 = \infty$, so that $k = 0$, these reduce to the Galilean transformations. And when $v_0 = c$, we get the actual Lorentz transformations. But note that we never had to appeal to any of the properties of light, or to any of Maxwell's equations. The equality of the fundamental speed v_0 to the speed of light is completely incidental. Moreover, it could just as well have been the case that there are no objects in the universe that travel at the fundamental speed.

Anyway, the point we take away from all this is that

$$\boxed{\text{Relativity has nothing to do with light!}}$$

[3] Taking the minus solution corresponds to spatial or time reversal, and it is my understanding that we reject this sort of transformation on physical grounds. See Weinberg (p. 28) for a similar line of reasoning; it's also worth looking at the proof that the LTs are the unique family of transformations that leave proper time invariant. Therefore, you can start (and this is what Weinberg does) by demanding that spacetime transformations be the ones possessing this invariant quantity.

3 Proper time and the Minkowski metric

We can discuss Lorentz transformations in a bit more detail. First we write these in the form

$$\begin{aligned} v_0\overline{t} &= \gamma(v_0 t - \beta x) \\ \overline{x} &= \gamma(x - \beta v_0 t), \end{aligned}$$

where $\beta = v/v_0$. Alternatively, by letting $x_0 = v_0 t$ (which in geometric units will correspond to our time component), we get

$$\begin{pmatrix} \overline{x}_0 \\ \overline{x}_1 \end{pmatrix} = \begin{pmatrix} \gamma & -\beta\gamma \\ -\beta\gamma & \gamma \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

We may generalize this to boosts in arbitrary directions, where $\vec{\beta} = \vec{v}/v_0$, with $\vec{v}$ the relative velocity between frames. We define $\vec{r} = (x_1, x_2, x_3)^T$, and we consider the components of $\vec{r}$ that are parallel and perpendicular to the direction of the boost $\vec{\beta}$.

$$\begin{aligned} \vec{r}'_\perp &= \vec{r}_\perp \\ \vec{r}'_\parallel &= \gamma(\vec{r}_\parallel - \vec{\beta} x_0) \\ x'_0 &= \gamma(x_0 - \vec{\beta} \cdot \vec{r}_\parallel) \end{aligned}$$

We know that

$$\begin{aligned} \vec{r}_\parallel &= (\vec{r} \cdot \vec{\beta}) \frac{\vec{\beta}}{|\vec{\beta}|^2} \\ \vec{r}_\perp &= \vec{r} - \vec{r}_\parallel \end{aligned}$$

But we will rarely use the details of these transformations. What's most important is what the structure of these transformations reveals to us.

> *Geometry* is properties or relations of objects that hold independent of either mode of representation or observer.

We have not yet defined what an observer is, but this will presumably be some Lorentz frame. Anyway, it is clear that physics must be geometrical, since it would exist without us being around to make measurements.

Let's take another look at Galilean transformations,

$$\begin{aligned} t' &= t, \\ \vec{x}' &= \vec{x} - \vec{v}_0 t. \end{aligned}$$

There is an obvious invariant here: all frames measure time in the same way, which is where Newton's idea of absolute time comes from. Defining the set $M = \{(t, x, y, z)\}$, we see that Galilean relativity implies the existence of a function $T : M \to \mathbb{R}$ which assigns to every event a certain number (time).

The level sets of T are called surfaces of simultaneity, since all events that take place on such a surface happen simultaneously in all frames. We say that M is foliated by these surfaces. Below is a picture of a worldline in M; it is the curvy thing moving upward in the t direction. We are forced to suppress z since 4-dimensional space is not easy to draw, but you get the idea.

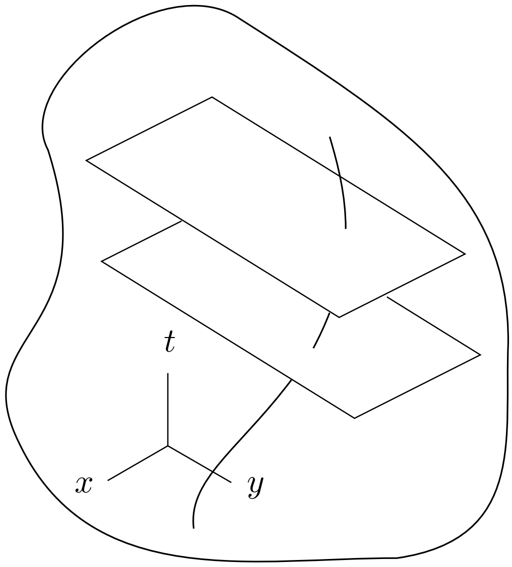


Figure 1: A picture of a worldline in Newtonian spacetime. The horizontal planes are surfaces of absolute time.

Notice what this implies about the causal structure of (pre-relativity) spacetime – given an event p, you can take the surface of simultaneity that it lies upon, and then say that everything above the surface is to the future of p, while everything underneath is to the past of p, and all observers will agree with their interpretation.

There's another thing that Galilean relativity comes with – an invariant distance,

$$d_E(\vec{x}_1, \vec{x}_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}.$$

The subscript E refers to the fact that this is the Euclidean notion of distance. It's not hard to verify that these distances are preserved by Galilean transformations, but intuitively this should be clear. We usually talk about 'the' length of a ruler, because at ordinary velocities, SR looks like Galilean relativity. But just to emphasize the point,

$$d_E(\vec{x}_1, \vec{x}_2) = d_E(\vec{x}'_1, \vec{x}'_2) = \text{constant}.$$

So we have two fixed structures: absolute time and Euclidean distance. We can draw a picture of Newtonian spacetime; see the next page. This happens to be the simplest example of something called a fiber bundle, in case you are interested in these things. Now, what about the Lorentz transformations?

In special relativity, we lose absolute time; space mixes with time, and so on. However, we still have a preserved quantity which we call the *invariant interval*,

$$\begin{aligned} \Delta s^2 &:= -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \\ &= \eta_{\alpha\beta} \Delta x^\alpha \Delta x^\beta. \end{aligned}$$

Here we are summing over repeated indices, and we follow the convention $\eta_{\alpha\beta} = \text{diag}(-1, 1, 1, 1)$. This provides us with the notion of light-cone structure, which is preserved by LTs. Special relativity is the statement that all physics must take place on this structure; it is a sort of meta-principle, since it constrains all our laws.

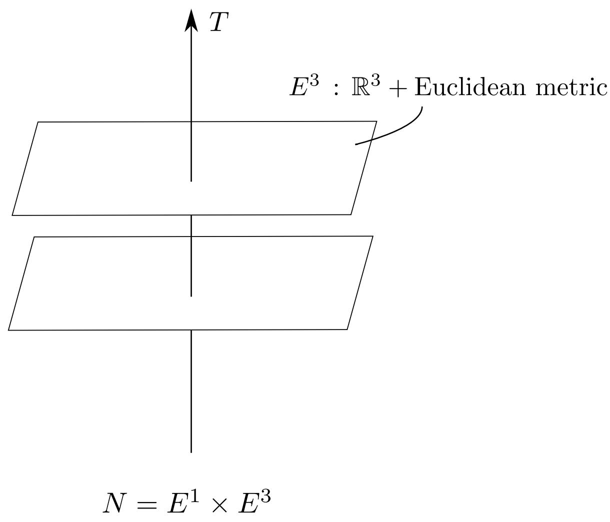


Figure 2: A picture of Newtonian spacetime.

Now let's think about geometric objects. The first such object in physics that you encountered was probably the 3-vector $\vec{F} = m\vec{a}$. The intuition we had here was that these corresponded to arrows in space, independent of any coordinate system. Once you assert something like $\vec{F} = (F_x, F_y, F_z)$, you lose this geometric meaning, since the numbers on the right have no physical meaning without the coordinate basis in which they are measured. In special relativity, our geometric objects are 4-vectors. An example is the displacement vector between two events,

$$\Delta x^\alpha = (c\Delta t, \Delta x, \Delta y, \Delta z).$$

This transforms according to the following rule:

$$\Delta x'^\mu = \Lambda^\mu{}_\nu \Delta x^\nu,$$

where $\Lambda^\mu{}_\nu$ is a member of the *Lorentz group*. This is defined to be (double check) the set of matrices satisfying

$$\Lambda^T \eta \, \Lambda = \eta,$$

and an equivalent way of expressing this condition is to write

$$\begin{aligned} & \Lambda^\rho{}_\mu \, \eta_{\rho\sigma} \, \Lambda^\sigma{}_\nu = \eta_{\mu\nu} \\ \iff \quad & \eta_{\rho\sigma} \, \Lambda^\rho{}_\mu \, \Lambda^\sigma{}_\nu = \eta_{\mu\nu}. \end{aligned}$$

An example of such a matrix is the one corresponding to a boost in the x^1-direction,

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

and you can show that this is indeed a member of the Lorentz group.

For now, we'll say that a 4-vector is an 'arrow' in Minkowski space (to be discussed in detail next lecture). It is easiest to define a 4-vector in the following way – we say that k^μ is a 4-vector if it transforms like a displacement vector: (under a Lorentz transformation?)

$$\boxed{k'_\mu = \Lambda^\mu{}_\nu k^\nu.} \qquad \text{(four-vector)}$$

We can consider general *Lorentz invariants*, which are scalars that are invariant wrt LTs.

$$\boxed{\begin{aligned} \vec{k}\cdot\vec{k} &:= \eta(\vec{k},\vec{k}) \\ &= k^\mu\,\eta_{\mu\nu}k^\nu \\ &= \eta_{\mu\nu}k^\mu k^\nu \end{aligned}}$$

To show that this is in fact an invariant, consider the same quantity in O'.

$$\begin{aligned} (\vec{k}\cdot\vec{k})' &= \eta_{\mu\nu}k'^\mu k'^\nu \\ &= \eta_{\mu\nu}(\Lambda^\mu{}_\alpha k^\alpha)(\Lambda^\nu{}_\beta k^\beta) \\ &= (\eta_{\mu\nu}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta)k^\alpha k^\beta \\ &= \eta_{\alpha\beta}k^\alpha k^\beta \\ &= \vec{k}\cdot\vec{k}. \end{aligned}$$

We used the definition of a 4-vector in the second line, and the fact that Λ is a member of the Lorentz group in the fourth line. Now notice that the invariant interval could have been defined, equivalently, as

$$\boxed{\Delta s^2 := \eta(\Delta\vec{x},\Delta\vec{x}).}$$

With the initial definition, we showed 'by hand' that this was an invariant, but this way there's no need to check; Δs^2 is a Lorentz invariant by definition.

In general, though, invariants do not immediately come with a physical interpretation. Whether these geometric objects are relevant to physics (for the most part) depends on whether you can attach such an interpretation to them.

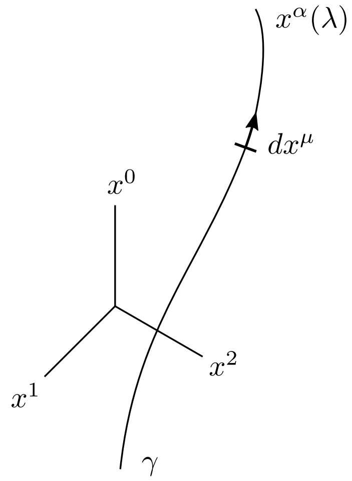


Figure 3: An timelike worldline γ.

Here is one invariant with a physical interpretation. Consider a 4-trajectory like the one above, and consider two infinitesimally close points on this trajectory. We have

$$ds^2 = \eta_{\mu\nu}\,dx^\mu\,dx^\nu$$

Then we introduce the definition

$$d\tau^2 := -\frac{ds^2}{c^2}.$$

This is an invariant multiplied by a constant, so this quantity is itself invariant. We can write

$$\boxed{\tau = \frac{1}{c}\int \sqrt{-ds^2} \qquad \text{(proper time)}}$$

Since this 'proper time' τ does not depend on who the observers are, it is a geometric construct. To see the physical meaning of this quantity, consider an observer O' (a set of basis vectors $\{\vec{e}'_\alpha\}$) that is instantaneously *comoving* with γ at some point P. Then we have

$$dx'^\alpha = (c\,dt', 0, 0, 0) \implies ds^2 = -c^2 dt'^2 \implies \tau = \int dt' = t' + B.$$

To describe what we did here, we started with the comoving observer (the first equation is effectively what comoving means), then we computed the invariant interval, and finally we used this to compute τ. The physical interpretation we take away from this is that τ is the time measured by a comoving clock.

We can look at what τ is according to an inertial observer O:

$$\begin{aligned} d\tau^2 &= -\frac{ds^2}{c^2} \\ &= -\eta_{\mu\nu}\frac{dx^\mu\, dx^\nu}{c^2} \\ &= dt^2 - \frac{d\vec{x}\cdot d\vec{x}}{c^2} \end{aligned}$$

There is a minor abuse of notation in the third line; we defined what the dot notation meant for 4-vectors, but the one here is a standard dot product between 3-vectors, so $d\vec{x}\cdot d\vec{x} \equiv \delta_{ij}\, dx^i\, dx^j$. It follows that

$$\begin{aligned} \left(\frac{d\tau}{dt}\right)^2 &= 1 - \frac{1}{c^2}\frac{d\vec{x}}{dt}\cdot\frac{d\vec{x}}{dt} \\ &\equiv 1 - \frac{1}{c^2}\left(\vec{u}_{(3)}\cdot\vec{u}_{(3)}\right) \end{aligned}$$

Note that $\vec{u}_{(3)}$, which is the 3-velocity of the particle with respect to O, is not a 4-vector; it only has meaning in the reference frame of O. We can invert this to get

$$\frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{\vec{u}_{(3)}\cdot\vec{u}_{(3)}}{c^2}}} = \gamma(u_{(3)}), \qquad \text{where } u_{(3)} = \sqrt{u_x^2 + u_y^2 + u_z^2}.$$

We'll use this to relate 3-velocities to 4-velocities in the next lecture.

4 Minkowski spacetime and spacetime diagrams

We showed in the previous lecture that the proper time τ, given by

$$\tau = \frac{1}{c}\int \sqrt{-\eta_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}}\,d\lambda,$$

is the time measured by a comoving clock. There are two pictures you can draw here.

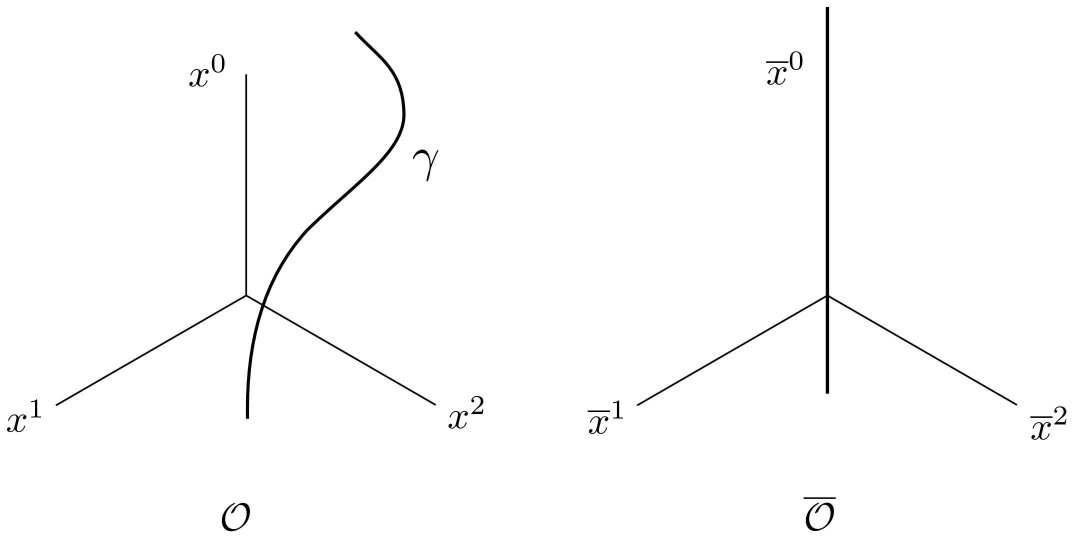


Figure 4: Two "coordinatizations" of the same worldline; the one on the left is with respect to some arbitrary observer, while the one on the right represents the coordinatization of the comoving observer. (check that this description is correct)

We can parametrize 4-trajectories in a number of ways, but in many cases we will eschew an arbitrary parameter λ in favor of τ. A closely related notion is that of a *four-velocity*

$$\boxed{u^\mu := \frac{dx^\mu}{d\tau}.}$$

Some worldlines will have τ undefined, but this will make sense for massive objects. Remember that the general approach in GR is to understand objects via geometry, and reserve components for actually computing things.

From the definition and our expression for $dt/d\tau$ (see prev. lecture), we can compute the four-velocity of a given particle if we know its 3-velocity with respect to an inertial observer O:

$$u^\mu = \gamma(u_{(3)})\begin{pmatrix} c \\ \dfrac{dt}{d\tau}\dfrac{d\vec{x}_{(3)}}{dt}\end{pmatrix} = \gamma(u_{(3)})\begin{pmatrix} c \\ \vec{u}_{(3)}\end{pmatrix}.$$

We write this in the form

$$\boxed{u^\mu = \gamma(u_{(3)})\,(c, \vec{u}_{(3)}).} \qquad \text{(for inertial observers)}$$

It's important to note that the spatial components of the 4-velocity do not correspond to those of the three-velocity $\vec{u}_{(3)}$, that is, $u^i \neq u^i_{(3)}$. The difference is a factor of γ. (There is a notational convention hidden here – some people like to reserve latin indices for spatial components, like in this case, where i runs from 1 to 3. We will use this quite often.) As another warning, the derivation above does not hold for arbitrary observers! If we want this relation between O and $\overline{O}$ to always hold, there are some subtleties we need to take into consideration (see lecture 7). But we'll find that this identity holds for inertial observers.

We've been mentioning Minkowski spacetime a lot, so now we ought to give a definition. First we need to know what an affine space is.

Definition (Affine space)**.** An *affine space* $\mathcal{E}$ is a nonempty set with a mapping V that satisfies the following properties.

(1) The mapping V takes two elements of $\mathcal{E}$ and returns an element of a vector space E:

$$V : \mathcal{E} \times \mathcal{E} \to E, \qquad (A, B) \mapsto V(A, B) =: \overrightarrow{AB}$$

(2) For any point $O \in \mathcal{E}$, the mapping $V_O : \mathcal{E} \to E$, which takes $M \mapsto V(O, M)$, is bijective. You can think of bijective as meaning that no information is lost by the mapping. Intuitively, this mapping corresponds to measuring things with respect to a fixed origin O.

(3) For any $A, B, C \in \mathcal{E}$,

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}.$$

Property (2) implies that if $\mathcal{E}$ is an affine space of dimension n, then E is isomorphic to $\mathbb{R}^n$. Note the difference between an affine space and a vector space – there is no notion of a fixed origin, in the sense that we cannot necessarily add two points in affine space.

> *Minkowski spacetime* is an affine space of dimension 4, together with a symmetric, nondegenerate bilinear form g of signature $(-, +, +, +)$; g is called the *metric tensor*. We say that elements of the affine space $\mathcal{E}$ are *events*, and that elements of the underlying vector space E are *four-vectors*.

There are a couple more things we need to specify to get a precise definition, but you can check Gourgoulhon for the details. Now, to motivate what was said about the metric tensor g, we can give the following provisional definition (it is incomplete): (should this part be removed? perhaps just say that we want to define a notion of orthonormality)

> An *inertial observer* is a timelike worldline (to be defined), together with an orthonormal basis $\{\vec{e}_\alpha\}$ of E (a *tetrad*, so-called because E is of dimension 4).

Ignore the mention of 'timelike worldline' for now, and focus on the fact that we need an orthonormal basis. As things stand, we have no notion of orthonormality – we only know how to form vectors given events. This notion is given to us by g, as we'll show in a moment.

In fact, this is what distinguishes Minkowski from Newtonian 'spacetime'. While $\mathcal{E}_{\text{Newt}}$ is also an affine space of dimension 4, it comes with a notion of absolute time. This provides a foliation of $\mathcal{E}_{\text{Newt}}$ by a family of 3-dimensional affine subspaces Σ_t which we can think of 'absolute space',

$$\mathcal{E}_{\text{Newt}} = \bigcup_{t \in \mathbb{R}} \Sigma_t.$$

The notion of absolute time $t : \mathcal{E}_{\text{Newt}} \to \mathbb{R}$ does not have a direct analogue in SR; instead, we work with the metric tensor $g : E \times E \to \mathbb{R}$. This trades one geometric structure for another. Now let us look at what g is, from a mathematical point of view:

(1) *Bilinear.* For any scalar λ and $\vec{u}, \vec{v}, \vec{w}$ in E,

$$g(\lambda\vec{u} + \vec{v}, \vec{w}) = \lambda g(\vec{u}, \vec{w}) + g(\vec{v}, \vec{w})$$
$$g(\vec{u}, \lambda\vec{v} + \vec{w}) = \lambda g(\vec{u}, \vec{v}) + g(\vec{u}, \vec{w})$$

(2) *Symmetric.*

$$g(\vec{u}, \vec{v}) = g(\vec{v}, \vec{u})$$

(3) *Nondegenerate.*

$$g(\vec{u}, \vec{v}) = 0 \text{ for all } \vec{v} \text{ only if } \vec{u} = 0.$$

(4) *Signature.* There exists a basis of E such that

$$g(\vec{u}, \vec{v}) = -u^0v^0 + u^1v^1 + u^2v^2 + u^3v^3.$$

This is an abstraction of the Minkowski metric we've dealt with thus far,

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1).$$

Recall that this is special because $\eta(\vec{k}, \vec{k})$ is preserved under Lorentz transformations for any four-vector $\vec{k}$. Now, we take $\vec{u} \cdot \vec{v}$ to mean the same thing as $g(\vec{u}, \vec{v})$.[4] Given any basis $\{\vec{e}_0, \vec{e}_1, \vec{e}_2, \vec{e}_3\}$, we define

$$g_{\alpha\beta} := g(\vec{e}_\alpha, \vec{e}_\beta).$$

The subscripts signal that we are using components. Once we know what g does to a basis, we can compute what it does to any two vectors:

$$\begin{aligned}\vec{u} \cdot \vec{v} &= g(\vec{u}, \vec{v}) \\ &= g(u^\alpha \vec{e}_\alpha, v^\beta \vec{e}_\beta) \\ &= u^\alpha v^\beta g(\vec{e}_\alpha, \vec{e}_\beta) \qquad \text{using bilinearity}\end{aligned}$$

In other words,

$$\vec{u} \cdot \vec{v} = g_{\alpha\beta}\, u^\alpha v^\beta.$$

The metric gives us our notion of orthogonality:

> Two vectors $\vec{u}$ and $\vec{v}$ are said to be *orthogonal* if $\vec{u} \cdot \vec{v} = 0$.

By Sylvester's law of inertia, the number of pluses and minuses is preserved for any diagonal writing of the metric (where there are no cross terms, such as u^0v^1). This means that there is no basis in which $\vec{e}_\alpha \cdot \vec{e}_\beta = \delta^\alpha{}_\beta$ for all values of α and β between 0 and 3. The one minus sign goes at the beginning:

> A basis $\{\vec{e}_\alpha\}$ of E is said to be *orthonormal* if $g_{\alpha\beta} = \eta_{\alpha\beta}$.

The norm of a vector is defined by Gourgoulhon as follows:

$$\|\vec{v}\|_g := \sqrt{|\vec{v} \cdot \vec{v}|} = \sqrt{|g(\vec{v}, \vec{v})|}$$

[4] The mostly-plus signature allows us to identify $\vec{u} \cdot \vec{v}$ with the Euclidean dot product on spacelike hyperplanes, which is something we don't get with mostly-minus. (The sign turns out wrong in this case.) Other reasons for this signature are (1) it is very common in GR books, and (2) there are fewer minus signs to juggle around.

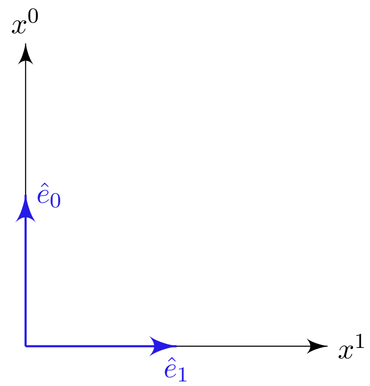
Being a bit imprecise with our language, we'll sometimes also use 'the norm of $\vec{u}$' to mean $g(\vec{u},\vec{u})$. But neither $g(\vec{u},\vec{u})$ nor $\|\vec{u}\|_g$ is a norm in the mathematical sense, because neither is positive definite. [That is, it's possible to have $g(\vec{u},\vec{u}) = 0$ but $\vec{u} \neq 0$.]

We can then classify all vectors in terms of their norm:

$$\begin{aligned} & g(\vec{v},\vec{v}) < 0 && (\vec{v} \text{ timelike}) \\ & g(\vec{v},\vec{v}) > 0 && (\vec{v} \text{ spacelike}) \\ \vec{v} \neq 0 \quad \text{and} \quad & g(\vec{v},\vec{v}) = 0 && (\vec{v} \text{ null, or lightlike}) \end{aligned}$$

The choice of names will make more sense in the context of the so-called light cone, which we will introduce in a bit.

Now, oftentimes in SR we rely on *spacetime diagrams* to understand things. Generally, we draw spacetime diagrams in two dimensions, so suppose that $\{\hat{e}_0, \hat{e}_1\}$ is an orthonormal basis. Then our diagram looks like the following (the convention is to have the timelike basis vector point vertically):



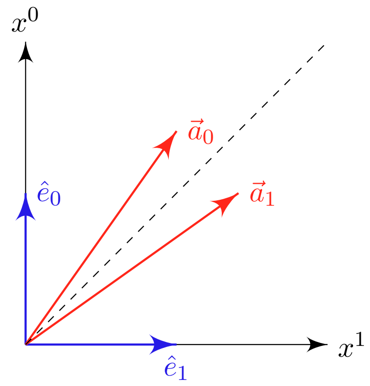
We draw $\hat{e}_0$ and $\hat{e}_1$ such that they are perpendicular. This is somewhat arbitrary, but it conforms to our intuition of what orthogonality means. Doing things this way also gives us a few useful geometric results – we will see this in a moment. To get a concrete idea of how these work, consider the vectors

$$\begin{aligned} \vec{a}_0 &= \sqrt{2}\,\hat{e}_0 + \hat{e}_1 \overset{O}{=} (\sqrt{2}, 1, 0, 0), \\ \vec{a}_1 &= \hat{e}_0 + \sqrt{2}\,\hat{e}_1 \overset{O}{=} (1, \sqrt{2}, 0, 0). \end{aligned}$$

It follows that

$$\begin{aligned} \vec{a}_0 \cdot \vec{a}_1 &= \eta_{\alpha\beta}\, a_0^\alpha a_1^\beta \\ &= -\sqrt{2} + \sqrt{2} \\ &= 0, \end{aligned}$$

so these vectors are orthogonal. But consider what these look like on our spacetime diagram:



This shows that vectors do not have to be perpendicular on a spacetime diagram for them to be orthogonal. Now notice that

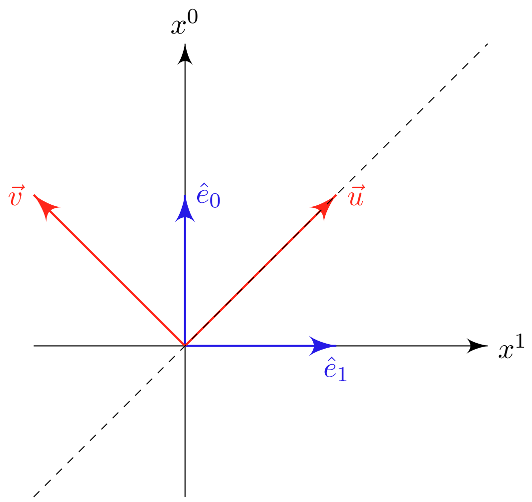
$$\begin{aligned}\vec{a}_0 \cdot \vec{a}_0 &= -1 \\ \vec{a}_1 \cdot \vec{a}_1 &= 1,\end{aligned}$$

so $\{\vec{a}_0, \vec{a}_1, \hat{e}_2, \hat{e}_3\}$ is also an orthonormal basis O'. As another example, consider the vectors

$$\vec{u} = \hat{e}_0 + \hat{e}_1 \overset{O}{=} (1, 1, 0, 0)$$

$$\vec{v} = \hat{e}_0 + \hat{e}_1 \overset{O}{=} (1, -1, 0, 0)$$

These are clearly perpendicular on our spacetime diagram.



However, $\vec{u}$ and $\vec{v}$ are not orthogonal:

$$\begin{aligned}\vec{u} \cdot \vec{v} &= \eta_{\alpha\beta}\, u^\alpha v^\beta \\ &= -(1)(1) + (1)(-1) \\ &= -2.\end{aligned}$$

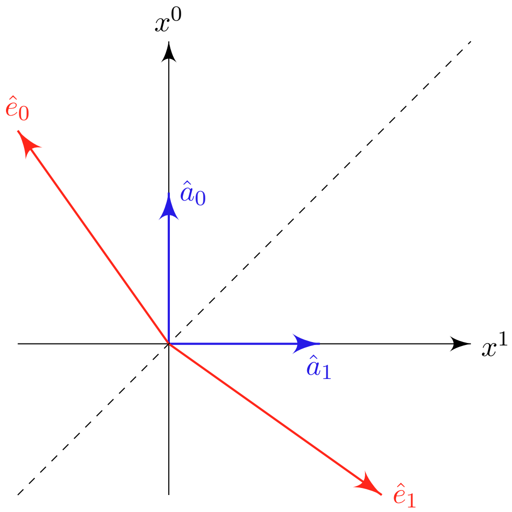
This shows that on a spacetime diagram, perpendicular $\not\Rightarrow$ orthogonal. Taken together with our previous example, we see that we cannot retain the idea of identifying perpendicular vectors with orthogonality. But it is just as easy to determine whether two vectors on a spacetime diagram are orthogonal, and we will see this now.

We could just as well have chosen to draw our spacetime diagram using the basis O', with $\{\hat{a}_0, \hat{a}_1, \hat{e}_2, \hat{e}_3\}$. It is straightforward to show that

$$\hat{e}_0 \overset{O'}{=} (\sqrt{2}, -1, 0, 0)$$

$$\hat{e}_1 \overset{O'}{=} (-1, \sqrt{2}, 0, 0),$$

so the corresponding picture is



By definition, the vectors $\hat{e}_0$ and $\hat{e}_1$ are orthogonal. Note what this picture has in common with the first picture we drew – it seems that two vectors are orthogonal when they are symmetric with respect to the dotted line we drew. It turns out that spacetime diagrams have the following two properties:

(1) Two vectors that are orthogonal with respect to g define directions that are symmetric with respect to one of the main bisectors of the figure.

(2) Null vectors are always drawn as arrows with slope $\pm 45°$.

We see that we just need to trade our notion of 'orthogonal $\iff$ perpendicular' with that of 'orthogonal $\iff$ symmetric with respect to bisectors'.

Now we prove assertions (1) and (2). Let $\{\vec{e}_0, \vec{e}_1\}$ be an orthonormal basis, and consider two arbitrary vectors of the form

$$\vec{u} = u^0 \vec{e}_0 + u^1 \vec{e}_1$$

$$\vec{v} = v^0 \vec{e}_0 + v^1 \vec{e}_1.$$

Then

$$\begin{aligned} & \vec{u} \cdot \vec{v} = 0 \\ \iff \quad & -u^0 v^0 + u^1 v^1 = 0 \\ \iff \quad & \frac{u^0}{u^1} = \frac{v^1}{v^0}. \end{aligned}$$

We can draw a picture to go with this:

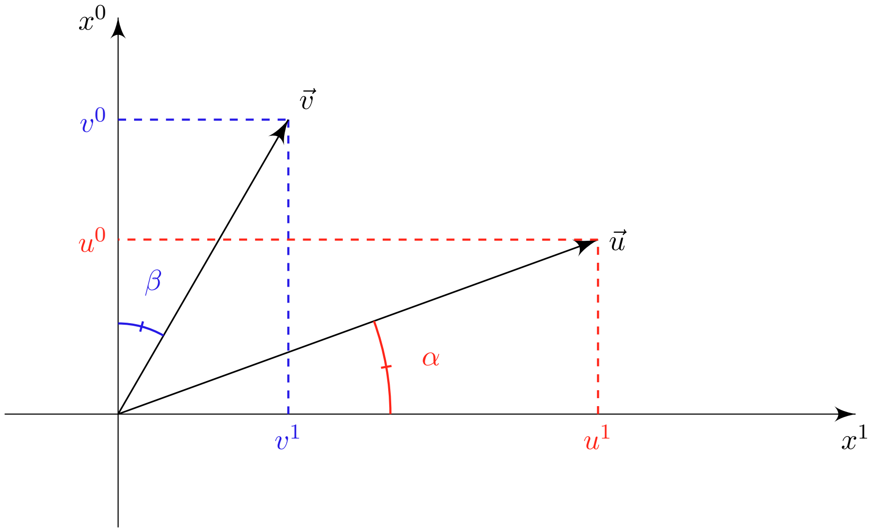


Figure 5: Two arbitrary vectors $\vec{u}$ and $\vec{v}$ on a spacetime diagram.

From the triangles in the figure, we see that

$$\frac{v^1}{v^0} = \tan\beta \qquad \text{and} \qquad \frac{u^0}{u^1} = \tan\alpha.$$

Therefore, $\vec{u}$ and $\vec{v}$ are orthogonal if and only if $\tan\alpha = \tan\beta$, which proves (1). From this, proving (2) is straightforward – by definition, any null vector is orthogonal to itself, and the only vectors that are symmetric with respect to reflections about one of the main bisectors are those with slope $\pm 45°$.

Notice that the only directions that can be canonically associated with the metric tensor g are the null ones; the metric tensor provides us with a notion of light-cone structure. (substantiate)

5 Metric duality; worldlines; four-accelerations

In today's Q&A, an important point is raised – we should detach ourselves from identifying norms with distances. Recall that our inner product is given by g, which is not positive definite. In particular, $g(\vec{u}, \vec{u})$ can be negative, and it can be zero for nonzero vectors $\vec{u}$.

This also serves as a reminder that proper time τ, which is given by

$$\tau = \frac{1}{c} \int \sqrt{-g\left(\frac{d\vec{x}}{d\lambda}, \frac{d\vec{x}}{d\lambda}\right)}\, d\lambda$$

is only well-defined when the vector being fed to g has a norm < 0. That is, it only makes sense when $d\vec{x}/d\lambda$ is a timelike vector.

We introduced Minkowski space as an affine space of dimension 4, together with a symmetric, nondegenerate bilinear form g of signature $(-,+,+,+)$. Let's review what these give us. The vector space E underlying the affine space means that there is a four-element basis $\{\vec{e}_i\}$ for E such that, for any $\vec{v} \in E$, we may write

$$\vec{v} = v^\alpha \vec{e}_\alpha.$$

Then to obtain an affine coordinate system (also called an *affine frame*), we choose some point O in $\mathcal{E}$ to serve as origin, and we measure every other point in $\mathcal{E}$ with respect to this. More precisely, for any $M \in \mathcal{E}$, we have

$$\overrightarrow{OM} \in E \quad \text{and} \quad \overrightarrow{OM} = x^\alpha \vec{e}_\alpha.$$

The numbers $\{x^\alpha\}$ are *affine coordinates* of M. Thus, an affine frame is defined by $(O, \{\vec{e}_\alpha\})$. All of this can be done without a metric. Recall that the metric gave us a notion of orthonormality on E. This then allows us to define an orthonormal basis, which is a set of vectors with the property

$$\vec{e}_\alpha \cdot \vec{e}_\beta = \eta_{\alpha\beta} = \mathrm{diag}(-1, 1, 1, 1).$$

Now, the metric g also furnishes a duality between vectors and one-forms, which we discuss next.

One-forms

A one-form is a linear mapping that takes vectors to numbers; we denote these with an underline.

$$\underline{\omega} : E \to \mathbb{R}, \qquad \vec{u} \mapsto \underline{\omega}(\vec{u}),$$

and

$$\forall\, \lambda \in \mathbb{R},\ \forall\, (\vec{u}, \vec{v}) \in E \times E, \qquad \underline{\omega}(\lambda\vec{u} + \vec{v}) = \lambda\underline{\omega}(\vec{u}) + \underline{\omega}(\vec{v}).$$

As an (optional) exercise, you can show that the set of all linear forms on E, written E^*, is also a vector space, and that it has the same dimension as E.[5] This is called the *dual space* of E, and $\underline{\omega} \in E^*$ are called *dual vectors*.

Given a basis $\{\vec{e}_\alpha\}$ of E, there exists a unique 4-tuple of linear forms $\{\underline{e}^\alpha\}$ such that

$$\underline{e}^\alpha(\vec{e}_\beta) = \delta^\alpha{}_\beta,$$

[5] To prove the second assertion, you'll need to exhibit a basis for E^*; you can use the one we introduce here. Of course, you first need to show that this is indeed a basis.

and this constitutes a basis of E^*. There are many possible choices for a basis, but this is an especially convenient one. Once we have this basis, we may write an arbitrary linear form $\underline{\omega}$ as

$$\underline{\omega} = \omega_\alpha \, \underline{e}^\alpha,$$

where $\omega_\alpha := \underline{\omega}(\vec{e}_\alpha)$. (These are the components of the one-form in this basis.) Then the action of $\underline{\omega}$ on an arbitrary vector $\vec{u}$ is

$$\begin{aligned}\underline{\omega}(\vec{u}) &= (\omega_\alpha \underline{e}^\alpha)(u^\beta \vec{e}_\beta) \\ &= \omega_\alpha u^\beta \underline{e}^\alpha(\vec{e}_\beta) \\ &= \omega_\alpha u^\alpha,\end{aligned}$$

where we used the linearity of the mapping in the second line, and the special property of our dual basis in the third line.

Metric duality

There is no *a priori* mapping between elements of E^* and elements of E, but our metric provides us with an isomorphism[6] between the two spaces which we will use for this purpose.

$$\Phi_g : E \to E^*, \qquad \vec{u} \mapsto \underline{u}, \qquad \text{where} \qquad \underline{u} : E \to \mathbb{R}, \qquad \vec{v} \mapsto g(\vec{u}, \vec{v}).$$

This allows us to identify every vector with a dual vector, and vice versa. Well, it's clear how to get a one-form given a vector; we simply give the vector to our metric. The reverse process is not so obvious, so we describe it now.

Since Φ_g is bijective, for any element $\underline{\omega}$ of E^* there is a unique vector in E, which we will denote by $\vec{\omega}$, such that

$$\underline{\omega} = \Phi_g(\vec{\omega}). \tag{5.1}$$

We want an explicit expression for $\vec{\omega}$. To do this, we first pick a basis $\{\vec{e}_\alpha\}$ of E, so that for any vector $\vec{v}$ in E, we have $\vec{v} = v^\beta \vec{e}_\beta$. It then follows that $\underline{\omega}(\vec{v}) = \omega_\beta v^\beta$, where ω_β denote the components of $\underline{\omega}$ in the $\{\underline{e}^\alpha\}$ basis. Now, take $\vec{\omega} \in E$ to be the vector whose components ω^α in the basis $\{\vec{e}_\alpha\}$ are given by

$$\boxed{\omega^\alpha := g^{\alpha\beta} \omega_\beta,}$$

with $g^{\alpha\beta}$ the inverse of the matrix of g's components in the basis $\{\vec{e}_\alpha\}$.[7] We claim that this is the vector we're looking for. To see this, multiply both sides of the boxed equation by the matrix $g_{\alpha\gamma}$ to get $g_{\alpha\gamma}\omega^\alpha = \omega_\gamma$. The equation $\underline{\omega}(\vec{v}) = \omega_\beta v^\beta$ then becomes

$$\underline{\omega}(\vec{v}) = g_{\alpha\beta}\omega^\alpha v^\beta = g(\vec{\omega}, \vec{v}),$$

and since $\vec{v}$ is arbitrary, this is precisely Eq. (5.1). We therefore have

$$\Phi_g^{-1} : E^* \to E, \qquad \underline{\omega} \mapsto \vec{\omega}, \quad \text{where } \vec{\omega} \text{ is the vector s.t. } \forall\, \vec{v} \in E, \;\; \underline{\omega}(\vec{v}) = g(\vec{\omega}, \vec{v}).$$

This vector has the components given above.

As a final note, Φ_g and Φ_g^{-1} are sometimes referred to as *musical isomorphisms*, so-called because the symbols $\flat$ and $\sharp$ are used, and for instance $\Phi_g(v) \equiv v^\flat$. The use of flat and sharp symbols might make more sense once we discuss abstract index notation, where these two mappings raise and lower indices.

[6] A proof that this is an isomorphism can be found in Gourgoulhon.

[7] Invertibility follows from the fact that g is nondegenerate! This is one reason we impose the nondegeneracy condition on our metric.

Worldlines

We identify a particle with its worldline, which is a curve in $\mathcal{E}$. Here we make a distinction between the physics and the mathematics; these curves correspond to real-life particles. Indeed, not all curves on $\mathcal{E}$ are cut out to be worldlines.

> Any *massive particle* is represented by a smooth curve $\mathscr{L}$ of Minkowski spacetime such that any vector tangent to $\mathscr{L}$ is timelike.

One can take this to be the definition of a massive particle. Given a parametrization $\varphi : \mathbb{R} \to \mathcal{E}$ of $\mathscr{L}$, where $\lambda \mapsto A(\lambda)$, we may define the associated tangent vector field $\vec{v}(\lambda)$ by

$$\vec{v}(\lambda) := \lim_{\epsilon \to 0} \frac{1}{\epsilon}\, \overrightarrow{A(\lambda)A(\lambda+\epsilon)}.$$

The object on the right is clearly a four-vector, and $\overrightarrow{A(\lambda)A(\lambda+\epsilon)}$ is often written $d\vec{x}$, so that

$$\boxed{\vec{v}(\lambda) := \frac{d\vec{x}}{d\lambda}.}$$

With an affine frame $(\mathcal{O}, \{\vec{e}_\alpha\})$, we can make the identification

$$\varphi(\lambda) \longrightarrow x^\alpha(\lambda)$$

so instead of thinking in terms of an abstract mapping, we can work with four components,

$$\vec{v}(\lambda) = \frac{dx^\alpha}{d\lambda}\vec{e}_\alpha.$$

Note that all this discussion relied on the affine structure of Minkowski space; curved space comes with its own subtleties.

Proper time along a worldline

Given two events A, B on a worldline with $A = \varphi(\lambda_1)$ and $B = \varphi(\lambda_2)$, the proper time between the two is

$$\tau(A,B) := \int_A^B d\tau = \frac{1}{c}\int_{\lambda_1}^{\lambda_2} \sqrt{-g\big(\vec{v}(\lambda), \vec{v}(\lambda)\big)}\, d\lambda,$$

where $\vec{v}(\lambda)$ is the tangent vector field associated with φ. If we want to work with components, we may pick a basis to get

$$\begin{aligned} g\big(\vec{v}(\lambda), \vec{v}(\lambda)\big) &= g\left(\frac{dx^\alpha}{d\lambda}\vec{e}_\alpha, \frac{dx^\beta}{d\lambda}\vec{e}_\beta\right) \\ &= \frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} g(\vec{e}_\alpha, \vec{e}_\beta), \end{aligned}$$

so by recalling that $g_{\alpha\beta} = g(\vec{e}_\alpha, \vec{e}_\beta)$, we obtain the alternative expression

$$\tau(A,B) = \frac{1}{c}\int_{\lambda_1}^{\lambda_2} \sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}}\, d\lambda.$$

One can think of a timelike worldline as carrying its own ideal clock, which indicates the proper time along that worldline.

Four-acceleration

Recall that we had

$$\vec{u} = \frac{d\vec{x}}{d\tau}.$$

Mathematically speaking, these are the tangent vectors you get when you parametrize a worldline by its proper time. This was not defined in the context of a frame or observer – this is because the 4-velocity is an absolute quantity that depends only on the worldline. We now introduce the notion of *four-acceleration*,

$$\boxed{\vec{a} := \frac{d\vec{u}}{d\tau}.}$$

Before we look at properties of $\vec{a}$, we quickly note that $\vec{u} \neq \vec{v}$, since the former represents a very specific parametrization. In fact,

$$\vec{v} \cdot \vec{v} = g_{\alpha\beta}\, v^\alpha v^\beta,$$

$$\begin{aligned} \text{while} \quad \vec{u} \cdot \vec{u} &= g_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \\ &= -c^2. \end{aligned}$$

To see why this is true, we look at the definition of $d\tau$:

$$\begin{aligned} d\tau &= \left(-g_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} \right)^{1/2} d\lambda \\ &= \left(-g_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \right)^{1/2} d\tau \\ &= (-g_{\alpha\beta}\, dx^\alpha dx^\beta)^{1/2} \end{aligned}$$

where we chose our parameter to be τ in the second line. From here, we just need to recognize the expression in parentheses as -1 times the square of the norm of $d\vec{x}$.

It is convenient to work in units where $c = 1$ (a.k.a geometric units), so that

$$\boxed{\vec{u} \cdot \vec{u} = -1.}$$

This shows that $\vec{u}$ is always normalized. This choice should be reminiscent of taking arc length as parameter; it affords us simpler formulas. Note that we can get $\vec{u}$ from an arbitrary $\vec{v}$ via normalization:

$$\vec{u} = \frac{\vec{v}}{\sqrt{-g(\vec{v}, \vec{v})}}.$$

Now we establish a couple of properties of the 4-acceleration $\vec{a}$.

Proposition.

(1) $\vec{a}$ is orthogonal to $\vec{u}$.

(2) $\vec{a}$ is either the zero vector or a spacelike vector.

Proof. Proposition (1) follows from a direct computation:

$$\vec{a} \cdot \vec{u} = \frac{d\vec{u}}{d\tau} \cdot \vec{u} = \frac{1}{2} \frac{d}{d\tau}(\vec{u} \cdot \vec{u}) = \frac{1}{2} \frac{d}{d\tau}(-1) = 0.$$

For (2), suppose that $\vec{a} \neq 0$. We will show that $\vec{a}$ must then be spacelike. Take $\vec{u}$ and $\vec{a}$, which are orthogonal by (1), and complete to an orthogonal basis $\{\vec{u}, \vec{a}, \vec{e}_1, \vec{e}_2\}$. Then the matrix representation of g with respect to this basis is

$$(g_{\alpha\beta}) = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \vec{a}\cdot\vec{a} & 0 & 0 \\ 0 & 0 & \vec{e}_1\cdot\vec{e}_1 & 0 \\ 0 & 0 & 0 & \vec{e}_2\cdot\vec{e}_2 \end{pmatrix}.$$

By Sylvester's law of inertia, the lower three diagonal terms must be positive – in particular, we have $\vec{a}\cdot\vec{a} > 0$. □

This result shows that

$$\boxed{\vec{a}\cdot\vec{a} \geq 0}$$

where equality holds if and only if $\vec{a} = 0$. Now, there is a more general result hiding in the proof of (2). All we needed to use were the orthogonality of $\vec{u}$ and $\vec{a}$, as well as the fact that $\vec{u}$ was timelike, so we conclude that

Any vector orthogonal to a timelike vector is necessarily spacelike or zero.

One more remark – photons and other massless particles are represented by *null geodesics*. In Minkowski, we can think of these as "straight lines" whose tangent vectors are null. Not all null curves are geodesics! For an example, consider the curve with parametrization

$$\begin{aligned} x^0(\lambda) &= r\lambda, \\ x^1(\lambda) &= r\cos\lambda, \\ x^2(\lambda) &= r\sin\lambda, \\ x^3(\lambda) &= 0. \end{aligned}$$

We then have

$$\begin{aligned} \vec{v}(\lambda)\cdot\vec{v}(\lambda) &= -r^2 + (-r\sin\lambda)^2 + (r\cos\lambda)^2 + 0^2 \\ &= -r^2 + r^2 \\ &= 0, \end{aligned}$$

so all the tangent vectors are null, but this curve isn't a "straight line".

6 Simultaneity; observers; time dilation

In this lecture, we talk about the SR notion of simultaneity, we define what an observer is, and then we give a geometric derivation of the time dilation effect in special relativity. The final expression we get is rather general – maybe more general than we need for most purposes, but the lesson here is in seeing how this result is obtained.

So far, we have not given a formal definition of what an observer is. We can give a heuristic one for now: observers are physical frames associated with worldlines. This differs from the purely mathematical notion of an affine frame.

Simultaneity

The only measure along a worldline we've discussed so far is proper time τ. We saw that parametrizing by τ allows us to get a normalized, timelike tangent vector, which we call the four-velocity $\vec{u}$. So here is the next question – how do we define 'time' away from the worldline? We do not want to introduce any new structures if we can avoid it, so we want to work with the affine properties of Minkowski, as well as the invariant light cone provided to us by the metric g.

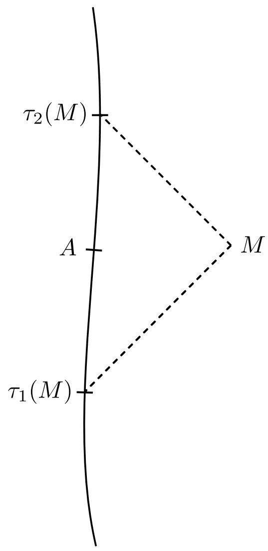


Figure 6: Defining times away from the worldline using the light cone.

Here is how we'll do this. Take an arbitrary worldline $\mathscr{L}^0$. Imagine an arbitrary event M and its associated light cone. Consider the points of the cone that intersect the worldline to the past and to the future; call these $\tau_1(M)$ and $\tau_2(M)$, respectively. Then for this arbitrary point M, we let

$$\boxed{t(M) = \frac{1}{2}\Big(\tau_1(M) + \tau_2(M)\Big).}$$

This gives us our notion of simultaneity – all events M such that

$$t(M) = \tau =: \tau(A)$$

are said to define a so-called *simultaneity hypersurface* $\Sigma_{\vec{u}}(A)$ for A. (The subscript $\vec{u}$ here denotes the four-velocity at A.) Since different observers correspond to different simultaneity hypersurfaces, we see that there is no notion of 'absolute time' in special relativity.

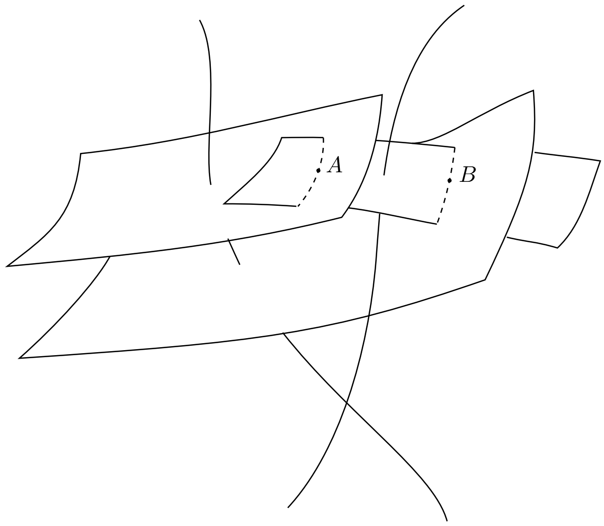


Figure 7: The events A and B are simultaneous according to the observer on the right, but not according to the one on the left. (Don't look too closely – the simultaneity hypersurfaces are almost certainly drawn inaccurately.) (this is cute, but observers assign coordinates using their local rest spaces, not their simultaneity hypersurfaces. use the figure from Gourgoulhon instead)

Let us see what we can do with this new quantity $t(M)$. In what follows, we will be referencing Figure 8. Take a point A on an arbitrary worldline $\mathscr{L}_0$, and consider a point B that is 'close' to A. Then draw the light cone associated with B and mark off the intersections A_1 and A_2. 'Close' has a precise mathematical meaning, but we'll get to this in a bit. For now, we can take this to mean that the curvature $a = \sqrt{\vec{a} \cdot \vec{a}} =: \|\vec{a}\|$ can be neglected from A_1 to A_2.[8]

Now consider $\overrightarrow{A_1 A}$ and $\overrightarrow{A_2 A}$. By the previous assumption, these are approximately collinear, so

$$\overrightarrow{A_1 A} \approx (\tau - \tau_1)\vec{u}(A),$$
$$\overrightarrow{A_2 A} \approx (\tau - \tau_2)\vec{u}(A),$$

since $\|\vec{u}(A)\| = 1$. Also, $\overrightarrow{A_1 B}$ is null by construction, and from the drawing we see that

$$\overrightarrow{A_1 B} = \overrightarrow{A_1 A} + \overrightarrow{AB}.$$

Using this last fact, we have

$$\begin{aligned} & \overrightarrow{A_1 B} \cdot \overrightarrow{A_1 B} = 0 \\ \iff \quad & \overrightarrow{A_1 A} \cdot \overrightarrow{A_1 A} + 2\overrightarrow{A_1 A} \cdot \overrightarrow{AB} + \overrightarrow{AB} \cdot \overrightarrow{AB} = 0 \\ \iff \quad & -(\tau - \tau_1)^2 + 2(\tau - \tau_1)\vec{u}(A) \cdot \overrightarrow{AB} + \overrightarrow{AB} \cdot \overrightarrow{AB} = 0. \end{aligned}$$

We can do the same for $\overrightarrow{A_2 B}$, which is also null by construction, to obtain

$$-(\tau - \tau_2)^2 + 2(\tau - \tau_2)\vec{u}(A) \cdot \overrightarrow{AB} + \overrightarrow{AB} \cdot \overrightarrow{AB} = 0.$$

[8] This would be exact if the worldline were in fact a straight line.

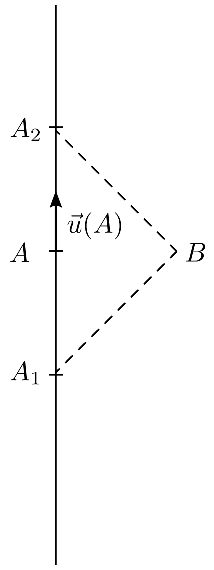


Figure 8: Examining points 'close to' the worldline. This has been drawn as a vertical line, but the derivation only requires that the worldline's curvature be sufficiently small; the line could be tilted at an arbitrary angle $< 45°$. Also, while this picture has been drawn with A_1 to the past and A_2 to the future of A, this is not necessary for the derivation. (Note that this picture corresponds to a case where $\overrightarrow{AB}$ is spacelike; this ensures that Synge's formula is well-defined.)

Then we can solve to get the following:

$$\vec{u}(A) \cdot \overrightarrow{AB} = \tau - \frac{1}{2}(\tau_1 + \tau_2)$$

$$\overrightarrow{AB} \cdot \overrightarrow{AB} = (\tau - \tau_1)(\tau_2 - \tau)$$

So this is quite neat – we've found a way to express both of these quantities solely in terms of proper times measurable by the observer. (But remember our initial assumption that the curvature was negligible.) This tells us a couple of things. First, if B is simultaneous to A, then

$$\vec{u}(A) \cdot \overrightarrow{AB} = 0.$$

In other words, all events B in the neighborhood of A that are simultaneous to A will define a hyperplane called the *local rest space* $\mathcal{E}_{\vec{u}}(A)$ of $\mathcal{O}$ at A. You'll see a similar notion in O'Neill. Note, however, that this local plane is not the same as the simultaneity hypersurface $\Sigma_{\vec{u}}(A)$; it is only tangent to it.

Here we can also revisit what we meant by 'close' – by retaining higher-order correction terms, you can show that this means

$$\frac{\|\overrightarrow{AB}\|}{R_a} \ll 1 \qquad \Longleftrightarrow \qquad \|\overrightarrow{AB}\|\,\|\vec{a}\| \ll 1.$$

Loosely speaking, you need to go out a certain distance to 'notice' that the curve is not straight.

The second thing we get from this is *Synge's formula*,

$$\|\overrightarrow{AB}\| = \sqrt{(\tau - \tau_1)(\tau_2 - \tau)},$$

which appears often in many guises. The expression on the right is only well-defined when $\overrightarrow{AB}$ is spacelike, since this ensures that the two quantities under the square root are positive. An important takeaway from this is that length is a *derived* notion; we start with time, which is a geometric object.

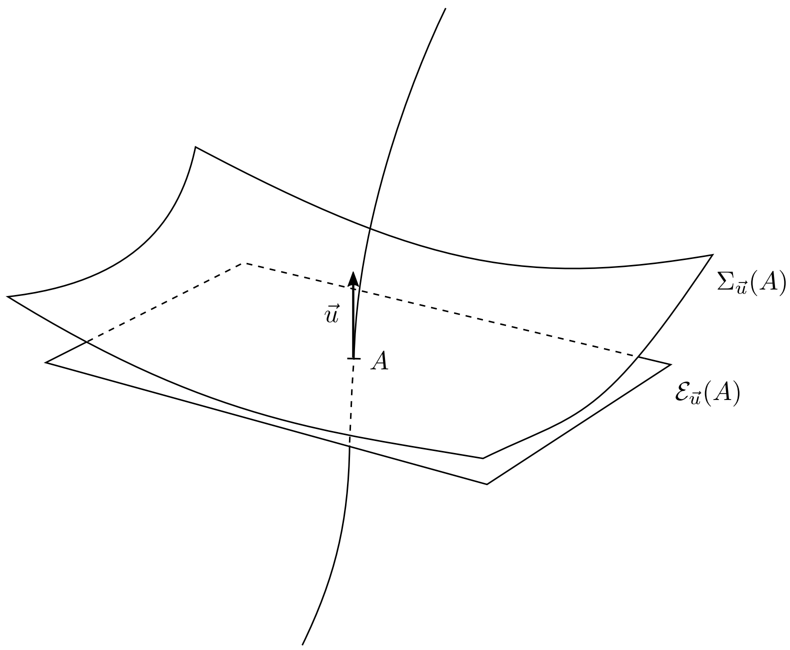


Figure 9: A schematic picture of the local rest space $\mathcal{E}_{\vec{u}}(A)$ and simultaneity hypersurface $\Sigma_{\vec{u}}(A)$ of a point A on a worldline.

Local frame of an observer

A general affine frame is a mathematical artifice; a local frame differs from this in that coordinates are measured by an observer. A timelike observer corresponds to a timelike worldline, say $\mathscr{L}_0$. We require an orthonormal basis $\{\vec{e}_\alpha(\tau)\}$ on $\mathscr{L}_0$,[9] with a special choice for the first basis vector:

$$\boxed{\vec{e}_0(\tau) = \vec{u}(\tau).}$$

That is, the time direction is in the same direction as the four-velocity. To summarize,

$$\boxed{\text{An observer “=” } \mathscr{L}_0 \ + \ \{\vec{e}_\alpha(\tau)\} \text{ on } \mathscr{L}_0.}$$

The remaining three basis vectors are harder to find in curved spacetimes, but the idea is here. (In Minkowski space, we can just use the Gram-Schmidt procedure.) Another condition we require is that the $\vec{e}_\alpha(\tau)$ be differentiable, but we won't need this now.

How do we assign coordinates using a local frame? (See Figure 10.) Consider an event M that is sufficiently close that it only lies in one local rest space. For the first coordinate, we choose the time attached to the local plane intersecting M,

$$x^0(M) = t(M) = \tau.$$

For the remaining three, we consider the vector in the local rest space of $\mathcal{O}(\tau)$ to get

$$\overrightarrow{\mathcal{O}(\tau)M} \in \mathcal{E}_{\vec{u}}(\tau) \quad \text{and} \quad \overrightarrow{\mathcal{O}(\tau)M} = x^i \vec{e}_i(\tau).$$

These numbers (t, x^i) are called *Fermi normal coordinates.* This construction is very useful for any spacetime.[10] More generally, when we assign coordinates in GR, it will always be in the context of projections onto local rest spaces.

[9] One also needs to specify an orientation, but we won't concern ourselves with that right now.

[10] The more general lesson here is that SR is not just about inertial observers. Our discussion has dealt with arbitrary worldlines – not just straight lines in Minkowski.

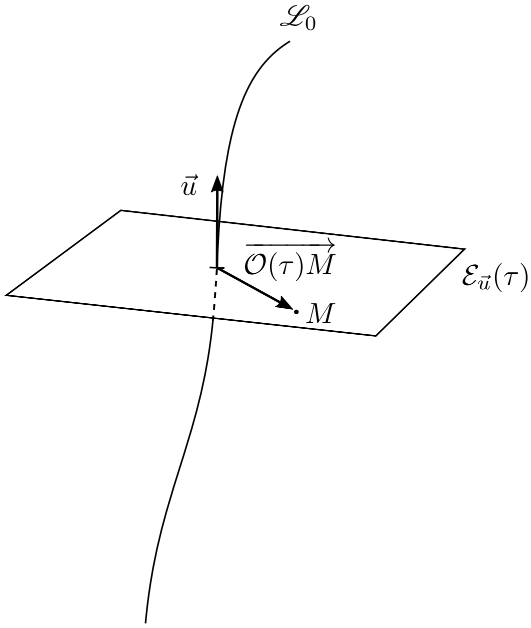


Figure 10: Assigning (Fermi normal) coordinates to an event using a local frame.

Time dilation

Now we'll use this formalism in an example. Suppose we have two worldlines $\mathscr{L}$ and $\mathscr{L}'$ close enough that curvature can be neglected (see Figure 11). Consider two events $\mathcal{O}(\tau)$ and $\mathcal{O}(\tau+d\tau)$ that occur along $\mathscr{L}$, and consider the events simultaneous to these along $\mathscr{L}'$, which we call $M(\tau)$ and $M(\tau+d\tau)$ respectively. Observer $\mathcal{O}$ measures a proper time of $d\tau$ between the events, but observer P is equipped with her own clock, which she uses to measure the proper time between $M(\tau)$ and $M(\tau+d\tau)$. This time interval $d\tau'$ is not in general equal to $d\tau$. Instead,

$$\boxed{d\tau = \Gamma\, d\tau'.}$$

What we want is to determine the factor Γ relating the proper times measured by the two.

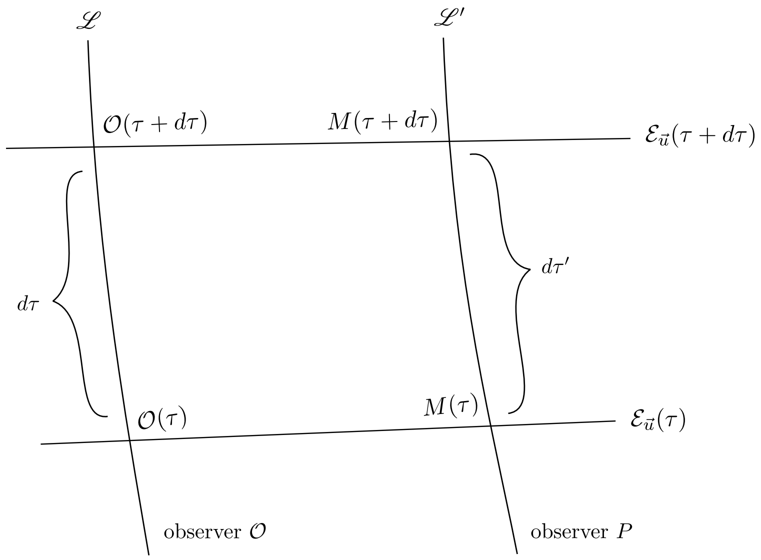


Figure 11: Relating the proper times measured by two observers. Note that the events $\mathcal{O}(\tau)$ and $M(\tau)$ are not necessarily simultaneous according to observer P, and the same goes for $\mathcal{O}(\tau + d\tau)$ and $M(\tau + d\tau)$.

First we can consider a familiar special case. Suppose both worldlines are straight lines in Minkowski space. Then the diagram (tilting your head appropriately) just looks like this:

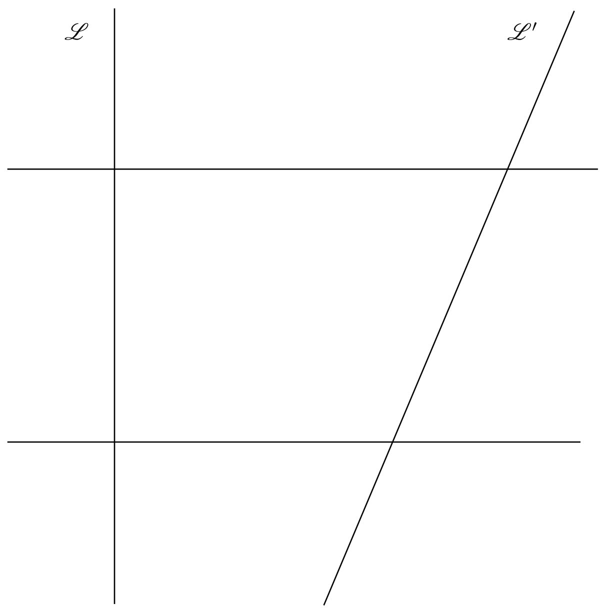


Figure 12: Two simply related observers; their worldlines are straight lines in Minkowski spacetime.

We may parametrize $\mathscr{L}'$ by the proper time measured by $\mathscr{L}$ to get

$$x(t) = vt,$$
$$y(t) = 0,$$
$$z(t) = 0.$$

Then from τ to $\tau + d\tau$, the corresponding displacement along $\mathscr{L}'$ is

$$dx^\alpha = (dt, v\,dt, 0, 0).$$

Thus, the proper time measured by P is

$$\begin{aligned} d\tau' &= \sqrt{(dx^0)^2 - (dx^1)^2 - \cancel{(dx^2)^2} - \cancel{(dx^3)^2}} \\ &= \sqrt{dt^2 - v^2\,dt^2} \\ &= dt\,\sqrt{1 - v^2}. \end{aligned}$$

So in this case, we have $\Gamma = 1/\sqrt{1-v^2} > 1$. In particular, this means that $d\tau' < d\tau$. But we would like to generalize, since this derivation was very coordinate-based. So let's revisit the previous diagram (Figure 11). From our work earlier in this lecture, we know that

$$\vec{u}(\tau + d\tau) \cdot \overrightarrow{\mathcal{O}(\tau + d\tau)M(\tau + d\tau)} = 0.$$

We can find alternative ways of writing these two vectors. First, we have

$$\vec{u}(\tau + d\tau) = \vec{u}(\tau) + d\tau\,\vec{a}(\tau) + \mathcal{O}(d\tau^2).$$

Also observe that

$$\begin{aligned} \overrightarrow{\mathcal{O}(\tau + d\tau)M(\tau + d\tau)} &= \overrightarrow{\mathcal{O}(\tau + d\tau)\mathcal{O}(\tau)} + \overrightarrow{\mathcal{O}(\tau)M(\tau)} + \overrightarrow{M(\tau)M(\tau + d\tau)} \\ &= -d\tau\,\vec{u}(\tau) + \overrightarrow{\mathcal{O}M} + d\tau'\,\vec{u}'(\tau), \end{aligned}$$

where the first line corresponds to going down, right, then up on the diagram, and where the second line follows from the same reasoning used to obtain the previous equation. Note that $\vec{u}'(\tau)$ refers to the four-velocity of $\mathscr{L}'$ at $M(\tau)$, as parametrized by $\mathcal{O}$'s proper time.

Substituting both of these into the equation expressing orthogonality, we get

$$\begin{aligned} & \left[\vec{u}(\tau) + d\tau\,\vec{a}(\tau)\right] \cdot \left[-d\tau\,\vec{u}(\tau) + \overrightarrow{\mathcal{O}M} + d\tau'\,\vec{u}'(\tau)\right] = 0 \\ \iff \quad & d\tau + \cancel{\vec{u}(\tau) \cdot \overrightarrow{\mathcal{O}M}} + d\tau\,\vec{a}(\tau) \cdot \overrightarrow{\mathcal{O}M} + d\tau'\,\vec{u} \cdot \vec{u}' = 0 \\ \iff \quad & d\tau(1 + \vec{a} \cdot \overrightarrow{\mathcal{O}M}) = -d\tau'\vec{u} \cdot \vec{u}', \end{aligned}$$

where we've used the facts that $\vec{u} \cdot \vec{u} = -1$ and $\vec{a}(\tau) \cdot \vec{u}(\tau) = 0$, and where we've discarded terms higher than first order in $d\tau$. Thus, we see that

$$\boxed{\Gamma = -\frac{\vec{u} \cdot \vec{u}'}{1 + \vec{a} \cdot \overrightarrow{\mathcal{O}M}}.}$$

Everything here is expressed in terms of four-vectors and our metric (as embodied by the inner product). Therefore, this is a geometric formula, which is good. There are two special cases to take note of here – these are (1) when the worldlines intersect, so that $\overrightarrow{\mathcal{O}M}$ vanishes, and (2) when $\vec{a} = 0$, which is what you dealt with in P108. Both of these lead to

$$\boxed{\Gamma = -\vec{u} \cdot \vec{u}' \quad \text{(special case)}}$$

Case (1) is what we'll see in GR, since measurements are always made locally. It is important to note that this factor is symmetric in $\vec{u}$ and $\vec{u}'$. This implies that both observers will see the other observer's clock run at a rate differing by the same factor Γ. We will use this fact extensively in the next couple of lectures.

A final comment – the geometric formula we obtained is usually only of interest to geometers and relativists, and we won't have use for this extra-general result. But it is not the result that matters; the lesson here is in seeing how it was derived.

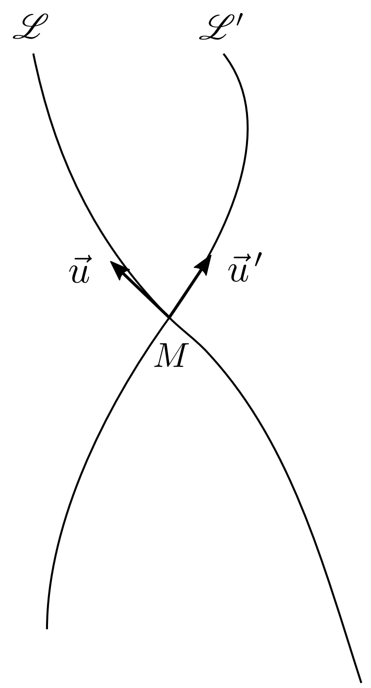


Figure 13: If the worldlines intersect ($\mathcal{O} = M$), the time dilation factor takes on an especially simple form: $\Gamma = -\vec{u} \cdot \vec{u}'$.

7 Relating velocities measured by observers

Recall that in the special case we considered last lecture (intersecting worldlines), the expression we got for Γ reduced to

$$\Gamma = -\vec{u} \cdot \vec{u}'.$$

This is usually what we'll work with in GR. Now, we started the derivation by saying that we would obtain the 'time dilation factor' in a geometric manner, but we didn't actually show that there was any dilation going on. To do this, we can express $\vec{u}'$ in terms of the tetrad carried by $\mathscr{L}$ to get $\vec{u}' = u'^0\,\vec{u} + u'^i\,\vec{e}_i$. Then we have

$$\begin{aligned}\Gamma &= -\vec{u} \cdot (u'^0\,\vec{u} + u'^i\,\vec{e}_i)\\ &= u'^0.\end{aligned}$$

We may also write $\vec{u}' \cdot \vec{u}' = -1$ in terms of these components to get

$$-\left(u'^0\right)^2 + \sum \left(u'^i\right)^2 = -1,$$

and since $u'^0 > 0$, we have

$$\Gamma = u'^0 = \sqrt{1 + \sum (u'^i)^2} \geq 1.$$

This explains why we say 'dilation'. Again, we made no assumptions here about the worldlines, other than that they were timelike.

Now we want to determine the relative velocities measured by observers in a geometric way.

Reference space of an observer

First we'll introduce the notion of the *reference space* R_O of an observer. Given an event M that is sufficiently close to the observer's worldline so that only one local space intersects it, the observer O may assign to it coordinates (t, x^i) in the manner discussed earlier (see lecture 6). Then there is a mapping φ that identifies the last three components with a vector in R_O:

$$\begin{aligned}\varphi : \mathcal{E} &\to R_O &&\text{(a 3-dimensional Euclidean space)},\\ M(t, x^i) &\mapsto \vec{x} = x^i\,\vec{e}_i &&\text{(a 3-vector)}.\end{aligned}$$

To get a better idea of what this is, note that the three spacelike vectors of O's frame remain fixed in R_O, whereas they evolve with t in Minkowski space. This is sort of like sitting in a box and making measurements using a coordinate grid fixed to the box; for all I know, the box and its grid could be twirling around in space, but I can still measure spatial coordinates and motion.

Projecting onto local rest spaces

We want to find a formula for projecting onto local rest spaces, so we start by writing an arbitrary vector $\vec{v}$ as the sum of a vector that is orthogonal to $\vec{u}$ and a vector parallel to $\vec{u}$.

$$\forall \vec{v}, \qquad \vec{v} = \perp_{\vec{u}} \vec{v} + \alpha \vec{u}$$

We'll define the first vector on the right to be such that $\perp_{\vec{u}} \vec{v} \in \mathcal{E}_{\vec{u}}(\tau)$. Then we have

$$\vec{u} \cdot \vec{v} = \cancel{\vec{u} \cdot \left(\perp_{\vec{u}} \vec{v}\right)} + \alpha\,\vec{u} \cdot \vec{u},$$

which implies that $\alpha = -\vec{u}\cdot\vec{v}$. Thus, in general we have

$$\vec{v} = \perp_{\vec{u}}\vec{v} - (\vec{u}\cdot\vec{v})\vec{u}.$$

Rearranging, this tells us what our projection operator does:

$$\boxed{\perp_{\vec{u}}\vec{v} = \vec{v} + \vec{u}(\vec{u}\cdot\vec{v}).}$$

We may then write

$$\perp_{\vec{u}} = \mathrm{Id} + \vec{u}\, g(\vec{u},\,\cdot\,).$$

Abstract index notation

This last formula is rather clunky; we need to specify that this operator accepts vectors. There is an alternative writing that many mathematical physicists and members of the Chicago school[11] prefer – this is called *abstract index notation* and is due to Penrose:

$$\begin{aligned} g(\,\cdot\,,\,\cdot\,) &\longrightarrow g_{ab} \\ \underline{u}(\,\cdot\,) &\longrightarrow u_a \\ \vec{v}(\,\cdot\,) &\longrightarrow v^b \end{aligned}$$

Downstairs indices signify that the object accepts vectors in those slots, while upstairs indices signify that it accepts one-forms. The number of indices tells you the number of arguments the object accepts to return a real number.

For example, in abstract index notation, our projection operator becomes

$$\perp_{\vec{u}} \longrightarrow \left(\perp_{\vec{u}}\right)^a{}_b = \delta^a{}_b + u^a u_b.$$

This also makes it easier to verify certain identities. For example, if we want to show that the composition of a projection operator with itself equals itself, we have

$$\begin{aligned} \left(\perp_{\vec{u}}\right)^a{}_b\left(\perp_{\vec{u}}\right)^b{}_c &= (\delta^a{}_b + u^a u_b)(\delta^b{}_c + u^b u_c) \\ &= \delta^a{}_c + u^a u_c + u^a u_c + u^a \underbrace{u_b\, u^b}_{=-1} u_c \\ &= \delta^a{}_c + u^a u_c \\ &= \left(\perp_{\vec{u}}\right)^a{}_c \end{aligned}$$

If this takes time to sink in, don't worry! Eventually, the notation will come to feel quite natural.

Relating velocities

As was mentioned earlier, we may view the vector in Fig. 14 as part of the reference space of O:

$$\overrightarrow{O(\tau)M(\tau)} = x^i(\tau)\,\vec{e}_i(\tau).$$

[11] Which was practically defined by Robert Wald and Robert Geroch (actually, double-check), both relativists associated with the University of Chicago.

Now we define the velocity of M with respect to O to be

$$\vec{v}_{(3)} := \frac{dx^i}{d\tau}\,\vec{e}_i(\tau).$$

We've previously used this notation to denote a 3-vector, so note here that $\vec{v}_{(3)}$ is a four-vector, which you can see because it is a linear combination of four-vectors.[12] Note also that in general,

$$\vec{v}_{(3)} \neq \frac{d\overrightarrow{OM}}{d\tau}.$$

This is because the basis vectors $\vec{e}_\alpha(\tau)$ are (generally) changing with time, so when $d/d\tau$ acts on these, there are additional terms resulting from this motion. We may say that

$$\frac{d\overrightarrow{OM}}{d\tau} = \vec{v}_{(3)} + \text{“non-inertial terms”}.$$

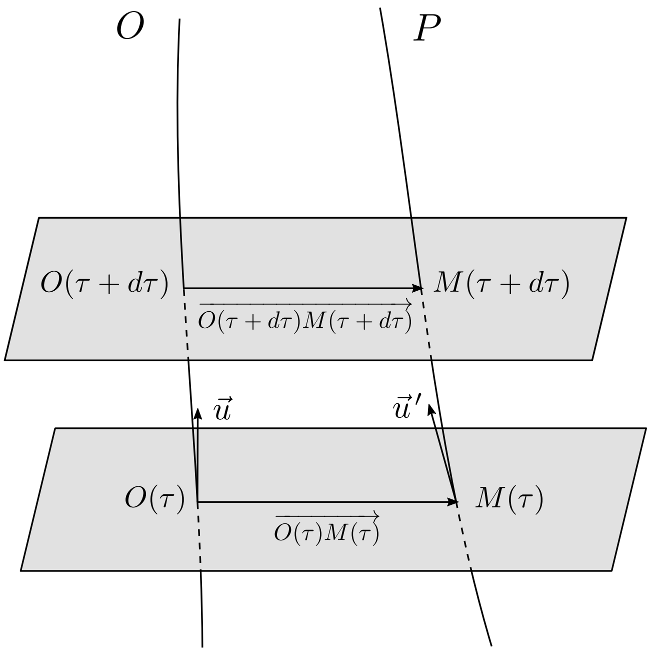


Figure 14: Diagram for relating the velocities of observer P to those of observer O, with the simultaneity hyperplanes of O marked in grey.

However, $\vec{v}_{(3)}$ and the $\overrightarrow{OM}$ derivative *are* equal in the case of inertial observers, and also in the case where $\overrightarrow{OM} = 0$.[13] These happen to be the cases of interest for us, so we can take the two to be equal, but remember that this equality does not hold for more general observers.

Now consider the situation in Fig. 14, where we have observers O and P. One can see that

$$\begin{aligned} d\overrightarrow{OM} &= \overrightarrow{O(\tau + d\tau)M(\tau + d\tau)} - \overrightarrow{O(\tau)M(\tau)} \\ &= \overrightarrow{O(\tau + d\tau)O(\tau)} + \cancel{\overrightarrow{O(\tau)M(\tau)}} + \overrightarrow{M(\tau)M(\tau + d\tau)} - \cancel{\overrightarrow{O(\tau)M(\tau)}} \\ &= -\vec{u}(\tau)\,d\tau + \vec{u}\,'(\tau)d\tau', \end{aligned}$$

[12] But the subscript makes sense because $\vec{v}_{(3)}$ sits in a three-dimensional subspace of E, namely $\mathcal{E}_{\vec{u}}$. This will be clearer in a moment.

[13] That is, if the worldlines of O and P coincide initially. (It might be better to call this ‘observer M’ for consistency. After all, O also stands for an event on O's worldline. . . but M is an event on P's worldline?)

and by writing $d\tau = \Gamma\, d\tau'$ we find

$$\frac{d\overrightarrow{OM}}{d\tau} = \frac{\vec{u}'}{\Gamma} - \vec{u}$$

which we can rearrange to get

$$\vec{u}' = \Gamma\left(\vec{u} + \frac{d\overrightarrow{OM}}{d\tau}\right).$$

As mentioned earlier, in the cases we're interested in, this then becomes

$$\boxed{\vec{u}' = \Gamma(\vec{u} + \vec{v}_{(3)}).} \qquad \text{(4-vel in terms of } \vec{v}_{(3)}\text{)}$$

Taking the norms of both sides, we get

$$\begin{aligned} -1 &= \Gamma^2(\vec{u} + \vec{v}_{(3)}) \cdot (\vec{u} + \vec{v}_{(3)}) \\ &= \Gamma^2(-1 + 2\cancel{\vec{u} \cdot \vec{v}_{(3)}} + \vec{v}_{(3)} \cdot \vec{v}_{(3)}) \end{aligned}$$

The term in the middle is zero because $\vec{v}_{(3)}$ is a linear combination of $\vec{e}_i$'s. Finally, then, we get

$$\boxed{\Gamma = \frac{1}{\sqrt{1 - \vec{v}_{(3)} \cdot \vec{v}_{(3)}}},} \qquad (\Gamma \text{ in terms of } \vec{v}_{(3)})$$

which is a familiar formula. Again, this expression is only valid when either $M \in \mathscr{L}$ or when O is an inertial observer. In terms of coordinates (t, x^i), we have[14]

$$u^\alpha \overset{*}{=} (1, 0, 0, 0)$$

$$v^\alpha_{(3)} \overset{*}{=} (0, v^1, v^2, v^3)$$

The two formulas we've derived can then be written

$$u'^\alpha = \left(\Gamma,\, \Gamma\, v^i_{(3)}\right)$$

and

$$\Gamma = \left(1 - \sum_{i=1}^{3} (v^i)^2\right)^{-1/2}$$

both of which should be familiar from more elementary treatments of SR. Note that we derived this same expression in lecture 4. Back then, we were careful to point out that it was true in the case of inertial observers. Now, we can generalize a bit and say that it is also applicable when the measurements are made locally by O.

⊛ Maximum relative velocity

We found earlier that $-1 = \Gamma^2(-1 + \vec{v}_{(3)} \cdot \vec{v}_{(3)})$. It follows immediately that $-1 + \vec{v}_{(3)} \cdot \vec{v}_{(3)} < 0$. Reintroducing units, this implies

$$\boxed{\|\vec{v}_{(3)}\| < c.}$$

Therefore,

> The constant c is a strict upper bound for the relative velocity of any massive particle observed locally ($\overrightarrow{OM} = 0$) by any given observer. Moreover, for inertial observers, this result can be extended to distant observations ($\overrightarrow{OM} \neq 0$).

[14]The star signals that this equality only holds in a particular chart; in this case, the comoving frame of O.

⊛ Photon motion

(fill in)

Comparing relative velocities

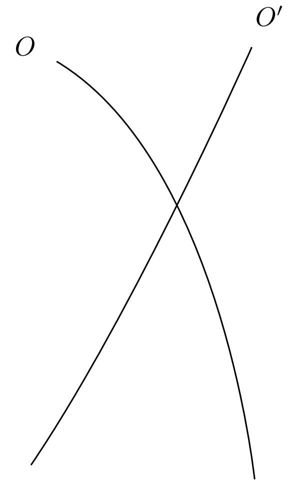


Figure 15: Intersecting worldlines.

Now consider the two worldlines in Fig. 15. We can write the relative velocity of O' with respect to O as

$$\vec{v}_{(3)} = v^i_{(3)}\vec{e}_i,$$

and similarly, the relative velocity of O with respect to O' is

$$\vec{v}'_{(3)} = v'^i_{(3)}\vec{e}'_i.$$

We emphasize that $\vec{v}_{(3)}$ and $\vec{v}'_{(3)}$ live in distinct vector spaces $\mathcal{E}_{\vec{u}}$ and $\mathcal{E}_{\vec{u}'}$, respectively. (In particular, we cannot write something like $\vec{v}_{(3)} = -\vec{v}'_{(3)}$.) Now, by following the procedure we carried out earlier, we get

$$\begin{aligned} \vec{u}' &= \Gamma_0(\vec{u} + \vec{v}_{(3)}) \\ \vec{u} &= \Gamma_0(\vec{u}' + \vec{v}'_{(3)}) \end{aligned}$$

Notice what we've done here – by symmetry, to deal with the other observer we simply had to swap primed and unprimed vectors. But $\Gamma = -\vec{u}\cdot\vec{u}'$ is symmetric with respect to $\vec{u}$ and $\vec{u}'$, so the same factor Γ_0 comes up in both equations.

Using the other boxed equation from earlier, we get

$$\begin{aligned} \Gamma_0 &= \left(1 - \vec{v}_{(3)}\cdot\vec{v}_{(3)}\right)^{-1/2} \\ &= \left(1 - \vec{v}'_{(3)}\cdot\vec{v}'_{(3)}\right)^{-1/2} \end{aligned}$$

and this tells us that $\vec{v}_{(3)}\cdot\vec{v}_{(3)} = \vec{v}'_{(3)}\cdot\vec{v}'_{(3)}$. While we are here, we also observe that

$$1 - \Gamma_0^2 = -\Gamma_0^2\,\vec{v}_{(3)}\cdot\vec{v}_{(3)}, \tag{7.1}$$

which just follows from a bit of algebra. Now we plug in our expression for $\vec{u}'$ from the first equation into the second one to get (we'll drop the subscript on Γ)

$$\begin{aligned}\vec{u} &= \Gamma\Big(\Gamma(\vec{u}+\vec{v}_{(3)}) + \vec{v}'_{(3)}\Big)\\ &= \Gamma^2(\vec{u}+\vec{v}_{(3)}) + \Gamma\vec{v}'_{(3)}\end{aligned}$$

which we can rearrange a bit to get

$$\begin{aligned}\Gamma\vec{v}'_{(3)} &= (1-\Gamma^2)\vec{u} - \Gamma^2\vec{v}_{(3)}\\ &= -\Gamma^2(\vec{v}_{(3)}\cdot\vec{v}_{(3)})\vec{u} - \Gamma^2\vec{v}_{(3)},\end{aligned}$$

where we've simplified using Eq. (7.1). Finally, then, we see that

$$\boxed{\vec{v}'_{(3)} = -\Gamma\Big(\vec{v}_{(3)} + (\vec{v}_{(3)}\cdot\vec{v}_{(3)})\vec{u}\Big)}$$

One can also obtain $\vec{v}_{(3)}$ in terms of $\vec{v}'_{(3)}$ and $\vec{u}'$. But now let's check that this formula makes sense in the non-relativistic limit. Restoring the dimensions of c, we have $\vec{v}_{(3)}\cdot\vec{v}_{(3)} \to (\vec{v}_{(3)}\cdot\vec{v}_{(3)})/c^2$, so as $c\to\infty$ we have $\Gamma\to 1$, and the second term goes to zero. Therefore, in this limit we get

$$\vec{v}'_{(3)} = -\vec{v}_{(3)},$$

which is familiar.

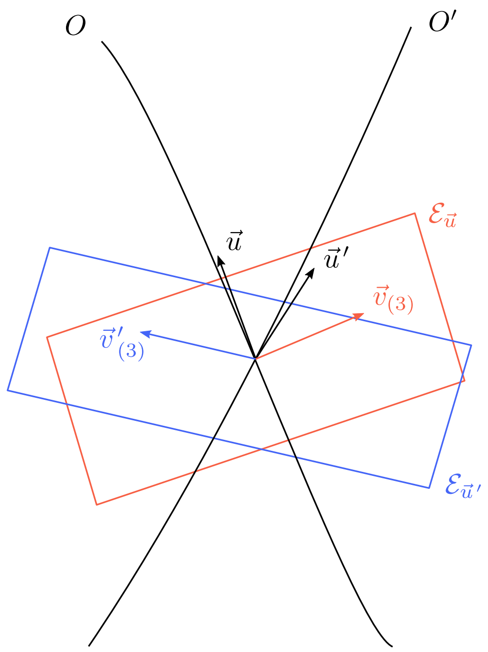


Figure 16: A reminder that the vectors $\vec{v}_{(3)}$ and $\vec{v}'_{(3)}$ live in rest spaces associated with O and O', respectively. In particular, this tells us that $\vec{u}\cdot\vec{v}_{(3)} = 0$, and likewise for the primed observer. Also remember that these relative velocities have the same magnitude!

However, at a glance this formula doesn't seem to tell us much; we see that we can obtain $\vec{v}'_{(3)}$ given $\vec{v}_{(3)}$ and $\vec{u}$, but there's no immediate geometric interpretation here. We can find such an interpretation by looking at the projection operator discussed at the beginning of this lecture.[15]

$$
\begin{aligned}
\perp_{\vec{u}'} \vec{v}_{(3)} &= \vec{v}_{(3)} + (\vec{u}' \cdot \vec{v}_{(3)})\vec{u}' \\
&= \vec{v}_{(3)} + \big[\Gamma(\vec{u} + \vec{v}_{(3)}) \cdot \vec{v}_{(3)}\big]\Gamma(\vec{u} + \vec{v}_{(3)}) \\
&= \vec{v}_{(3)} + \Gamma^2(\vec{v}_{(3)} \cdot \vec{v}_{(3)})(\vec{u} + \vec{v}_{(3)}) \\
&= \vec{v}_{(3)}\Big(1 + \Gamma^2\, \vec{v}_{(3)} \cdot \vec{v}_{(3)}\Big) + \Gamma^2(\vec{v}_{(3)} \cdot \vec{v}_{(3)})\vec{u} \\
&= \Gamma^2\Big(\vec{v}_{(3)} + (\vec{v}_{(3)} \cdot \vec{v}_{(3)})\vec{u}\Big)
\end{aligned}
$$

The second line uses $\vec{u}' = \Gamma(\vec{u} + \vec{v}_{(3)})$, the third line uses $\vec{u} \cdot \vec{v}_{(3)} = 0$, the fourth line is just a regrouping of terms, and the last line follows from Eq. (7.1).

The last thing we want to do is compare this to the formula we obtained earlier. We get

$$\boxed{\vec{v}'_{(3)} = -\frac{1}{\Gamma} \perp_{\vec{u}'} \vec{v}_{(3)}}$$

and by symmetry,

$$\boxed{\vec{v}_{(3)} = -\frac{1}{\Gamma} \perp_{\vec{u}} \vec{v}'_{(3)}}$$

There are a few benefits of using this formula instead of the other one. First, we see that relative velocities are related by a rescaling (by Γ) and a projection onto a local space. This is not at all obvious from our first formula. A second, more practical reason is that it is shorter to write.

[15] And it does not seem unreasonable that projection operators should be involved, because both $\vec{v}_{(3)}$ and $\vec{v}'_{(3)}$ live in rest spaces.

8 Length contraction; velocity addition

In this lecture, we use the geometric formalism developed so far to derive length contraction and a law for velocity composition. These derivations will rely heavily on results from the previous lecture. Grey boxes indicate a main result.

Length contraction

Recall the useful identity from the previous lecture,

$$\vec{u}' = \Gamma_0\big(\vec{u} + \vec{v}_{(3)}\big).$$

Now, $\vec{v}_{(3)}$ is a spatial vector, and we can define a unit vector $\vec{e}$ pointing in the same direction:

$$\vec{e} = \frac{\vec{v}_{(3)}}{\|\vec{v}_{(3)}\|}.$$

Then using the notation $v_{(3)} \equiv \|\vec{v}_{(3)}\|$, we may write

$$\vec{v}_{(3)} = v_{(3)}\,\vec{e}.$$

and by the main result from the previous lecture, we have

$$\vec{v}'_{(3)} = -\frac{v_{(3)}}{\Gamma}\perp_{\vec{u}'}\vec{e}$$

Now, note that we can write (this is totally general)

$$\vec{v}'_{(3)} = v'_{(3)}\,\vec{e}'.$$

By comparing these two expressions, we see that we can make the idenfication

$$\begin{aligned} v'_{(3)} &= -v_{(3)}, \\ \vec{e}' &:= \frac{1}{\Gamma}\perp_{\vec{u}'}\vec{e}. \end{aligned}$$

You might wonder why we use the symbol $\vec{e}'$. Is this also a unit vector? The answer is yes, and this has to do with the fact that these two relative velocities have the same norm. That is, since

$$\vec{v}'_{(3)}\cdot\vec{v}'_{(3)} = \vec{v}_{(3)}\cdot\vec{v}_{(3)} = v^2_{(3)},$$

we may take dot products to get

$$v^2_{(3)} = v^2_{(3)}\,\vec{e}'\cdot\vec{e}'$$

which in turn implies that $\vec{e}'\cdot\vec{e}' = 1$. Now, we'd like to find an explicit expression for $\vec{e}'$ in terms of $\vec{e}$ to use in our derivation. Recall that

$$\vec{v}'_{(3)} = -\Gamma\Big(v_{(3)}\,\vec{e} + v^2_{(3)}\vec{u}\Big)$$

and from what we just discussed, we know that

$$\vec{v}'_{(3)} = -v_{(3)}\,\vec{e}'.$$

Comparing these two, we immediately find

$$\boxed{\vec{e}' = \Gamma\Big(\vec{e} + v_{(3)}\,\vec{u}\Big),}$$

and by symmetry, we also have

$$\boxed{\vec{e} = \Gamma\left(\vec{e}' + v'_{(3)}\,\vec{u}'\right).}$$

Notice what we have – both $\vec{u}'$ and $\vec{e}'$ can be expressed as linear combinations of $\vec{u}$ and $\vec{e}$. (The expression for $\vec{u}'$ was given at the start of this lecture.) So now consider the plane spanned by the unprimed vectors, and consider a ruler being carried by the primed observer.

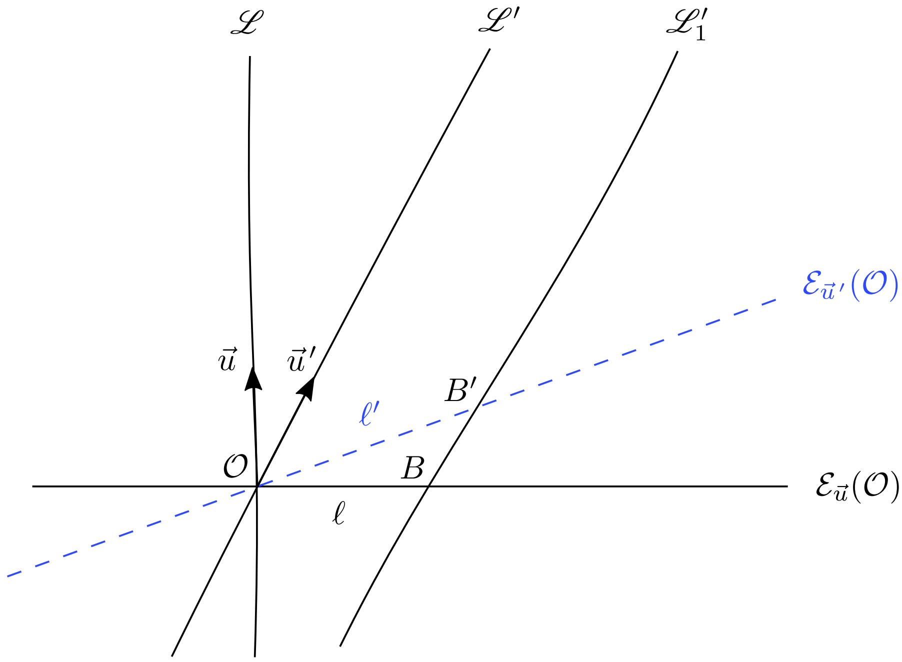


Figure 17: A ruler being carried by $\mathscr{L}'$ appears shorter in the frame of $\mathscr{L}$. (this figure is missing some information)

The two worldlines $\mathscr{L}'$ and $\mathscr{L}'_1$ correspond to the two ends of the ruler. Now, one can see from the picture that

$$\overrightarrow{\mathcal{O}B'} = \ell'\vec{e}'$$
$$\overrightarrow{\mathcal{O}B} = \ell\,\vec{e}$$
$$\overrightarrow{BB'} = \alpha\vec{u}'$$

Then we write the vector $\overrightarrow{\mathcal{O}B'}$ in two different ways:

$$\begin{aligned}\overrightarrow{\mathcal{O}B'} &= \overrightarrow{\mathcal{O}B} + \overrightarrow{BB'}\\ &= \ell\,\vec{e} + \alpha\vec{u}'\end{aligned}$$

and also

$$\begin{aligned}\overrightarrow{\mathcal{O}B'} &= \ell'\vec{e}'\\ &= \ell'\left(\frac{1}{\Gamma}\,\vec{e} + v_{(3)}\vec{u}'\right)\\ &= \frac{\ell'}{\Gamma}\,\vec{e} + \ell' v_{(3)}\vec{u}'\end{aligned}$$

Comparing these two, we see that

$$\boxed{\ell = \ell'/\Gamma \quad \text{(length contraction)}}$$

and since we know from previous discussions that $\Gamma \geq 1$, it follows that the ruler is shorter for $\mathcal{O}$ than it is for $\mathcal{O}'$. This is what we wanted to derive.

Velocity addition

Here we will be talking about three objects, since we already know how to relate velocities measured by two observers. This means that we will need to introduce a bunch of symbols:

$$\begin{aligned}
\vec{v}_{(3)} \quad &— \quad \text{rel. vel. of } \mathcal{P} \text{ with respect to } \mathcal{O} \\
\vec{v}'_{(3)} \quad &— \quad \text{rel. vel. of } \mathcal{P} \text{ with respect to } \mathcal{O}' \\
\vec{u}_{(3)} \quad &— \quad \text{rel. vel. of } \mathcal{O}' \text{ with respect to } \mathcal{O} \\
\vec{u}'_{(3)} \quad &— \quad \text{rel. vel. of } \mathcal{O} \text{ with respect to } \mathcal{O}'
\end{aligned}$$

To keep track of everything, note that a prime denotes a measurement made by the primed observer, that a u denotes a measurement of an observer velocity, and that a v refers to a measurement of the particle velocity. (In principle, one can think of $\mathcal{P}$ as an observer, too, but in what follows, we won't refer to 'measurements made by $\mathcal{P}$'.)

We reemphasize this point to avoid confusion: $\vec{v}_{(3)}$ does not mean the same thing as it did in the previous lecture. Before, we used $\vec{v}_{(3)}$ to refer to the relative velocity of $\mathcal{O}'$ with respect to $\mathcal{O}$, but now we will be using $\vec{u}_{(3)}$ for this.

Our goal will be to express $\vec{v}'_{(3)}$ in terms of $\vec{v}_{(3)}$ and $\vec{u}'_{(3)}$. First we collect a few equations that we will be referring back to often. We know that

$$\vec{v} = \Gamma(\vec{u} + \vec{v}_{(3)}),$$

where

$$\Gamma = -\vec{u} \cdot \vec{v} = (1 - \vec{v}_{(3)} \cdot \vec{v}_{(3)})^{-1/2}.$$

Recall also that (as we established before; an analogous statement holds for the primed vectors)

$$\vec{u} \cdot \vec{v}_{(3)} = 0.$$

Similarly, we can relate the particle 4-velocity to the primed observer via

$$\vec{v} = \Gamma'(\vec{u}' + \vec{v}'_{(3)}),$$

where

$$\Gamma' = -\vec{u}' \cdot \vec{v} = (1 - \vec{v}'_{(3)} \cdot \vec{v}'_{(3)})^{-1/2}.$$

Finally, recall that

$$\Gamma_0 = -\vec{u} \cdot \vec{u}'. \tag{8.1}$$

Now, notice that we've written down two expressions for $\vec{v}$. Here is an idea – why not just equate the two, and then solve for $\vec{v}'_{(3)}$? This gives

$$\vec{v}'_{(3)} = \frac{\Gamma}{\Gamma'}(\vec{u} + \vec{v}_{(3)}) - \vec{u}', \tag{8.2}$$

but this is not what we want, since this mixes rest space vectors with 4-velocities.[16] In search of a geometric formula, we project onto the local rest space $\mathcal{E}_{\vec{u}'}$. The projection operator $\perp_{\vec{u}'}$ does nothing to $\vec{v}'_{(3)}$, so we get

$$\vec{v}'_{(3)} = \frac{\Gamma}{\Gamma'}\Big(\perp_{\vec{u}'} \vec{u} + \perp_{\vec{u}'} \vec{v}_{(3)}\Big)$$

To deal with the first term in parentheses on the right side, recall that we can write

$$\vec{u} = \Gamma_0(\vec{u}' + \vec{u}'_{(3)})$$

so we see immediately that

$$\perp_{\vec{u}'} \vec{u} = \Gamma_0\, \vec{u}'.$$

Next, we want to simplify the factor of Γ/Γ' in front. Recalling that $\vec{u}' \cdot \vec{v}'_{(3)} = 0$, we dot both sides of Eq. (8.2) with $\vec{u}'$ and apply Eq. (8.1) to get

$$0 = \frac{\Gamma}{\Gamma'}(-\Gamma_0 + \vec{u}' \cdot \vec{v}_{(3)}) + 1, \tag{8.3}$$

where we've used $\vec{u}' \cdot \vec{u}' = -1$. Using what we know about the unprimed vectors,

$$\vec{u} \cdot \vec{v}_{(3)} = 0 \quad \text{and} \quad \vec{u}' = \Gamma_0(\vec{u} + \vec{u}_{(3)}),$$

we obtain

$$\begin{aligned}\vec{u}' \cdot \vec{v}_{(3)} &= \Gamma_0(\vec{u} + \vec{u}_{(3)}) \cdot \vec{v}_{(3)} \\ &= \Gamma_0 \vec{u}_{(3)} \cdot \vec{v}_{(3)},\end{aligned}$$

and we substitute this into Eq. (8.3) to obtain

$$0 = \frac{\Gamma}{\Gamma'}(-\Gamma_0 + \Gamma_0 \vec{u}_{(3)} \cdot \vec{v}_{(3)}) + 1$$

It follows immediately that

$$\boxed{\Gamma' = \Gamma\Gamma_0(1 - \vec{u}_{(3)} \cdot \vec{v}_{(3)}).}$$

The last thing we want to do is eliminate $\vec{u}_{(3)}$ from this expression. Recall the two equations

$$\vec{u}'_{(3)} = -\Gamma_0\big[\vec{u}_{(3)} + (\vec{u}_{(3)} \cdot \vec{u}_{(3)})\vec{u}\big]$$

and

$$\vec{u}'_{(3)} = -\frac{1}{\Gamma_0} \perp_{\vec{u}'} \vec{u}_{(3)}.$$

We can dot $\vec{v}_{(3)}$ to both sides of the first equation (recall that $\vec{u} \cdot \vec{v}_{(3)} = 0$) to get

$$\vec{u}'_{(3)} \cdot \vec{v}_{(3)} = -\Gamma_0 \vec{u}_{(3)} \cdot \vec{v}_{(3)},$$

which we can substitute into our boxed equation to get

$$\frac{\Gamma'}{\Gamma} = \Gamma_0\left(1 + \frac{1}{\Gamma_0}\vec{u}'_{(3)} \cdot \vec{v}_{(3)}\right).$$

[16] Well, if we're being pedantic, the final expression we'll get here will also depend on these 4-velocities. If you prefer, you could say that this formula is not ideal because it has no immediate geometric interpretation – recall that we were motivated by similar considerations when deriving the final result of lecture 7.

Finally, since $\vec{u}'_{(3)}$ lies in $\mathcal{E}_{\vec{u}}$, we note that $\vec{u}'_{(3)} \cdot \vec{v}_{(3)} = \vec{u}'_{(3)} \cdot \left(\perp_{\vec{u}'} \vec{v}_{(3)}\right)$.[17] This gives us everything we need – returning to the equation we wanted to simplify, we see that we have

$$\boxed{\vec{v}'_{(3)} = \frac{1}{\Gamma_0 + \vec{u}'_{(3)} \cdot \perp_{\vec{u}'} \vec{v}_{(3)}} \left(\perp_{\vec{u}'} \vec{v}_{(3)} + \Gamma_0 \vec{u}'_{(3)}\right)} \qquad \text{(vel. comp. law)}$$

where $\Gamma_0 = (1 - \vec{u}_{(3)} \cdot \vec{u}_{(3)})^{-1/2}$. Immediately, we observe that the linear combination in parentheses belongs to the rest space of $\mathcal{O}'$, and that this vector gets scaled by the weird factor in front.

To check that this is sensible, we can work out the implications of this formula in the non-relativistic limit. Restoring dimensions, we get $u'_{(3)} \to u'_{(3)}/c$, so as we let $c \to \infty$, we have $u'_{(3)}/c \to 0$ and $\Gamma_0 \to 1$. Note also that

$$\begin{aligned} \perp_{\vec{u}'} \vec{v}_{(3)} &= \vec{v}_{(3)} + \overbrace{(\vec{u}' \cdot \vec{v}_{(3)})}^{\text{goes to } 0} \vec{u}' \\ &= \vec{v}_{(3)}, \end{aligned}$$

so our formula becomes

$$\vec{v}'_{(3)} = \vec{v}_{(3)} + \vec{u}'_{(3)}$$

which is the Newtonian velocity addition rule.

We are close to wrapping up our very brief discussion of SR. Notice that all of our derivations have relied heavily on the affine structure of Minkowski spacetime; in curved spacetimes, we will need to work with a 'connection' in order to compare vectors. These will be introduced in a couple of lectures.

The last question we want to address is this – how valid is a given set of coordinates? Recall the way in which an observer assigns coordinates to an event M. We had to assume that M was 'close' enough to the worldline such that it intersected only one local space, for the following reason: one can imagine a worldline with two (or more) rest spaces intersecting M. In this case, the prescription to assign M a time corresponding to the proper time of the rest space becomes ambiguous.

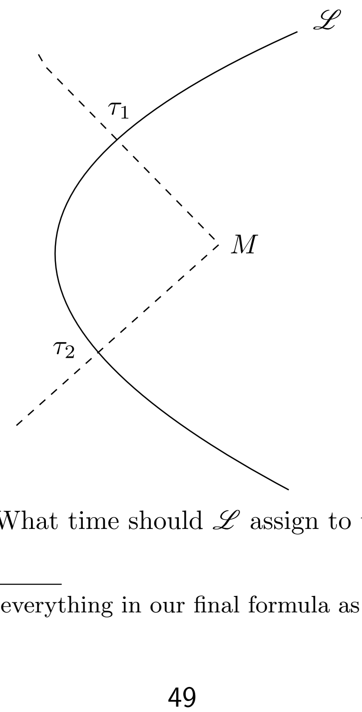


Figure 18: What time should $\mathscr{L}$ assign to the event M?

[17] This allows us to identify (almost) everything in our final formula as a vector living in $\mathcal{E}_{\vec{u}'}$.

When this happens, one has to ask 'what time should the observer assign to M?' In general, this sort of question turns out to be very difficult question to answer. (In GR terminology, we usually assume a 'normal neighborhood' such that all points within are only intersected by one local rest space.) In the next lecture, we'll address this problem more carefully, carrying out a computation that will tell us where our local coordinates are 'good'. The same question comes up in GR – 'until when are my local coordinates valid?'

9 Validity of local coordinates in SR; four-momenta

Coordinate breakdown

When discussing how an observer can set up coordinates (t, x^i), we assumed that this procedure was well-defined, in that we could assign a unique time to events in this manner. It turns out that this isn't always possible!

Now, a misconception we want to address here is that

$$\boxed{\text{flat space} \quad \not\Rightarrow \quad \begin{array}{c}\text{coordinate systems carried by} \\ \text{observers are valid everywhere.}\end{array}}$$

One might think this to be the case because flat space is supposed to be nicer and easier than curved spacetime. While this may be true, one shouldn't think that all issues in curved spacetime can be resolved by moving to Minkowski.

Remember that our coordinates were based on $\mathcal{E}_{\vec{u}}$. But it's possible for these hyperplanes to intersect, giving us two representations for one point. Let us construct one such problematic point and give this the name π. (revise)

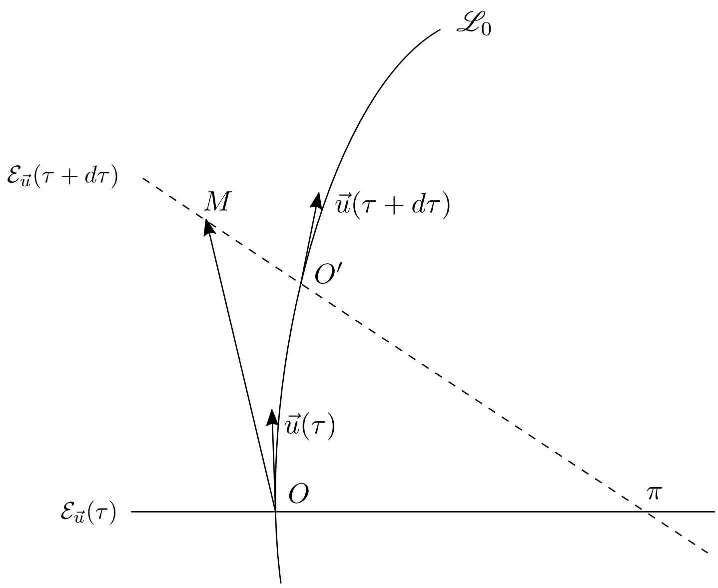


Figure 19: We're suppressing all but one spatial dimension here; M doesn't have to lie in the same plane as $\vec{u}$ and $\vec{\varepsilon}_1$. (this hyperplane slopes in the wrong direction!)

We want to be able to characterize what is happening here in terms of our worldline (observer). Again, we're suppressing all but one spatial coordinate in our drawings. Now consider the tetrad $\{\vec{u}(\tau), \vec{e}_1(\tau), \vec{e}_2(\tau), \vec{e}_3(\tau)\}$, where for the second basis vector we take

$$\vec{e}_{(1)} = \vec{\varepsilon}_1(\tau) := \frac{\vec{a}(\tau)}{\|\vec{a}(\tau)\|_g}.$$

Let us 'freeze' this tetrad at the instant τ, turning it into an affine coordinate system. We'll drop the arguments from here on; just keep this in mind. Next, consider any event M lying in the hyperplane $\mathcal{E}_{\vec{u}}(\tau + d\tau)$, and note that we can write

$$\overrightarrow{OM} = x^0\vec{u} + x^i\vec{\varepsilon}_i.$$

Now, in terms of the coordinates associated with our worldline (more accurately, in terms of the tetrad frozen at τ), what is the equation for the rest space $\mathcal{E}_{\vec{u}}(\tau)$? This is not too hard – it is

$$\boxed{x^0 = 0}$$

by definition. How about the equation for the neighboring hyperplane associated with proper time $\tau + d\tau$? The point O gets swept over to O', which sits in $\mathcal{E}_{\vec{u}}(\tau + d\tau)$. Recall what we know about local rest spaces:

(1) The 4-velocity $\vec{u}(\tau + d\tau)$ must be normal to $\mathcal{E}_{\vec{u}}(\tau + d\tau)$.

(2) A point P belongs to $\mathcal{E}_{\vec{u}}(\tau + d\tau)$ if and only if $\vec{u}(\tau + d\tau) \cdot \overrightarrow{O'P} = 0$.

Now notice that on our diagram,

$$\overrightarrow{OO'} + \overrightarrow{O'M} = \overrightarrow{OM},$$

and since $M \in \mathcal{E}_{\vec{u}}(\tau + d\tau)$, we can use condition (2) to get

$$\begin{aligned}\vec{u}(\tau + d\tau) \cdot \overrightarrow{OM} &= \vec{u}(\tau + d\tau) \cdot \overrightarrow{OO'} \\ &= -d\tau + \mathcal{O}(d\tau^2),\end{aligned}$$

where the second line follows because $\vec{u}(\tau+d\tau) = \vec{u}(\tau)+\vec{a}(\tau)\,d\tau+\mathcal{O}(d\tau^2)$, and also $\overrightarrow{OO'} = \vec{u}(\tau)\,d\tau$. Recalling our definition for $\vec{\varepsilon}_1$, we see that

$$(\vec{u}(\tau) + a\vec{\varepsilon}_1\,d\tau) \cdot \overrightarrow{OM} = -d\tau,$$

and since $\overrightarrow{OM} = x^0\vec{u} + x^i\vec{\varepsilon}_i$, we find

$$\boxed{-x^0 + ax^1\,d\tau = -d\tau.}$$

Now we are in the position to ask where our coordinates break, at least in the manner we described at the start. The point M will lie along the intersection of the two local rest spaces, just like our 'problematic point' π, when

$$\boxed{x^1 = -\frac{1}{a}.}$$

Notice that the components corresponding to the other spatial basis vectors do not enter into this condition. Furthermore, note that this is a local condition, since this is expressed in terms of the 4-acceleration at a given point. We can now characterize the point π (there is really a whole line (actually a surface) of problematic points):

$$\pi \overset{*}{=} (0, -a^{-1}, x^2, x^3)$$

The minus sign here is just an artifact of our coordinate system. We see that

$$|x^1| \quad \longrightarrow \quad \text{"distance" between } O \text{ and } \pi,$$

so our coordinates will be valid as long as

$$r \ll \frac{1}{a} = \|\vec{a}\|_g^{-1}.$$

To reinsert units, we note that $a \sim L/T^2$, so

$$c^2 a^{-1} \sim \left(\frac{L}{T}\right)^2 \frac{T^2}{L} \sim L,$$

and in non-geometric units we get

$$\boxed{r \ll \frac{c^2}{a}.}$$

Notice, in particular, that *the coordinates carried by an inertial observer are valid everywhere.* In order to get an idea of how restrictive the requirement on a general observer is, we can estimate some quantities. For a typical acceleration, say $a \sim 10\,\mathrm{m/s^2}$, we have

$$r_{\max} \sim 9 \times 10^{15}\,\mathrm{m} \sim 1\,\mathrm{ly}.$$

Momentarily comoving reference frames

(fill in)

Four-momenta

A relativist would argue that the 'four' is unnecessary – why add another word, when momentum really is 'four'? – but we'll stick to standard usage.

$$\begin{aligned} \vec{p} &= m\vec{u} \\ &= m\frac{d\vec{x}}{d\tau} \end{aligned}$$

Here it will be convenient to introduce $\xi = \tau/m$, so that

$$= \frac{d\vec{x}}{d\xi}$$

This is covariant, that is, it does not depend on the coordinate system. We emphasize that this only works for structureless particles describable by a single number m. Also, four-momentum is best thought of as a one-form (written $\underset{\sim}{p}$) instead of as a four-vector, but we'll have more to say about this later. Now clearly,

$$\vec{p} \cdot \vec{p} = -m^2.$$

Also, with a basis,

$$\vec{p} = p^\mu \vec{e}_{(\mu)} = m \underbrace{\gamma(v_{(3)})}_{=\frac{dt}{d\tau}} \big(\vec{e}_{(0)} + v^i_{(3)}\vec{e}_{(i)}\big),$$

where $v^i_{(3)}$ is the 3-velocity. The 0th component is then

$$\begin{aligned} p^0 &= mu^0 \\ &= m\gamma \\ &= \frac{m}{\sqrt{1-v^2}} \approx m + \frac{1}{2}mv^2 + \cdots \end{aligned}$$

where in the last line we've assumed $\|\vec{v}_{(3)}\| \ll 1$. This last expression can be thought of as energy, which is why it's common to say that

$$E =: p^0.$$

But this is frame-dependent, because $\vec{v}_{(3)}$ is. Now, for the remaining components,

$$\begin{aligned} p^j &= mu^j \\ &= m\gamma v^j_{(3)} \\ &= Ev^j_{(3)}, \end{aligned}$$

so we have

$$p^j \approx mv^j_{(3)} + \frac{1}{2}mv^j_{(3)}v^2 + \cdots$$

Finally, note that in any frame

$$\begin{aligned} \vec{p}\cdot\vec{p} &= -E^2 + \delta_{ij}p^ip^j \\ &= -m^2 \end{aligned}$$

so we have

$$\boxed{E^2 = m^2 + \delta_{ij}p^ip^j.}$$

In particular, if the object is not moving relative to the observer, so that $v^i_{(3)} = 0$, then we get $E = m$ (in nongeometric units, $E = mc^2$). Since $E := p^0$, this equation is really nothing but a statement of the normalization of $\vec{p}$.

How do we make sense of massless particles? It turns out that in the limit where $\tau \to 0$, we have $m \to 0$ as well, and $\xi = \tau/m$ remains finite; this is why we introduced ξ earlier. Taking a page from QM, we have

$$p^0 = E = \hbar\omega$$

for any photon in an inertial frame, so we can rewrite

$$-(p^0)^2 + \delta_{ij}p^ip^j = 0$$

in the form

$$\boxed{p^i = En^i = \hbar\omega n^i,}$$

where the n^i's can be thought of as direction cosines. To summarize, we always have

$$\vec{p} \overset{\mathcal{O}}{=} (E, p^i),$$

and depending on whether the particle is massive or massless, either

$$\boxed{\vec{p} = m\gamma(1, v^i_{(3)}) \quad \text{or} \quad \vec{p} = E(1, n^i).}$$

Now notice that

$$\begin{aligned} p^0 = E &= -g(\vec{p}, \vec{e}_{(0)}) \\ &= -g(\vec{p}, \vec{u}_{(\text{obs})}), \end{aligned}$$

where the minus sign comes in because $\vec{e}_{(0)} \cdot \vec{e}_{(0)} = -1$. This is a very important formula that should not be forgotten:

$$\boxed{E = -g(\vec{p}, \vec{u}_{(\text{obs})})}$$

Equivalently, one can think of this as the action of $\underset{\sim}{p}$ on a four-vector, as in

$$\boxed{E = -\underset{\sim}{p}(\vec{u}_{(\text{obs})}) \quad \text{or} \quad E = -\underset{\sim}{p}(\vec{e}_{(0)}).}$$

Four-momentum is best thought of as a one-form

We introduced the four-momentum using the relation $\vec{p} = m\vec{u}$ because this is a more familiar way of writing things; it's what you would see in an introductory class. Shortly afterwards, we said that the four-momentum was best thought of as a one-form. But why? We're familiar with the concept of metric duality, so can't we think of *any* four-vector as a one-form, or vice versa?

The answer has to do with how the four-momentum appears in the context of our theories. All our physical laws will say something about the four-momentum, not the four-velocity. In particular, we have

$$\begin{aligned} \vec{u} \quad &— \quad \text{kinematics} \\ \vec{p} \quad &— \quad \text{dynamics} \end{aligned}$$

So the process usually goes

$$\text{laws} \implies \vec{p} \implies \vec{u}.$$

This tells us that $\vec{p}$ is not just something we get by multiplying the four-velocity by a constant; instead, this sort of process usually happens in the reverse direction. More fundamentally, all theories are represented by a Lagrangian. Remember that

$$\boxed{p_\alpha = \frac{\partial L}{\partial \dot{x}^\alpha},}$$

so we reemphasize the point: we deal with four-momenta because the laws are expressed in terms of these. To work out a simple, concrete example, consider the free particle Lagrangian,

$$L = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}}.$$

It follows that

$$\begin{aligned} p_\alpha &= \frac{\partial L}{\partial \dot{x}^\alpha} \\ &= m\,\frac{2g_{\alpha\beta}\dfrac{dx^\beta}{d\lambda}}{\sqrt{-g_{\alpha\beta}\dfrac{dx^\alpha}{d\lambda}\dfrac{dx^\beta}{d\lambda}}} \\ &= m g_{\alpha\beta} u^\beta, \end{aligned}$$

where we took our arbitrary parameter to be $\lambda = \tau$ in the last line. The point of this calculation was to stress that the four-momentum naturally shows up as a one-form; you can see this from the indices in the last expression. Another example is that of a particle in a Maxwell field,

$$L = -m\sqrt{-g_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda}} + qA_\beta \dot{x}^\beta.$$

One finds that

$$\underline{p} = m\underline{u} + q\underline{A},$$

and then we need to infer what $\underline{u}$ is from $\underline{p}$. (Then we can turn this into a four-vector using our metric.) This process is usually only 'easy' for structureless particles. So for now, we can get away with thinking of the four-momentum as a four-vector, but it is more rigorous to say that it is a one-form.

We'll wrap up this lecture by noting that an operation like

$$E = -\underline{p}(\vec{e}_{(0)}),$$

where we feed a geometric object basis vectors, is the most common thing we'll do in GR. This allows us to go from geometric objects to components – the latter are what are relevant in the context of measurement. Components in an orthonormal frame are going to correspond to measurements made by an observer.

10 Tensors

Soon we'll move on to curved spactime. Flat spacetime (what we've been dealing with so far) is described by the instructor as inert – it is interesting, but the theory is not as rich as what you get with curvature. We start with the observation that physics in spacetime is described by geometric objects (tensors). We'll define what tensors are in a moment, but note that we've already encountered a few – the four-velocity and four-momentum are both geometric objects that live on spacetime. Another example that you might've encountered is the Faraday tensor F.

A recurring question will be the distinction between the measurements we make to probe geometric objects, and the geometric objects themselves. In physics, we tend to express things in terms of components because these are what we can measure, but bear in mind that all laws should be expressible in terms of geometric objects.

A review of four-vectors and one-forms

To review, a four-vector transforms like a displacement vector. If we have two frames (represented by barred and unbarred indices), then we can write the same displacement vector $\vec{A}$ as

$$\vec{A} = A^\alpha \vec{e}_\alpha,$$
$$\vec{A} = A^{\overline{\alpha}} \vec{e}_{\overline{\alpha}}.$$

The Lorentz transformations relate the components in the following manner:

$$\boxed{\Lambda^{\overline{\alpha}}{}_\beta A^\beta \vec{e}_{\overline{\alpha}} = A^\alpha \vec{e}_\alpha.}$$

An example of such a transformation is a boost in one direction,

$$\Lambda^{\overline{\alpha}}{}_\beta = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now, we can relabel the dummy indices on the right side and reorder terms on the left to get

$$A^\beta \Lambda^{\overline{\alpha}}{}_\beta \vec{e}_{\overline{\alpha}} = A^\beta \vec{e}_\beta \implies A^\beta(\Lambda^{\overline{\alpha}}{}_\beta \vec{e}_{\overline{\alpha}} - \vec{e}_\beta) = 0.$$

Then since A^β was arbitrary, this tells us how the basis vectors are related:

$$\vec{e}_\beta = \Lambda^{\overline{\alpha}}{}_\beta \vec{e}_{\overline{\alpha}}.$$

We can multiply by a matrix inverse to get

$$\boxed{\vec{e}_{\overline{\alpha}} = \Lambda_{\overline{\alpha}}{}^\beta \vec{e}_\beta.}$$

Now we turn our attention to one-forms. Take, for instance, the one-form $\underset{\sim}{p}$. (There's nothing special about this choice; think of this as an arbitrary element of V^*.) Suppose that $\{\vec{e}_\alpha\}$ is a basis of V. We define the components of $\underset{\sim}{p}$ in this basis to be

$$\boxed{p_\alpha := \underset{\sim}{p}(\vec{e}_\alpha).}$$

As with any linear transformation, these components fully describe $\underset{\sim}{p}$:

$$\begin{aligned}\underset{\sim}{p}(\vec{A}) = \underset{\sim}{p}(A^\alpha \vec{e}_\alpha) &= A^\alpha \underset{\sim}{p}(\vec{e}_\alpha) \\ &= A^\alpha p_\alpha. \end{aligned} \tag{10.1}$$

(Note that the expression $A^\alpha p_\alpha$ doesn't depend on the choice of basis, since $\underset{\sim}{p}(\vec{A})$ doesn't. This is an example of a *contraction.*)

How do the components of a one-form transform? With primes to denote the new coordinate system, we have

$$\begin{aligned} p_{\beta'} = \underset{\sim}{p}(\vec{e}_{\beta'}) &= \underset{\sim}{p}(\Lambda_{\beta'}{}^\alpha \vec{e}_\alpha) \\ &= \Lambda_{\beta'}{}^\alpha p_\alpha, \end{aligned}$$

that is,

$$\boxed{p_{\beta'} = \Lambda_{\beta'}{}^\alpha p_\alpha.}$$

Notice that the components of a one-form transform in exactly the same way that basis vectors do. Old-fashioned books (and Arfken) like to define tensors according to how they transform, instead of doing things geometrically. One-forms are called *covariant vectors*, while four-vectors are called *contravariant vectors*. We won't be using this terminology very much.

Suppose we want to write $\underset{\sim}{p}$ in terms of a basis $\{\underset{\sim}{e}^{(\alpha)}\}$ of V^*, as in

$$\underset{\sim}{p} = p_\alpha \underset{\sim}{e}^{(\alpha)}.$$

Consider feeding an arbitrary vector to $\underset{\sim}{p}$. We get

$$\begin{aligned}\underset{\sim}{p}(\vec{A}) &= (p_\alpha \underset{\sim}{e}^\alpha)(\vec{A}) \\ &= p_\alpha \underset{\sim}{e}^\alpha (A^\beta \vec{e}_\beta) \\ &= p_\alpha A^\beta \underset{\sim}{e}^\alpha (\vec{e}_\beta). \end{aligned}$$

If this is to be equivalent to Eq. (10.1), our basis vectors for the dual space need to satisfy the property

$$\underset{\sim}{e}^{(\alpha)}(\vec{e}_{(\beta)}) = \delta^\alpha{}_\beta,$$

which we encountered previously. The parentheses in the sub- and superscripts are meant to indicate that these are not components; to denote these you would write, say, $(\vec{e}_{(\beta)})^\mu$.

Gradient of a function

We tend to think of the gradient of a function as a vector, but this is actually a prototypical example of a one-form. Consider a scalar function ϕ defined on flat space, and take some curve $x^\alpha = x^\alpha(\tau)$. We then have

$$u^\alpha = \frac{dx^\alpha}{d\tau}.$$

By considering the values that ϕ takes along this curve, we get

$$\begin{aligned}\phi(\tau) &= \phi(t(\tau), x^i(\tau)) \\ &= \phi(x^\alpha(\tau)). \end{aligned}$$

By the chain rule,

$$\frac{d\phi}{d\tau} = \frac{\partial\phi}{\partial x^\alpha} u^\alpha.$$

Now ordinarily, we would think of $\partial\phi/\partial x^\alpha$ as the components of the gradient of ϕ. But consider what is happening here – we have a number on the left, but we have a vector u^α on the right, so the gradient is really a one-form, which acts in the following way:

$$\underline{d}\phi(\vec{u}) = \frac{d\phi}{d\tau}.$$

So while we're used to writing things like $\vec{\nabla} f$, we should keep in mind that this is afforded to us by metric duality. On arbitrary manifolds, we cannot make this identification.

$$\boxed{\underline{d}\phi = \frac{\partial\phi}{\partial x^\alpha}\,\underline{e}^{(\alpha)}}$$

Let's summarize. Given $\{\vec{e}_{(\alpha)}\}$, we have

$$\vec{V} = V^\alpha \vec{e}_{(\alpha)},$$

and to find these components, we have

$$V^\alpha = \underline{e}^{(\alpha)}(\vec{V}) =: \vec{V}(\underline{e}^{(\alpha)}).$$

This last notion, of feeding a one-form to a vector, is needed to make vectors and one-forms completely symmetric. This might seem weird, but in a moment we'll see that both vectors and one-forms are examples of linear maps.[18] Similarly, we can write

$$\underline{\omega} = \omega_\alpha \underline{e}^{(\alpha)},$$

where the components are given by

$$\omega_\alpha = \underline{\omega}(\vec{e}_{(\alpha)}).$$

All this relies on a special choice of basis for the dual space: we need

$$\underline{e}^{(\alpha)}(\vec{e}_{(\beta)}) = \delta^\alpha{}_\beta.$$

Tensors

Tensors are multilinear functions that map vectors and one-forms into $\mathbb{R}$. A tensor of rank (type) (m, n) maps m one-forms and n vectors into $\mathbb{R}$ in a linear manner. Therefore, a vector is a $(1, 0)$ tensor, and a one-form is a $(0, 1)$ tensor. The space of (m, n) tensors is itself a vector space and is denoted E^m_n. Special examples are

$$V \in E^1_0 = E$$
$$\omega \in E^0_1 = E^*$$

Now consider two tensors $T \in E^0_p$ and $S \in E^0_q$. We define their *tensor product* $T \otimes S$ by

$$(T \otimes S)(\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p, \vec{v}_1, \ldots, \vec{v}_q) := T(\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_p) S(\vec{v}_1, \ldots, \vec{v}_q).$$

[18] In a more formal sense, this is true because we can identify V with its 'double dual' space V^{**}, which is the set of linear functionals on V^*.

You should be able to show that this is also a tensor. From this definition, you can see that $T \otimes S \neq S \otimes T$, since the order in which the arguments go matters. But both have the same rank; they are both elements of E^0_{p+q}.

As an example, we can consider

$$\vec{u} \otimes \vec{v} \in E^2_0.$$

We then have

$$\begin{aligned}(\vec{u} \otimes \vec{v})(\underline{\alpha}, \underline{\beta}) &= \vec{u}(\underline{\alpha})\vec{v}(\underline{\beta}) \\ &= u^\mu \alpha_\mu v^\nu \beta_\nu,\end{aligned}$$

where to get to the second line we chose some basis for E; this is how we actually compute things.

Bases for tensors

In physics, we don't measure geometric objects themselves; we measure components. Therefore, we would like to establish bases for tensors. Consider a tensor $R \in E^q_0$. This $(q, 0)$ tensor is uniquely defined by its action on all elements of $\underbrace{E^* \times E^* \times \cdots \times E^*}_{q \text{ times}} = (E^*)^q$.

So consider some arbitrary collection of q one-forms. We have

$$\begin{aligned}R(\underline{\omega}^{(1)}, \underline{\omega}^{(2)}, \ldots, \underline{\omega}^{(q)}) &= R(\omega^{(1)}_{\alpha_1}\underline{e}^{(\alpha_1)}, \omega^{(2)}_{\alpha_2}\underline{e}^{(\alpha_2)}, \ldots, \omega^{(q)}_{\alpha_q}\underline{e}^{(\alpha_q)}) \\ &= R(\underline{e}^{(\alpha_1)}, \underline{e}^{(\alpha_2)}, \ldots, \underline{e}^{(\alpha_q)})\, \omega^{(1)}_{\alpha_1}\omega^{(2)}_{\alpha_2}\cdots\omega^{(q)}_{\alpha_q},\end{aligned}$$

which allows us to write

$$R = R^{\alpha_1\alpha_2\cdots\alpha_q}\, \underbrace{e_{(\alpha_1)} \otimes e_{(\alpha_2)} \otimes \cdots \otimes e_{(\alpha_q)}}_{\text{basis tensor for } {E^q}_0}$$

where

$$R^{\alpha_1\alpha_2\cdots\alpha_q} := R(\underline{e}^{(\alpha_1)}, \underline{e}^{(\alpha_2)}, \ldots, \underline{e}^{(\alpha_q)}).$$

Similarly, if we have $S \in E^0_q$, we can write[19]

$$S = S_{\alpha_1\alpha_2\cdots\alpha_q}\, \underbrace{e^{(\alpha_1)} \otimes e^{(\alpha_2)} \otimes \cdots \otimes e^{(\alpha_q)}}_{\text{basis tensor for } {E^0}_q},$$

where

$$S_{\alpha_1\alpha_2\cdots\alpha_q} := R(\vec{e}_{(\alpha_1)}, \vec{e}_{(\alpha_2)}, \ldots, \vec{e}_{(\alpha_q)}).$$

Now, while we've been working with fully co/contravariant tensors, we can generalize this discussion to (p, q) tensors. For $T \in E^q_p$, we have

$$T = T^{\alpha_1\alpha_2\cdots\alpha_q}{}_{\beta_1\beta_2\cdots\beta_p}\, e_{(\alpha_1)} \otimes e_{(\alpha_2)} \otimes \cdots \otimes e_{(\alpha_q)} \otimes e^{(\beta_1)} \otimes \cdots \otimes e^{(\beta_p)},$$

where

$$T^{\alpha_1\alpha_2\cdots\alpha_q}{}_{\beta_1\beta_2\cdots\beta_p} := T(e^{(\alpha_1)}, \ldots, e^{(\alpha_q)}, e_{(\beta_1)}, \ldots, e_{(\beta_p)}).$$

Note that in writing the tensor product here, the ordering of vectors relative to one-forms and vice versa doesn't matter – we just need to avoid swapping two one-forms or two vectors, as this changes the meaning of the tensor.

[19]Notice that we can drop the arrow/underline notation, since our new convention will be to indicate vectors and one-forms by downstairs and upstairs indices, respectively.

Some examples of tensors

We can see that the metric tensor g is an element of E^0_2; it is multilinear, and accepts two vectors to spit out a real number. Instead of working with the fully geometric object, we can work with components by picking some basis.

$$g = g_{\alpha\beta}\, e^{(\alpha)} \otimes e^{(\beta)}$$

Here we can see that the definition $g_{\alpha\beta} := g(e_{(\alpha)}, e_{(\beta)})$ we introduced before was just a special case of this more general discussion. In Minkowski, where $g(e_{(\alpha)}, e_{(\beta)}) = \eta_{\alpha\beta}$, we have

$$\boxed{g = -e^{(0)} \otimes e^{(0)} + e^{(1)} \otimes e^{(1)} + e^{(2)} \otimes e^{(2)} + e^{(3)} \otimes e^{(3)},}$$

and this is how a mathematician might write things out. But it is common for physicists to speak of something called a line element: they will write

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2,$$

and they will say that this is the metric. This line of thinking is heuristic at best – what they have in mind is something like

$$d\vec{r} = dx^\alpha e_{(\alpha)},$$

so that

$$\begin{aligned} ds^2 = g(d\vec{r}, d\vec{r}) &= g(dx^\alpha \vec{e}_{(\alpha)}, dx^\beta \vec{e}_{(\beta)}) \\ &= g(\vec{e}_{(\alpha)}, \vec{e}_{(\beta)}) dx^\alpha dx^\beta \\ &= \eta_{\alpha\beta} dx^\alpha dx^\beta \\ &= -dt^2 + dx^2 + dy^2 + dz^2. \end{aligned}$$

This talk of ‘infinitesimal’ spacetime intervals tends to make mathematicians uncomfortable.

Another example that you may have encountered in your electromagnetism classes is the Faraday tensor F. This is completely antisymmetric, so that

$$F_{\mu\nu} = -F_{\nu\mu}.$$

One can define this by[20]

$$\begin{aligned} F &= F_{\mu\nu}\, dx^\mu \wedge dx^\nu \\ &= F_{\mu\nu}\, e^{(\mu)} \wedge e^{(\nu)}, \end{aligned}$$

where the new symbol denotes a *wedge product*, defined by

$$e^{(\mu)} \wedge e^{(\nu)} := \tfrac{1}{2}(e^{(\mu)} \otimes e^{(\nu)} - e^{(\nu)} \otimes e^{(\mu)}).$$

Recall that $F_{\mu\nu} = F(e_{(\mu)}, e_{(\nu)})$. In physics, these components have interpretations – using metric duality, we can identify

$$F^{\mu\nu} = \overline{F}(e^{(\mu)}, e^{(\nu)}) = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & B_z & -B_y \\ -E_y & -B_z & 0 & B_x \\ -E_z & B_y & -B_x & 0 \end{pmatrix}$$

[20]The second line is how a mathematician would write things.

There are only 6 independent components here because the tensor is antisymmetric. The E-field measured by a particular observer is given by

$$E^i = F^{0i}.$$

Note that the observer enters into this equation via their basis, which specifies the components of the Faraday tensor.

11 Manifolds; charts; curves; tangent vectors

We continue to emphasize the following point: components are important because they are actually measurable, but the laws themselves are expressed in terms of geometric objects.

More mathematically inclined relativists can be allergic to components, but this becomes a disadvantage, because this means that you will not be able to calculate things that are interesting. (At the very least, this becomes inordinately difficult.) On the other hand, a focus on components can obscure your understanding, in a geometric sense, of what is actually going on. It's important to find a healthy balance between the two.

The appropriate structure for curved spacetime

According to GR, gravitational phenomena are the result of 'stuff' moving in 'spacetime'. We can make both of these notions more precise – 'spacetime' is a Lorentzian manifold, while 'stuff' refers to tensors, which are geometric objects that live on the manifold. Today we'll focus on manifolds. This will probably be a boring review for those who have already encountered this material, and will probably go by too quickly for those who haven't.[21]

Manifolds

We begin by stating that

> A *manifold* is a topological space that is locally coordinatizable.

A topological space M is a set with a specification of what an open set is. One then writes something like $U_\alpha \subset M$. There are certain restrictions on what you can declare to be an open set, but this description will suffice for us.

When we say locally coordinatizable, we are speaking of the notion of a *chart* on a manifold. These allow us to identify points on an n-dimensional manifold with coordinates in $\mathbb{R}^n$, and these are defined on open sets. The 'local' comes from the fact that we cannot (in general) rely on a single chart to provide a coordinate description for the entire manifold.

Stated more precisely, a chart is a pair (U, ϕ), where U is the open set on which the chart is defined, and

$$\phi : M \to \mathbb{R}^n, \qquad \phi(p) = (x^1(p), x^2(p), \ldots, x^n(p))$$

Note that the x's are in themselves functions mapping from M to $\mathbb{R}$ – later we'll have more to say about this. To get a feel for how this works, we can draw a picture; see Fig. 20. We've introduced another chart (U_2, ψ), and we've drawn things so that there is some overlap between U_1 and U_2. In this region we can speak of a *transition function*

$$\phi \circ \psi^{-1} : \mathbb{R}^n \to \mathbb{R}^n, \qquad x^\alpha = x^\alpha(y^\beta)$$

Similarly, one can consider $\psi \circ \phi^{-1}$. You will probably be more familiar with calling these coordinate transformations. The bare minimum, in a mathematical sense, is that these mappings be continuous (C^0) – but this is too little for us, since we want to do calculus. Beyond differentiability, though, we will rarely need to be careful with these things, and we can get away with demanding that our mappings be C^∞.

We will not be relying on any special or distinguished set of local coordinates, but in many applications, it's possible to deal only with the charts.

[21] A recurring question is whether this is secretly a differential geometry class. I'm not in the position to give an answer, but there's no denying that DG is a large (and important) part of the course.

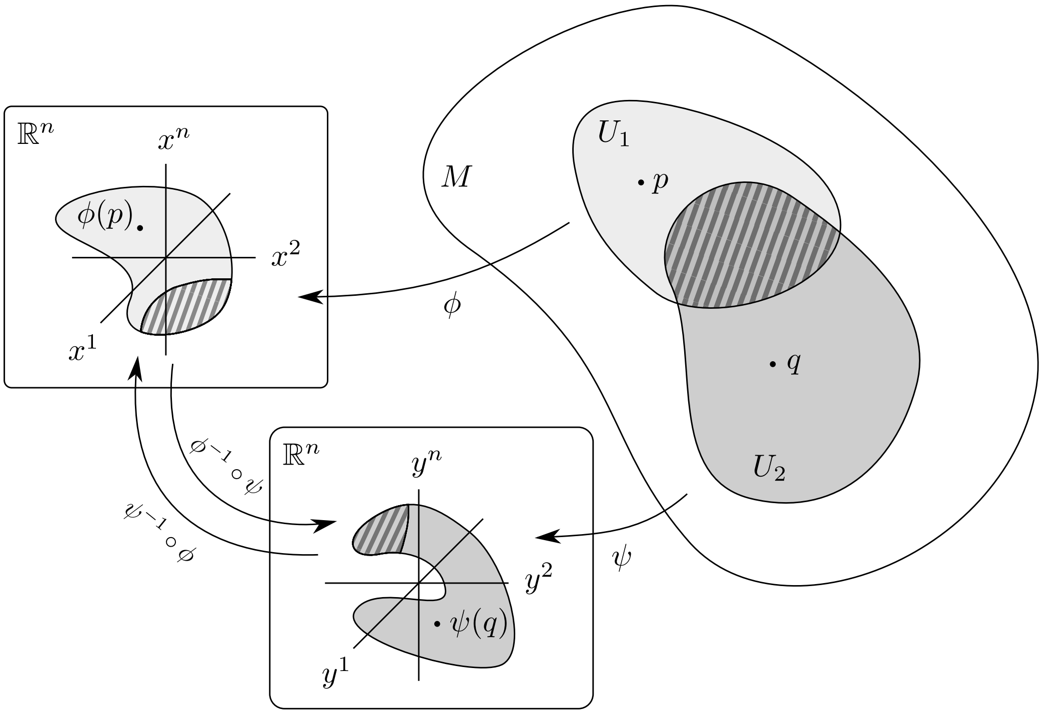


Figure 20: A schematic picture of a manifold. We'll assume that the transition functions which take us from one coordinate description to the other in the region $U_1 \cap U_2$ are class C^∞.

Coordinates

Consider $\mathbb{R}^2$, which is itself a manifold. One coordinatization is $\phi = \text{Id}$, which happens to work everywhere. This is an exception, since most manifolds will not have a global coordinate system. Another way to coordinatize $\mathbb{R}^2$ is to assign polar coordinates to points. This sort of procedure is fairly intuitive – we've been coordinatizing space and time ever since we were born. But there are weird sets where we cannot do this, which is just something to keep in mind.

Now consider the unit sphere S^2. One way to assign coordinates is via stereographic projection, but there is a well-known issue with this – the north pole cannot be included in this way. We can get around this issue by introducing another chart, say, stereographic-projecting from the south pole onto another plane.

Yet another example of a coordinate system that will not work everywhere is spherical coordinates on S^2. But we'll rarely need global coordinates, and you can't really get these except for with the most trivial manifolds, anyway.

Curves on manifolds

Consider a mapping

$$\gamma : I \to M, \qquad I \subseteq \mathbb{R}.$$

If this lies in some open set with a chart, then we can coordinatize this curve:

$$(\phi \circ \gamma)(\lambda) = (x^1(\lambda), \ldots, x^n(\lambda)).$$

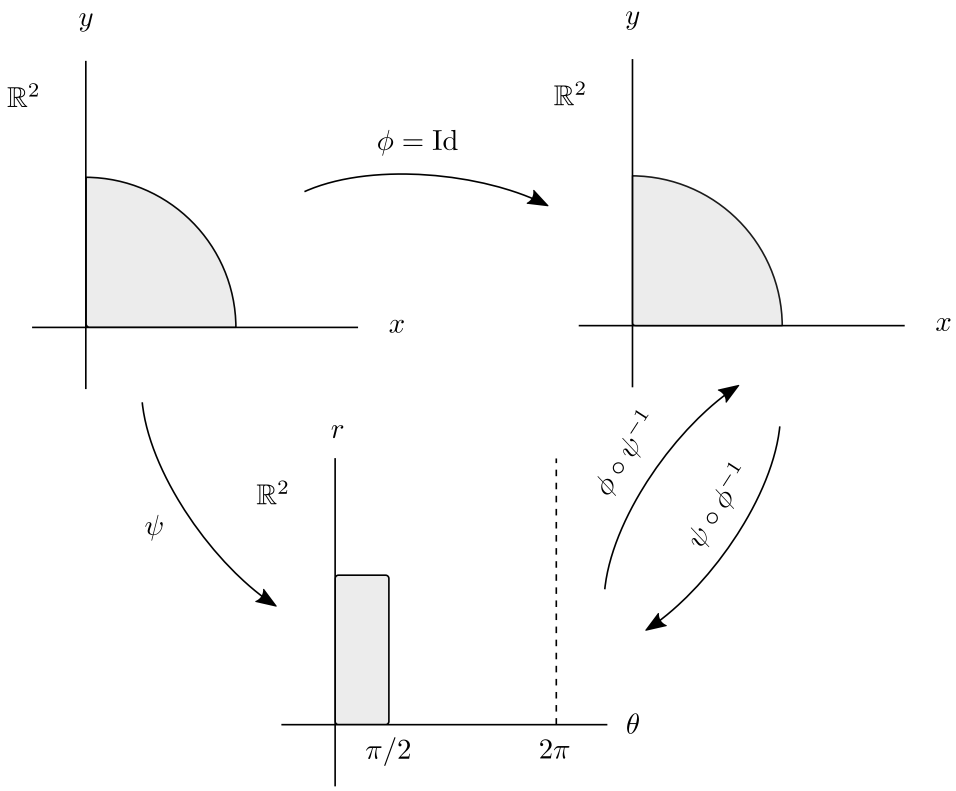


Figure 21: Two ways to assign coordinates to the manifold $M = \mathbb{R}^2$. One of these is simply the identity mapping, while ψ assigns polar coordinates to points.

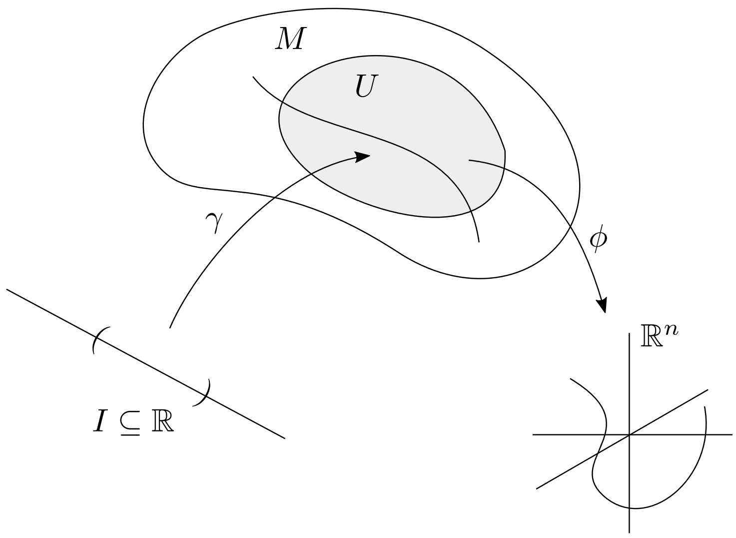


Figure 22: Working with the mapping $\phi \circ \gamma$ allows us to use the methods of advanced calculus.

Tangent vector on a curve

This is a familiar notion in $\mathbb{R}^3$ – given some parametrized curve

$$\vec{r}(\lambda) = \big(x(\lambda), y(\lambda), z(\lambda)\big),$$

we can consider the associated tangent vectors

$$\vec{t}(\lambda) = \frac{d\vec{r}}{d\lambda} = \left(\frac{dx}{d\lambda}, \frac{dy}{d\lambda}, \frac{dz}{d\lambda}\right).$$

Now take the same prescription, but regard this as the coordinatization of a curve on a manifold:

$$\vec{r}(\lambda) \longrightarrow x^i(\lambda) = (\phi \circ \gamma)(\lambda) \quad \text{and} \quad t^i(\lambda) = \frac{dx^i}{d\lambda}$$

A natural question is to ask what happens to these tangent vectors if we choose a different coordinatization. We then have some transition function $y^i = y^i(x^j)$, and in the new coordinates

$$y^i(\lambda) = (\psi \circ \gamma)(\lambda) \quad \text{with} \quad t^{\bar{i}}(\lambda) = \frac{dy^i}{d\lambda}.$$

We've used the same letter to indicate the tangent vector, which is meant to be suggestive – if the tangent vector is to be understood geometrically, then these two objects must represent the same thing. Let's work out what happens:

$$\begin{aligned}\frac{dy^i}{d\lambda} = \frac{d}{d\lambda}(\psi \circ \gamma) &= \frac{d}{d\lambda}\big(y^i(x^j(\lambda))\big)\\ &= \frac{d}{d\lambda}\Big[(\underbrace{\psi \circ \phi^{-1}}_{\equiv y^i(x^j)}) \circ (\underbrace{\phi \circ \gamma}_{\equiv x^j(\lambda)})\Big]\end{aligned}$$

Thus, by the chain rule, we get

$$\frac{dy^i}{d\lambda} = \frac{\partial y^i}{\partial x^j}\frac{dx^j}{d\lambda},$$

which tells us that a tangent vector transforms like

$$\boxed{t^{\bar{i}} = \frac{\partial y^i}{\partial x^j}\, t^j.}$$

Arfken loves this kind of thing, but this makes for an odd definition, because it only tells you about the components, and not what the object really is.

The geometric meaning of a tangent vector

Let f be a scalar function on M, i.e., $f : M \to \mathbb{R}$. To consider the values that f attains on a curve in M, we can take some $\gamma : \mathbb{R} \to M$ and define $f(\gamma(\lambda)) =: f(\lambda)$. Then $df/d\lambda$ is well-defined and independent of coordinate system, but now suppose we introduce some charts.

For example, given two charts ϕ and ψ, we can write[22]

$$f(\lambda) = f\big(\phi^{-1} \circ \phi(\lambda)\big) = f(x^j(\lambda)),$$
$$f(\lambda) = f\big(\psi^{-1} \circ \psi(\lambda)\big) = f(y^j(\lambda)).$$

We can take these two representations and compute the same thing:

$$\frac{df}{d\lambda} \overset{\text{“}\phi\text{”}}{=} \frac{\partial f}{\partial x^i}\frac{dx^i}{d\lambda} = \frac{dx^i}{d\lambda}\frac{\partial f}{\partial x^i} = t^i \frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial y^j} f,$$

(the last equality follows from the definition of t^i and the chain rule)

$$\frac{df}{d\lambda} \overset{\text{“}\psi\text{”}}{=} \frac{\partial f}{\partial y^i}\frac{dy^i}{d\lambda} = \frac{dy^i}{d\lambda}\frac{\partial f}{\partial y^i} = t^{\bar{i}} \frac{\partial}{\partial y^i} f.$$

Since f was arbitrary, we conclude

$$\frac{d}{d\lambda} = t^{\bar{i}}\frac{\partial}{\partial y^i} = t^i \frac{\partial}{\partial x^i}, \quad \text{where} \quad t^{\bar{j}} = \frac{\partial y^j}{\partial x^i} t^i.$$

Therefore, we see that a tangent vector is basically a derivative operator that has different components t^i depending on your coordinates. Now, $d/d\lambda$ is a derivative operator on functions whose components in $\{x^i\}$ are $dx^i/d\lambda$:

$$\frac{d}{d\lambda} = \dot{x}^i \frac{\partial}{\partial x^i}$$

If $\{x^i\}$ is specified, then $\partial/\partial x^i$ is also a derivative operator on functions. Consider, for example, some curve on M along which only x^1 changes.[23] Then we have

$$x^i(\lambda) = (\lambda, \underbrace{a_2, a_3, \ldots, a_n}_{\text{constants}})$$

and it follows that

$$\frac{d}{d\lambda} = \dot{x}^i \frac{\partial}{\partial x^i} = \frac{\partial}{\partial x^1}.$$

Similarly, $\partial/\partial x^2$, etc., are also tangent vectors. Now notice that

$$\frac{d}{d\lambda} = \dot{x}^j \partial_j.$$

This sheds light on the chain rule from calculus – recall that

$$\frac{\partial}{\partial x^i} = \frac{\partial y^k}{\partial x^i}\frac{\partial}{\partial y^k}.$$

Viewed in this context, the chain rule is nothing but a change of basis! Our next claim is that the sets $\{\frac{\partial}{\partial x^i}\}$ and $\{\frac{\partial}{\partial y^j}\}$, associated with $\{x^i\}$ and $\{y^j\}$ respectively, are both bases for the space of derivative operators. But we did this (did we?) when we wrote

$$\frac{d}{d\lambda} = \frac{dx^i}{d\lambda}\partial_i.$$

[22] To make the equivalence clear, note that $\phi(\lambda)$ denotes the same function as $\phi \circ \gamma$, and similarly for $\psi(\lambda)$.
[23] Again, curves on M exist prior to coordinatization; this notion of a constant-x^1 curve comes from our chart.

The last thing we want to do is show that the space of derivative operators forms a vector space. Consider two curves like the ones indicated below.

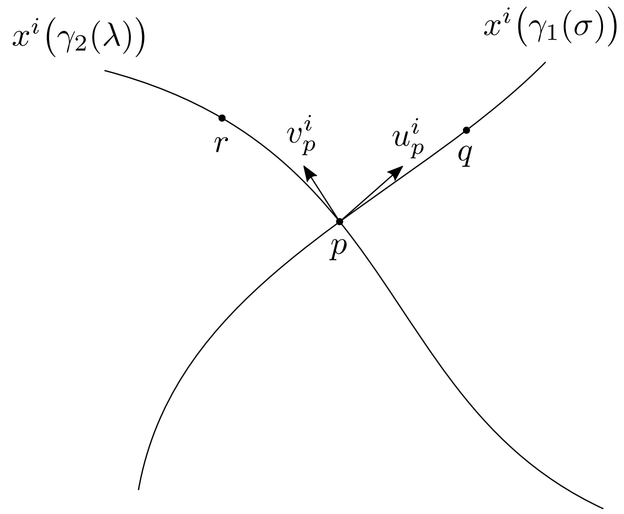


Figure 23: Two tangent vectors living in the same space (they belong to T_pM); these come from the curves γ_1 and γ_2.

We first want to point out that

$$\left(\frac{d}{d\sigma}\right)_p \neq \left(\frac{d}{d\sigma}\right)_q$$

even though these are associated with the same curve. Similarly,

$$\left(\frac{d}{d\lambda}\right)_r \neq \left(\frac{d}{d\lambda}\right)_p$$

and this is our first encounter with the notion that these are local objects. This is a very important point:

$$\boxed{\textit{A priori}\text{, we cannot add a tangent vector at } q \text{ to one at } r.}$$

However, this is not an issue if they live at the same point. Then we have

$$\begin{aligned}\left[a\left(\frac{d}{d\sigma}\right)_p + b\left(\frac{d}{d\lambda}\right)_p\right] f &= a\left(\frac{df}{d\sigma}\right)_p + b\left(\frac{df}{d\lambda}\right)_p \\ &\stackrel{?}{=} \left(\frac{d}{d\chi}\right)_p f\end{aligned}$$

To establish that this is a vector space, we need to find a curve whose directional derivative is this combination of tangent vectors. We can construct this – first note that

$$\begin{aligned}\gamma_1(\sigma): \qquad x^i(\gamma_1(\sigma)) &= x_p^i + \left(\frac{dx^i}{d\sigma}\right)_p \sigma + \mathcal{O}(\sigma^2) \\ &= x_p^i + u_p^i\sigma + \mathcal{O}(\sigma^2)\end{aligned}$$

and

$$\gamma_2(\lambda): \qquad x^i\big(\gamma_2(\lambda)\big) = x^i_p + \left(\frac{dx^i}{d\lambda}\right)_p \lambda + \mathcal{O}(\lambda^2)$$
$$= x^i_p + v^i_p\lambda + \mathcal{O}(\lambda^2)$$

Then define $\gamma_3(\chi)$ to be the curve

$$x^i\big(\gamma_3(\chi)\big) := x^i_p + (au^i_p + bv^i_p)\chi + \mathcal{O}(\chi^2).$$

Does such a curve exist? The answer is yes – and there are many such curves! The $\mathcal{O}(\chi^2)$ terms give us lots (an infinite amount) of freedom. For example, in Fig. 24 we've indicated a curve γ_3 with the properties we want, but we could just as well have chosen any of the dashed curves to get the same tangent vector.

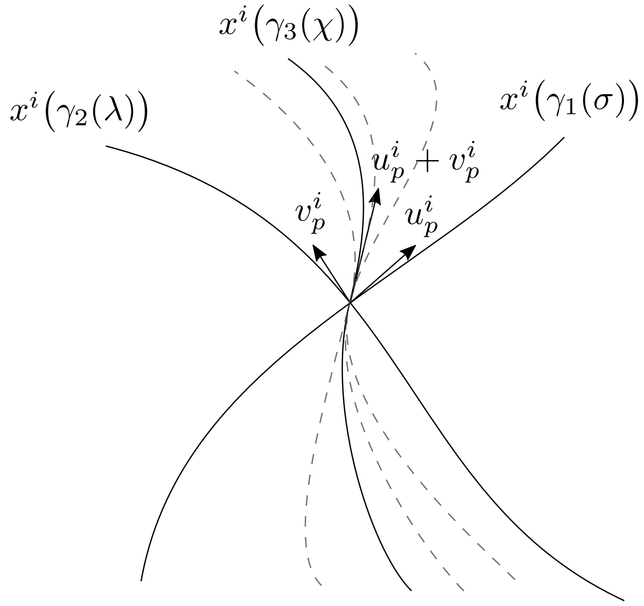


Figure 24: Tangent vectors can also be defined as equivalence classes of curves; this will be mentioned (briefly) in lecture 12.

In the same sense, you don't need to specify a particular curve to get a tangent vector. There are an infinite number of curves associated with a vector; in fact, another way to define these are as equivalence classes of curves. To summarize, then, a tangent vector is a derivative operator with respect to curves.

A very important concept is that of a *tangent space* T_pM, which is the space of all tangent vectors at p. Note that this is an intrinsic description – we start from curves living on the manifold. We do not need to think of manifolds as embedded in another space; this is a crutch.

As a final comment, while you might be used to thinking of a tangent vector on S^2 as being something that sits in $\mathbb{R}^3$, we would say (in our new language) that

$$\frac{d}{d\lambda} = v^\theta \frac{\partial}{\partial\theta} + v^\phi \frac{\partial}{\partial\phi}.$$

12 Coordinate bases; tangent and cotangent bundles; vector fields

The notion that emerged in the previous lecture was that a tangent vector is to be understood as a derivative operator on functions on M:

$$\text{For } f : M \to \mathbb{R}, \qquad X_p f = \left.\frac{dx^i}{dt}\right|_{t=t_0} \left.\partial_i f\right|_{\vec{x}=\vec{x}_0}$$

The tangent vector has a meaning of its own,

$$\boxed{X_p = \frac{dx^i}{dt}\partial_i,}$$

and this is independent of what coordinates you choose. Now, oftentimes differential geometry books will provide the following definition without any motivation. Hopefully, you can now see why these properties are reasonable.

Definition. A *tangent vector at* $p \in M$ is a real-valued function $v : F(M) \to \mathbb{R}$ that

(1) is $\mathbb{R}$-linear: $v(af + bg) = av(f) + bv(g)$ for all $f, g \in F(M)$; $a, b \in \mathbb{R}$,

(2) obeys the Leibniz rule: $v(fg) = f(p)v(g) + g(p)v(f)$.

One can show from these properties that if $h \in F(M)$ is constant, then $v(h) = 0$.

Sometimes, these are referred to as 'derivations', but we won't use this. You shouldn't need to hesitate about representing tangent vectors by arrows, but remember that these are bound to a point, and live in a particular tangent space. You don't usually get the 'sliding' property that we tacitly assume for $\mathbb{R}^n$.

Coordinate bases

Given a chart (U, ϕ), we will define

$$(\partial_\mu)_p := \left.\frac{\partial}{\partial x^\mu}\right|_p$$

and we showed that these form a basis on T_pM. For $f : M \to \mathbb{R}$, then, we have

$$(\partial_\mu)_p f = \left.\frac{\partial(f \circ \phi^{-1})}{\partial x^\mu}\right|_{x^\alpha = \phi^\alpha(p)}$$

More generally, we may write

$$X_p f := \left.\frac{df\big(c(t)\big)}{dt}\right|_{t=0}$$

where $c(t)$ is such that $c : I \subseteq \mathbb{R} \to M$, and where the $t = 0$ is general, since we can always reparametrize our curves. There's nothing here that makes reference to coordinates, but if we do have a chart, then by the chain rule

$$X_p f = \left.\frac{\partial(f \circ \phi^{-1})}{\partial x^\mu}\right|_{\phi = \phi(p)} \left.\frac{dx^\mu(c(t))}{dt}\right|_{t=0}$$

This is quite long to write out, so we'll use the shorthand

$$X_p f = \left(X^\mu \frac{\partial}{\partial x^\mu}\right)_p f,$$

where

$$X_p^\mu = \left.\frac{dx^\mu(c(t))}{dt}\right|_{t=0}$$

Then we get the compact expression

$$\boxed{X_p = X_p^\mu \, \partial_\mu.}$$

If we consider another chart, then

$$X = X^{\overline{\mu}} \, \partial_{\overline{\mu}} = X^{\overline{\mu}} \frac{\partial x^\nu}{\partial \overline{x}^\mu} \, \partial_\nu,$$

by the chain rule. Thus

$$\boxed{\partial_{\overline{\mu}} = \frac{\partial x^\nu}{\partial \overline{x}^\mu} \, \partial_\nu,}$$

and this comes with the 'change of basis' interpretation that we discussed earlier. Similarly, we can ask how the components change:

$$\begin{aligned} X = X^{\overline{\mu}} \partial_{\overline{\mu}} &= X^\nu \partial_\nu \\ &= X^\nu \frac{\partial x^{\overline{\mu}}}{\partial x^\nu} \partial_{\overline{\mu}} \end{aligned}$$

This gives the old-fashioned, 'classical' definition of a vector,

$$\boxed{X^{\overline{\mu}} = \frac{\partial x^{\overline{\mu}}}{\partial x^\nu} X^\nu.}$$

Given a chart, the mapping $\phi : M \to \mathbb{R}^n$ comes with n *coordinate functions* $x^\mu : M \to \mathbb{R}$,

$$x^\mu = \phi^\mu(p).$$

In this way, one can think of

$$(f \circ \phi^{-1})(x^\nu) = f(x^\nu).$$

Consider X_p applied to the coordinate functions themselves. We get

$$\begin{aligned} X_p \, x^\mu = X[x^\mu] &= \left(\frac{dx^\nu(c(t))}{dt} \frac{\partial}{\partial x^\nu} \right) x^\mu \\ &= \frac{dx^\nu(c(t))}{dt} \delta^\mu_\nu \\ &= \left.\frac{dx^\mu(t)}{dt}\right|_{t=0} \end{aligned}$$

Written in the more compact notation, this is

$$\begin{aligned} X_p \, x^\mu &= \left(X_p^\nu \partial_\nu \right) x^\mu \\ &= X_p^\nu \delta^\mu_\nu \\ &= X_p^\mu. \end{aligned}$$

The takeaway is that you get a number,

$$\boxed{X_p^\nu = X_p\, x^\nu.}$$

The following idea is not so useful in GR, but can be seen in fields like mathematical relativity. One can make the identification

$$X_p \longleftrightarrow [c(t)],$$

where the brackets are standard notation for a so-called *equivalence class*,

$$[c(t)] := \left\{ \tilde{c}(t) \;\middle|\; \tilde{c}(0) = c(0) = p \;\text{ and }\; \left.\frac{dx^\mu((\tilde{c}(t))}{dt}\right|_{t=0} = \left.\frac{dx^\mu(c(t))}{dt}\right|_{t=0} \right\}.$$

So instead of thinking of X_p as associated with a single curve, it's best to associate it with a whole equivalence class of curves. We pointed this out earlier when we showed that there were an infinite number of curves that could give the same tangent vector.

⦿ Yes, the coordinate basis is a basis

In case you're uncertain, we can give a proof; this reproduces the one in section 2.2 of Wald. This also shows that $\dim V_p = n$.[24] To avoid confusion with our previous discussion, note that the coordinate vectors are denoted X_μ here, and that we're omitting the subscript p. Also, we wrote X_p to denote an arbitrary tangent vector at p, but we'll be using v for this in what follows.

Proof. Let ϕ be a chart, let $f \in \mathscr{F}$, and define

$$X_\mu(f) = \frac{\partial}{\partial x^\mu}(f \circ \phi^{-1})\Big|_{\phi(p)},$$

where the x^μ are the ordinary Cartesian coordinates of $\mathbb{R}^n$. It's straightforward to see from the properties of partial derivatives that $X_\mu \in V_p$. Note that these are just the coordinate basis vectors from earlier; we want to show that they comprise a basis for V_p.

To see that they are linearly independent, let $f \in \mathscr{F}$ and suppose

$$\sum_{\mu=1}^{n} \alpha^\mu \frac{\partial}{\partial x^\mu}(f \circ \phi^{-1})\Big|_{\phi(p)} = 0.$$

Since f is arbitrary, we can choose it so that (can you make this more convincing)

$$(f \circ \phi^{-1})(x) = x^\nu$$

for any $\nu = 1, \dots, n$. Then we have

$$0 = \sum_{\mu=1}^{n} \alpha^\mu \frac{\partial}{\partial x^\mu}(f \circ \phi^{-1})\Big|_{\phi(p)} = \sum_{\mu=1}^{n} \alpha^\mu \delta_\mu^\nu = \alpha^\nu,$$

and in this way we conclude that all the coefficients α^μ must be zero. To see that they span V_p, we use the fact that for each C^∞ function $F : \mathbb{R}^n \to \mathbb{R}$, we can write for any $a, x \in \mathbb{R}^n$

$$F(x) = F(a) + \sum_{\mu=1}^{n} (x^\mu - a^\mu) H_\mu(x),$$

[24] As a side comment, it seems unusual to frame this as a proof that $\dim V_p = n$, as is done in the book, since the notion of a coordinate basis is so important.

where each H_μ is a C^∞ function with the property

$$H_\mu(a) = \frac{\partial F}{\partial x^\mu}\bigg|_{x=a}.$$

Since $f \circ \phi^{-1}$ is C^∞ by assumption, we may put $F = f \circ \phi^{-1}$ and $a = \phi(p)$, so that for any $q \in M$

$$\big(f \circ \phi^{-1}\big)\big(\phi(q)\big) = \big(f \circ \phi^{-1}\big)\big(\phi(p)\big) + \sum_{\mu=1}^{n} \big(x^\mu(q) - x^\mu(p)\big) H_\mu\big(\phi(q)\big),$$

or equivalently,

$$f(q) = f(p) + \sum_{\mu=1}^{n} \big(x^\mu(q) - x^\mu(p)\big)(H_\mu \circ \phi)(q).$$

Here we're using the notation $\phi(p) = (x^1(p), \dots, x^n(p))$, that is, each x^μ is a coordinate function, and in particular $x^\mu \in \mathscr{F}$. Finally, let $v \in V_p$. Recalling that $v(h) = 0$ if h is constant, we get

$$\begin{aligned} v(f) &= \cancel{v[f(p)]} + \sum_{\mu=1}^{n} \cancel{\big(x^\mu(q) - x^\mu(p)\big)}\bigg|_{q=p} v(H_\mu \circ \phi) \\ &\qquad\qquad + \sum_{\mu=1}^{n} v\big[x^\mu(q) - x^\mu(p)\big](H_\mu \circ \phi)(p) \\ &= \sum_{\mu=1}^{n} v(x^\mu)(H_\mu \circ \phi)(p) \\ &\equiv \sum_{\mu=1}^{n} v^\mu X_\mu(f), \end{aligned}$$

where we've defined $v^\mu \equiv v(x^\mu)$, and where we've used the fact that

$$(H_\mu \circ \phi)(p) = \frac{\partial}{\partial x^\mu}(f \circ \phi^{-1})\bigg|_{\phi(p)} = X_\mu(f).$$

Thus the tangent vectors X_μ span V_p, as we wanted to show. □

Examples of a basis

On $\mathbb{R}^2$, two straightforward choices are

$$\left\{\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right\} \quad \text{and} \quad \left\{\frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}\right\}.$$

With coordinates $\{t, r, \theta, \phi\}$, we can pick

$$\left\{\frac{\partial}{\partial t}, \frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}, \frac{\partial}{\partial \phi}\right\},$$

and then any tangent vector can be expressed in the form $u = u^\alpha \partial_\alpha$. But basis vectors on a tangent space need not be coordinate arrows. For example, we could take our basis vectors to be arbitrary linear combinations of the form $e_i = A_i^\mu \partial_\mu$, where $A_i^\mu = \mathrm{GL}(n, \mathbb{R})$; these are still linearly independent and span the space.

Another example of basis vectors that are not coordinate arrows are the normalized versions of $\{\frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}\}$ – one can take as basis $\{\hat{r}, \hat{\phi}\}$, where

$$\begin{aligned}\hat{r} &= \frac{1}{\sqrt{g\left(\frac{\partial}{\partial r}, \frac{\partial}{\partial r}\right)}} \frac{\partial}{\partial r}, \\ \hat{\theta} &= \frac{1}{\sqrt{g\left(\frac{\partial}{\partial \theta}, \frac{\partial}{\partial \theta}\right)}} \frac{\partial}{\partial \theta}.\end{aligned}$$

One-forms

There also exist dual vectors to elements of T_pM, and the space of these constitutes T_p^*M, which is called the *cotangent space* at p. A concrete interpretation is to think of these elements as 'row vectors'. For our purposes, the most useful example of a dual vector will be a generalization of the gradient. We will define the *differential of* f to be

$$\underline{d}f : T_pM \to \mathbb{R}, \qquad \underline{d}f(V) := V_p[f],$$

and given a chart we have

$$\begin{aligned}\underline{d}f(V) &= V_p[f] \\ &= V_p^\mu \frac{\partial f}{\partial x^\mu}\bigg|_{x^\mu(p)} \in \mathbb{R}.\end{aligned}$$

It's not hard to show that this is indeed linear. Now, recalling that $V^\mu = V[x^\mu]$, we have

$$\begin{aligned}\underline{d}f[V] = V^\mu \frac{\partial f}{\partial x^\mu} &= V[x^\mu]\frac{\partial f}{\partial x^\mu} \\ &= \left(\frac{\partial f}{\partial x^\mu}\,\underline{d}x^\mu\right)[V].\end{aligned}$$

Then since V was arbitrary,

$$\boxed{\underline{d}f = \frac{\partial f}{\partial x^\mu}\,\underline{d}x^\mu.}$$

This expression will probably look very familiar from previous classes.

(there's probably a better place to put this proof) Now let us show that the one-forms $\{\underline{d}x^\mu\}$ comprise a basis for T_pM^*. Take an arbitrary one-form $\omega \in T_p^*M$, and take some chart $\{x^\mu\}$. Define the components of ω in this basis by

$$\omega_\mu \equiv \omega(\partial_\mu). \tag{12.1}$$

(These are just numbers, because elements of T_p^*M eat vectors and spit out numbers.) Then

$$\boxed{\omega = \omega_\mu\,\underline{d}x^\mu,} \tag{12.2}$$

that is, $\{\underline{d}x^\mu\}$ span the cotangent space at p. To see this, we simply feed basis vectors ∂_ν to both sides. The left-hand side of Eq. (12.2) becomes ω_ν by Eq. (12.1), while the right-hand side becomes

$$(\omega_\mu\,\underline{d}x^\mu)(\partial_\nu) = \omega_\mu\,\underline{d}x^\mu[\partial_\nu] = \omega_\mu\partial_\nu[x^\mu] = \omega_\mu\delta^\mu_\nu = \omega_\nu$$

by the definition of the differential. Since the two sides of Eq. (12.2) are linear maps that agree on a basis, they are the same mapping, which justifies the equality sign. Then to see that $\{\underline{d}x^\mu\}$ are linearly independent, we put

$$\alpha_\mu \, \underline{d}x^\mu = 0.$$

Feeding ∂_ν to both sides gives $\alpha_\nu = 0$, which shows that all the coefficients are zero.

To complete this description, we observe that (the first equality is a definition; see the construction of the coordinate basis)

$$\begin{aligned} V^\mu = V[x^\mu] &= \underline{d}x^\mu(V) \\ &= V[\underline{d}x^\mu], \end{aligned}$$

so there exists a symmetry between how we get components of a vector and the components of a one-form; compare Eq. (12.1). Note that all of this exists prior to the notion of a metric.

Another thing we can comment on is the physicist's notion of an 'infinitesimal'. This is the sort of thing that mathematicians find problematic. When a physicist talks about a quantity dr that is 'very small, but not zero', one can make this rigorous by making the mental substitution

$$dr \longrightarrow \underline{d}r.$$

How do the components of a one-form transform? Suppose I go from $\{\partial_\mu\} \to \{\partial_{\overline{\nu}}\}$. The components of the same one-form ω would then be given by

$$\omega_\mu = \omega[\partial_\mu]$$

and

$$\begin{aligned} \omega_{\overline{\nu}} = \omega[\partial_{\overline{\nu}}] &= \omega\left[\frac{\partial x^\mu}{\partial \overline{x}^\nu}\partial_\mu\right] \\ &= \frac{\partial x^\mu}{\partial \overline{x}^\nu}\omega[\partial_\mu], \end{aligned}$$

which tells us that

$$\boxed{\omega_{\overline{\nu}} = \frac{\partial x^\mu}{\partial \overline{x}^\nu}\omega_\mu.}$$

This is Arfken's definition of a 'covariant vector' – this is really a one-form.

Tangent and cotangent bundles

Consider an arbitrary manifold. Each point p on the manifold is associated with a tangent and cotangent space, T_pM and T_p^*M respectively. One can talk about the union of all these spaces, which are called

$$\boxed{TM = \bigcup_p T_pM} \qquad \text{(tangent bundle)}$$

and analogously,

$$\boxed{T^*M = \bigcup_p T_p^*M} \qquad \text{(cotangent bundle)}$$

We aren't just making an academic distinction here. For example, Hamiltonian mechanics takes place on a cotangent bundle![25] And in GR, the objects we'll be interested in will turn out to live on tangent or cotangent bundles as well.

Vector fields

One can think of vector fields and one-form (covector) fields as 'sections' of TM and T^*M. To make sense of this, consider a one-dimensional manifold M; we can think of tangent vectors 'sticking out' of the manifold, but these really live in one-dimensional space as well.

(figure here)

Now if we flatten this out into a horizontal line and look at each point on the manifold, we get a picture like this:

(figure here)

If we associate each point on the manifold with a vector in its corresponding tangent space, we get a sort of 'cross section'. This happens to be an example of a fiber bundle, so-called because there are 'fibers' associated with each point. Notice that TM is itself a two-dimensional manifold. Similarly, $T\mathbb{R}^3$ is a six-dimensional manifold. In this class, we'll be working with smooth vector fields; since the tangent bundle is itself a manifold, it is also coordinatizable, and smooth means the same thing.

Recall that vectors are denoted

$$X_p : F(M) \to \mathbb{R},$$

and now we will say that a vector field X is such that

$$X : F(M) \to F(M).$$

[25] Goldstein does a rather poor job of explaining the underlying mathematical structure of classical mechanics. If you want to do things geometrically, take a look at the books by Arnold or Jose and Saletan.

13 More tensors; the geodesic equation; connections

Previously, we discussed multilinear maps based on vector spaces and dual spaces; in a similar way, we can consider multilinear maps based on tangent and cotangent spaces. In particular, we can conceive of objects that receive multiple vectors, one-forms, or some combination of the two.

These multilinear maps are called tensors. If such a map accepts m one-forms and n vectors, then we call this a tensor of rank (m, n). This is exactly what we discussed previously; rank $(1, 0)$ tensors are vectors, and rank $(0, 1)$ tensors are one-forms. But now we're talking about tangent and cotangent spaces, and these are tied to particular points p on the manifold, so our notation will change slightly. We will say that

$$W \in T^m_{n,p}(M),$$

and we can get a coordinate expansion

$$W = W^{\mu_1 \cdots \mu_m}{}_{\nu_1 \cdots \nu_n} \partial_{\mu_1} \otimes \partial_{\mu_2} \otimes \cdots \otimes \partial_{\mu_m} \otimes dx^{\nu_1} \otimes \cdots \otimes dx^{\nu_n}$$

where

$$W^{\mu_1 \cdots \mu_m}{}_{\nu_1 \cdots \nu_n} = W(dx^{\mu_1}, dx^{\mu_2}, \dots, dx^{\mu_m}, \partial_{\nu_1}, \dots, \partial_{\nu_n}).$$

When we write this sort of coordinate expansion, we need to remember that it is local; this is chart-dependent. Some standard terminology is

$$T^1_{0,p}(M) = T_pM,$$
$$T^0_{1,p}(M) = T^*_pM,$$

and it is generally presumed that

$$T^0_0(M) = F(M),$$

where the right-hand side is the space of smooth functions on the manifold M. To see how this works out, we will revisit Minkowski and work through some examples.

Tensor products

If $g \in \mathcal{T}^0_2$, then for any vectors X, Y we have

$$\begin{aligned} g(X, Y) &= g(X^i \partial_i, Y^j \partial_j) \\ &= g(\partial_i, \partial_j) X^i Y^j \\ &= g_{ij} X^i Y^j \\ &= g_{ij}\, dx^i(X) dx^j(Y) \\ &= (g_{ij} dx^i \otimes dx^j)(X, Y), \end{aligned}$$

which shows that

$$g = g_{ij}\, dx^i \otimes dx^j,$$

where $g_{ij} := g(\partial_i, \partial_j)$. In particular, with a symmetric tensor g (so that $g_{ij} = g_{ji}$) in two dimensions, we obtain

$$\begin{aligned} g &= g_{11}\, dx^1 \otimes dx^1 + g_{12}(dx^1 \otimes dx^2 + dx^2 \otimes dx^1) + g_{22} dx^2 \otimes dx^2 \\ &= g_{11} \tfrac{1}{2}(dx^1 \otimes dx^1 + dx^1 \otimes dx^1) + 2 g_{12} \tfrac{1}{2}(dx^1 \otimes dx^2 + dx^2 \otimes dx^1) \\ &\qquad + g_{22} \tfrac{1}{2}(dx^2 \otimes dx^2 + dx^2 \otimes dx^2). \end{aligned}$$

The first line is straightforward, but the second line looks sort of unnecessary. However, there's a reason we wrote things this way – in many physics books, you won't even see the tensor product notation. Instead, they work with the following notation:

$$dx^i dx^j := \tfrac{1}{2}(dx^i \otimes dx^j + dx^j \otimes dx^i)$$

Then you get

$$\begin{aligned} g &= g_{ij}\, dx^i dx^j \\ &= g_{11}(dx^1)^2 + 2g_{12}\, dx^1 dx^2 + g_{22}(dx^2)^2 \end{aligned}$$

Physicists have always written things like this, but this notation tends to puzzle mathematicians. And it's important to note that something like $(dx^1)^2$ is not a literal square; it is actually a tensor product.

Now, just because these are tensor products does not mean that you should abandon your calculus intuition. The modern notation makes many operations look familiar. For example, a change of basis looks like

$$\begin{aligned} g &= g_{ij}\, dx^i dx^j \\ &= g_{ij} \frac{\partial x^i}{\partial y^k}\, dy^k \frac{\partial x^j}{\partial y^l}\, dy^l \\ &= g_{ij} \frac{\partial x^i}{\partial y^k} \frac{\partial x^j}{\partial y^l}\, dy^k dy^l \\ &= g'_{kl}\, dy^k dy^l, \end{aligned}$$

which tells us that the components transform like

$$\boxed{g'_{kl} = \frac{\partial x^i}{\partial y^k} \frac{\partial x^j}{\partial y^l}\, g_{ij}, \quad \text{where } x = x(y).}$$

If you want to be more explicit, this is

$$\boxed{g'_{kl}(y) = \frac{\partial x^i}{\partial y^k}(y) \frac{\partial x^j}{\partial y^l}(y) g_{ij}\big(x(y)\big)}$$

Many of the things we studied in SR will remain true in GR. For example, the metric is still the structure that we use to define proper time and proper length. On a Riemannian manifold, there exists a basis in which all the g_{ij} components come with positive signs. But what we have is a semi-Riemannian manifold, so we need to be a bit careful with signs. Now,

$$\ell(\gamma) = \int \sqrt{g(\dot{x}, \dot{x})}\, d\lambda$$

is something we can define in Riemannian geometry, where

$$\dot{x} = \frac{dx^\mu}{d\lambda} \partial_\mu,$$

so that

$$\ell(\gamma) = \int \sqrt{g_{\mu\nu}\, \dot{x}^\mu(\lambda) \dot{x}^\nu(\lambda)}\, d\lambda.$$

In Lorentzian geometry, we put a minus sign under the square root to define

$$\tau(\gamma) = \int \sqrt{-g_{\mu\nu}\, \dot{x}^\mu(\lambda)\dot{x}^\nu(\lambda)}\, d\lambda.$$

A key point we revisit is that, *a priori*, we cannot compare vectors living on different tangent spaces. There is no canonical correspondence between these spaces, even though they are isomorphic. This will become a problem when we try to define the notion of a derivative – if we want to be able to do this without relying on 'preferred' coordinates, then we will need to introduce new structures.

The geodesic equation

But even without this, we can already think of extremal paths. In particular, we can ask about paths that extremize τ. Since the square root function is monotonically increasing, we may as well extremize the quantity inside. The Euler-Lagrange equation gives us a solution to this extremization problem. In this case we have

$$L = g_{\mu\nu}\big(x(\lambda)\big)\dot{x}^\mu(\lambda)\dot{x}^\nu(\lambda),$$

so an extremal path will satisfy

$$\frac{\partial L}{\partial x^\mu} = \frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\mu}\right).$$

Let's examine both sides:

$$\begin{aligned}\frac{\partial L}{\partial x^\mu} &= \frac{\partial g_{\alpha\beta}}{\partial x^\mu}\dot{x}^\alpha \dot{x}^\beta \\ &\equiv g_{\alpha\beta,\mu}\dot{x}^\alpha\dot{x}^\beta\end{aligned}$$

(The comma is a more compact way of denoting a derivative.)

$$\begin{aligned}\frac{\partial L}{\partial \dot{x}^\mu} &= 2g_{\alpha\beta}\delta^\alpha{}_\mu \dot{x}^\beta \\ &= 2g_{\mu\beta}\dot{x}^\beta\end{aligned}$$

Therefore, applying the chain rule,

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\mu}\right) = 2g_{\mu\beta,\alpha}\, \dot{x}^\alpha\dot{x}^\beta + 2g_{\mu\beta}\, \ddot{x}^\beta.$$

Now we equate our expressions and divide by 2 on both sides to get

$$\frac{1}{2}g_{\alpha\beta,\mu}\dot{x}^\alpha\dot{x}^\beta = g_{\mu\beta,\alpha}\, \dot{x}^\alpha\dot{x}^\beta + g_{\mu\beta}\, \ddot{x}^\beta,$$

which implies

$$\begin{aligned}g_{\mu\beta}\ddot{x}^\beta &= \frac{1}{2}(-2g_{\mu\beta,\alpha}\, \dot{x}^\alpha\dot{x}^\beta + g_{\alpha\beta,\mu}\, \dot{x}^\alpha\dot{x}^\beta) \\ &= -\frac{1}{2}(2g_{\mu\beta,\alpha} - g_{\alpha\beta,\mu})\dot{x}^\alpha\dot{x}^\beta \\ &= -\frac{1}{2}(g_{\mu\alpha,\beta} + g_{\mu\beta,\alpha} - g_{\alpha\beta,\mu})\dot{x}^\alpha\dot{x}^\beta.\end{aligned}$$

The last line is our first instance of 'index gymnastics'; it follows because $g_{\mu\beta,\alpha} = g_{\mu\alpha,\beta}$ (this just swaps dummy indices) and $\dot{x}^\alpha \dot{x}^\beta = \dot{x}^\beta \dot{x}^\alpha$. We do this because it makes a symmetry more apparent, and also because this is the standard way of writing things, which we'll see in a moment. If we then multiply both sides by the inverse of g, we get

$$\underbrace{g^{\sigma\mu} g_{\mu\beta} \ddot{x}^\beta}_{=\delta^\sigma{}_\beta \ddot{x}^\beta} = -\frac{1}{2} g^{\sigma\mu}(g_{\mu\alpha,\beta} + g_{\mu\beta,\alpha} - g_{\alpha\beta,\mu}) \dot{x}^\alpha \dot{x}^\beta$$

so we end up with

$$\boxed{\ddot{x}^\sigma = -\frac{1}{2} g^{\sigma\mu}(g_{\mu\alpha,\beta} + g_{\mu\beta,\alpha} - g_{\alpha\beta,\mu}) \dot{x}^\alpha \dot{x}^\beta} \qquad \text{(geodesic equation)}$$

This is the equation for an extremal path; one thing you need to appreciate is that this is a Very Complicated Equation. The right-hand side is a highly nontrivial combination of a lot of things; as such, this equation is highly nonlinear, and in general is extremely challenging to solve. The only input here is the metric, and generally, the g's you see on the right will be functions of x.

The shorthand for this is to use *Christoffel symbols*,

$$\begin{aligned}\Gamma^\sigma_{\alpha\beta} &:= \frac{1}{2} g^{\sigma\mu}(g_{\mu\alpha,\beta} + g_{\mu\beta,\alpha} - g_{\alpha\beta,\mu}) \\ &= \left\{ \begin{matrix} \sigma \\ \alpha\beta \end{matrix} \right\}.\end{aligned}$$

The second line is what you'll sometimes see in older texts. This symbol may have been chosen to emphasize the fact that $\Gamma^\sigma_{\alpha\beta}$ is not a tensor! Also, the rewriting we did earlier shows that this is symmetric with respect to the two lower indices. If you're looking at a more modern book, you'll see Christoffel symbols:

$$\boxed{\ddot{x}^\sigma + \Gamma^\sigma_{\alpha\beta} \dot{x}^\alpha \dot{x}^\beta = 0} \qquad \text{(also geodesic equation)}$$

We emphasize that no structure beyond the metric was needed to carry this out.

Acceleration in $\mathbb{R}^3$

How can one define "straightness"? We're often not aware that one way to do this is the figure below.

(figure here)

Then if $\vec{v}(t) = \vec{v}_\parallel$, we say that the curve γ is not accelerating. OK. But this relied on our connection; that is, we made use of an implicit parallel transport rule, where Cartesian components are preserved. Looking at things from a slightly different perspective, we compared vectors living in two different tangent spaces. However, defining a parallel transport rule on more general manifolds is a nontrivial matter. We wouldn't want to carry the current prescription over to every manifold, since there's nothing special about Cartesian coordinates in these cases. So what should our new procedure be?

A parallel transport rule is completely arbitrary, in the sense that you can define any number of ways to parallel transport along a curve. But let's look for a parallel transport rule that respects the geometry of $\mathbb{R}^3$. Our requirements will be as follows:

(1) If $\vec{u}$ is parallel transported, then $\|\vec{u}(t)\| = \|\vec{u}(t+\epsilon)\|$. (Norms are preserved.)

(2) If $\vec{u}$ and $\vec{v}$ are parallel transported, then $\vec{u}(t) \cdot \vec{v}(t) = \vec{u}(t+\epsilon) \cdot \vec{v}(t+\epsilon)$. (Angles between vectors are preserved.)

In this regard, we are tying our transport rule to our metric. Now, a central result in differential geometry is that given a metric g, there exists a *unique* transport rule satisfying both of these requirements. And once you define a transport rule on some $\gamma(t)$, then you can define a derivative.

$$\frac{DV(t)}{dt} = \lim_{\epsilon \to 0} \frac{V_\epsilon^{\|} - V(t)}{\epsilon}$$

This absolute derivative depends on the rule that we choose for our parallel-transported vector $V_\epsilon^{\|}$, and we will refer to this transport rule as a connection.

Linear/affine connection, ∇

We'll start with a definition.

Definition. A *linear/affine connection* ∇ is a mapping

$$\nabla : TM \times TM \to TM$$

that can accept $X \in TM$ in its first slot,

$$\nabla_X : TM \to TM,$$

and possesses the following properties (X, Y, Z are vector fields and f, g are functions)

(1) $\nabla_{fX+gY} Z = f\nabla_X Z + g\nabla_Y Z$,

(2) $\nabla_X(Y+Z) = \nabla_X Y + \nabla_X Z$,

(3) $\nabla_X(fY) = f\nabla_X Y + (\nabla_X f)Y$,

and the action of ∇ on functions is defined by

$$\nabla_X f = X(f).$$

The first two properties show that ∇ is linear in its first slot, while the second two show that ∇_X satisfies a Leibniz property. (Notice that these are the properties of a directional derivative.) We emphasize that ∇ is not a tensor, due to property (3). But notice that (1) implies the map $\nabla Y : X \mapsto \nabla_X Y$ is a linear map from TM to itself – hence it defines a $(1,1)$ tensor.

By definition, $\nabla_X Y \in TM$. Suppose we have a chart

$$\{x^j\} \longleftrightarrow \partial_j.$$

Now suppose we put in our basis vectors. We know that $\nabla_{\partial_j} \partial_j \in TM$, so if we assume that our manifold is D-dimensional, then there exist D^3 functions Γ^k_{ij} (which depend on $\{x^j\}$) such that

$$\boxed{\nabla_j \partial_i = \Gamma^k_{ij}\, \partial_k.} \qquad \text{(connection coefficients)}$$

Here we've adopted the shorthand notation $\nabla_{\partial_j} = \nabla_j$. Put another way, choosing the set of D^3 functions Γ^k_{ij} is tantamount to specifying how ∇ acts on an arbitrary vector. This is a more concrete way of thinking about a connection; it will be important to keep this in mind when we start speaking about a certain *choice* of a linear connection. Finally, remember that a connection is an extra structure which, *a priori*, has nothing to do with our metric.

14 Covariant derivatives

During the Q&A portion of today's lecture, Dr. Vega talks about Weinberg's book. While general relativity is the easiest place to see how geometry applies to physics, this doesn't mean that all non-geometric treatments of the subject are bad! Weinberg is very good because it contains a lot of explicit calculations that cannot be found in elementary textbooks.

As another comment, abstract index notation (which we'll be seeing more of shortly) is not universally adopted – Dr. Vega attributes this in part to institutional inertia. Mathematicians tend to reserve indices exclusively for components; they fought hard to find a way of writing that sheds all dependence on components. But this gets clunky for what we'll be doing.

Recall that

$$\nabla : TM \times TM \to TM \quad \text{and} \quad \nabla_X : TM \to TM.$$

You can think of $\nabla_X = \nabla(X, \cdot\,)$ and $\nabla_X Y = \nabla(X, Y) \in TM$, but this is not how we'll write things, because this notation suggests that ∇ is a $(1, 2)$ tensor. And ∇ is *not* a tensor, because it is not multilinear.

Earlier we pointed out that ∇Y is a $(1, 1)$ tensor. For notation, we'll use

$$\boxed{\nabla Y(\omega, X) = \langle \omega, \nabla_X Y\rangle.}$$

While this may look like an inner product, it isn't! It's simply another way of indicating that a vector is being fed to a one-form, or vice versa; remember that both are linear maps. Thus $\langle \cdot\,, \cdot\rangle$ is linear in both slots, and $\langle \omega, v\rangle \equiv \omega(v) \equiv v(\omega)$. In the same way

$$\begin{aligned}(\nabla_X Y)(\omega) &= \omega(\nabla_X Y)\\ &= \langle \omega, \nabla_X Y\rangle.\end{aligned}$$

We say that ∇Y is the *covariant derivative* of Y. This name relates to the fact that the covariant derivative of a rank (r, s) tensor is a tensor of rank $(r, s + 1)$; that is, it adds a covariant index to the tensor. Indeed, in abstract index notation, we would write $\nabla_b Y^a$ or $Y^a{}_{;b}$.[26]

While we're here, we can also consider the covariant derivative of a scalar function. We have

$$\nabla f : X \mapsto \nabla_X f = X(f),$$

where the equality follows from our definition of ∇. Note, then, that $\nabla f = df$.

Our definitions so far have had to do with vector fields, but we can extend our discussion to general tensor fields by introducing some postulates. First, if A and B are tensors, then

$$\nabla_X(A \otimes B) = (\nabla_X A) \otimes B + A \otimes (\nabla_X B),$$

which is a very natural product rule property. We will also postulate that

$$\nabla_X \circ C = C \circ \nabla_X.$$

That is, ∇_X commutes with contractions.

[26] The expression $\nabla_b Y^a$ is a bit more prone to misinterpretation, since the two symbols are tied to each other. In particular, you'll never see a ∇_a sitting by itself in an equation, because ∇ is not a tensor. This isn't an issue with the semicolon notation.

Contractions

Consider an arbitrary tensor $Q \in T^m_n$. Then a contraction C is a mapping

$$C : T^m_n \to T^{m-1}_{n-1}.$$

One needs to specify which slots we are contracting on. We'll say that for $Q \in T^m_n$, we have

$$\boxed{C^k_j Q := \sum_i Q(\overbrace{-,-,\dots,\underset{\substack{\uparrow\\ k\text{th one-form slot}}}{dx^i},\dots,}^{m\text{ one-form slots}}\overbrace{-,\dots,\underset{\substack{\uparrow\\ j\text{th vector slot}}}{\partial_i},\dots}^{n\text{ vector slots}})}$$

Since Q is linear in its remaining slots (it's a tensor), it is clear that $C^k_j Q \in T^{m-1}_{n-1}$ as claimed. To see a concrete example of how this works, consider $\omega \in T^0_1$ and $Y \in T^1_0$. Then $\omega \otimes Y$ is a $(1,1)$ tensor, and there is only one way to contract, so we'll omit the indices in what follows.

$$\begin{aligned}
C(\omega \otimes Y) &:= \sum_i (\omega \otimes Y)(\partial_i, dx^i) \\
&= \sum_i \omega(\partial_i) Y(dx^i) \\
&= \sum_i \omega(\partial_i) dx^i(Y) \\
&= \sum_i \omega_i Y^i \\
&\equiv \omega_i Y^i.
\end{aligned}$$

In particular, $C(\omega \otimes Y) = \omega_i Y^i = \langle \omega, Y \rangle$. If we recall that

$$\begin{aligned}
\omega \otimes Y &= (\omega \otimes Y)_i{}^j dx^i \otimes \partial j \\
&= (\omega_i Y^j)\, dx^i \otimes \partial_j,
\end{aligned}$$

we see that this contraction can be thought of as a sum over "corresponding components". For $(1,1)$ tensors, you can think of a contraction as taking a trace, since these tensors can be interpreted as matrices. But more generally, the important thing to remember is that a contraction reduces the rank of a tensor in the way we defined.

While it isn't immediately obvious, this definition does not depend on the basis you choose. We won't give a proof, but you can see this in the example we just gave, since $C(\omega \otimes Y) = \omega_\alpha Y^\alpha = \omega(Y)$, and the thing on the right does not depend on the choice of basis.

Covariant derivatives of $(1,0)$ tensors

Let us now bring this into the context of equations that we can work with. (That is, let's now think about components.) A choice of ∇ defines parallel transport and allows us to define a derivative for geometric objects. Now suppose we've made a selection of a vector at each point on the manifold. Given a parametrization of a curve on the manifold, $\gamma : I \subseteq \mathbb{R} \to M$, where t is our parameter and $\dot\gamma$ a tangent vector, we define the *absolute derivative*

$$\boxed{\frac{DV}{dt} := \nabla \dot\gamma V.}$$

With a chart $\{x^j\}$, we get
$$\dot\gamma^j = \frac{dx^j}{dt}, \quad \text{because} \quad \dot\gamma = \frac{dx^j}{dt}\partial_j.$$
Let's see what the components of this look like. To make things neater, we first let W be an arbitrary vector and then substitute $W = \dot\gamma$ afterwards. We have
$$\begin{aligned} \nabla_W V &= \nabla_W(V^i\partial_i) && \text{using our basis} \\ &= (\nabla_W V^i)\partial_i + V^i\nabla_W\partial_i && \text{Leibniz property} \end{aligned}$$
and we can examine these two terms individually:
$$\begin{aligned} \nabla_W V^i &= W(V^i) && \text{since } \nabla_X f = Xf \\ &= (W^j\partial_j)(V^i) \\ \implies \nabla_{\dot\gamma} V^i &= \frac{dx^j}{dt}\frac{\partial V^i}{\partial x^j} \end{aligned}$$
and
$$\begin{aligned} \nabla_W &= \nabla_{W^i\partial_i} \\ &= W^i\nabla_{\partial_i} && \text{first-slot linearity of } \nabla_X \\ &\equiv W^i\nabla_i \\ \implies \nabla_{\dot\gamma} &= \frac{dx^i}{dt}\nabla_i. \end{aligned}$$
Recall also that
$$\nabla_{\partial_j}\partial_i = \nabla_j\partial_i = \Gamma^k_{ij}\partial_k,$$
where Γ^k_{ij} are D^3 functions that we choose in order to specify our connection. Thus
$$\begin{aligned} \nabla_{\dot\gamma} V &= \left(\frac{dx^i}{dt}\frac{\partial V^i}{\partial x^j}\right)\partial_i + V^i\frac{dx^j}{dt}\Gamma^k_{ij}\partial_k \\ &= \frac{dV^i\big(x(t)\big)}{dt}\partial_i + \Gamma^i_{jk}\big(x(t)\big)V^j\big(x(t)\big)\dot x^k\partial_i \end{aligned}$$
which tells us that
$$\boxed{\frac{DV}{dt} = \left[\frac{dV^i\big(x(t)\big)}{dt} + \Gamma^i_{jk}\big(x(t)\big)V^j\big(x(t)\big)\dot x^k\right]\partial_i.}$$

Notice that DV/dt is also a vector field, although we knew this without having to bring components in. The reason we care about this is that we will say that
$$\frac{DV}{dt} = 0 \iff V \text{ is parallel-transported along } \gamma.$$
This gives a set of D differential equations; all that one needs to solve these are the initial components of V^i and x^i. From the theory of ordinary differential equations, we know that a unique solution exists.

Following the steps from earlier, we see that for an arbitrary $X \in TM$,

$$\boxed{\nabla_X Y = \Big(X^j \partial_j Y^i + \Gamma^i_{kj} X^j Y^k\Big)\partial_i}$$

This shows that given a chart, it is not so hard to compute the components of $\nabla_X Y$. Note that setting $X = \dot\gamma$ recovers the previous boxed equation.

Covariant derivatives of $(0,1)$ tensors

We want to find analogous expressions for the components when working with $\nabla_X : T^*M \to T^*M$. Then by using the product rule property that we postulated for ∇_X, we will be able to find an expression for the components of $\nabla_X : T^r_s \to T^r_s$. First consider the tensor δ^a_b. Recall that given a chart, we have $dx^a(\partial_b) = \delta^a_b$. Now

$$\begin{aligned}
0 = \nabla_X \delta^a{}_b &= \nabla_X \langle dx^a, \partial_b\rangle && \text{since } \delta^a{}_b = dx^a(\partial_b) = \langle dx^a, \partial_b\rangle \\
&= \nabla_X C(dx^a \otimes \partial_b) && \text{since } C(\omega\otimes Y) = \langle \omega, Y\rangle \\
&= C\nabla_X(dx^a \otimes \partial_b) && \text{by the postulate } C\circ\nabla_X = \nabla_X \circ C \\
&= C\big[(\nabla_X dx^a)\otimes\partial_b + dx^a \otimes \nabla_X\partial_b\big] && \text{by the product rule postulate} \\
&= \langle \nabla_X dx^a, \partial_b\rangle + \langle dx^a, \nabla_X \partial_b\rangle && \text{linearity of } C \\
&= (\nabla_X dx^a)(\partial_b) + \langle dx^a, X^c\nabla_c\partial_b\rangle && \text{first-slot linearity of } \nabla_X \\
&= (\nabla_X dx^a)_b + \langle dx^a, X^c\Gamma^d_{bc}\partial_d\rangle && \text{using } \omega_b = \omega(\partial_b) \text{ and } \nabla_c\partial_b = \Gamma^d_{bc}\partial_d \\
&= (\nabla_X dx^a)_b + X^c\Gamma^d_{bc}\underbrace{dx^a(\partial_d)}_{=\delta^a_b} && \text{linearity of } \langle\,\cdot\,,\,\cdot\,\rangle
\end{aligned}$$

This implies that $0 = (\nabla_X dx^a)_b + \Gamma^a_{bc}X^c$, or

$$\boxed{(\nabla_X dx^a)_b = -\Gamma^a_{bc}X^c.}$$

Now observe that

$$\nabla_X\partial_a = X^b\nabla_b\partial_a = X^b\Gamma^c_{ab}\partial_c,$$

so we obtain the equivalent expressions

$$\boxed{\nabla_X\partial_a = \Gamma^c_{ab}X^b\partial_c}$$

and

$$\boxed{\nabla\partial_a = \Gamma^c_{ab}\,\partial_c\otimes dx^b.}$$

(To see why these are equivalent, note that giving X to the second expression gives $dx^b(X) = X^b$, and the object that's left accepts a one-form.) To find analogous expressions for one-forms, note that our expression from earlier implies

$$\boxed{\nabla_X dx^a = -\Gamma^a_{bc}X^c dx^b,}$$

so an equivalent expression (following the same line of reasoning we just used) is

$$\boxed{\nabla dx^a = -\Gamma^a_{bc}\, dx^b \otimes dx^c.}$$

Since we know what ∇ does to the basis one-forms, we know how it acts on arbitrary one-forms:

$$\begin{aligned}
\nabla_X \omega &= \nabla_X(\omega_a dx^a) \\
&= (\nabla_X \omega_a) dx^a + \omega_a \nabla_X dx^a \\
&= (X^b \omega_{a,b}) dx^a + \omega_a \left[-\Gamma^a_{bc} X^c dx^b\right] \\
&= (X^a \omega_{b,a}) dx^b - \Gamma^a_{bc} \omega_a X^c dx^b
\end{aligned}$$

Thus

$$\boxed{\nabla_X \omega = \Big(X^a \omega_{b,a} - \Gamma^a_{bc} \omega_a X^c\Big) dx^b.}$$

To see why the third line follows, recall that

$$X = \frac{dx^i}{dt} \partial_i,$$

which implies (recall that ω_a is a function!)

$$\nabla_X \omega_a = X \omega_a = \frac{dx^b}{dt} \frac{\partial \omega_a}{\partial x_b} \equiv X^b \omega_{a,b}.$$

The expression we've obtained makes it clear that $\nabla_X \omega$ is another one-form. Moreover, it is not hard to compute the components – we simply take a derivative and then put together some terms. (Computing these is very similar to what we do for $\nabla_X Y$.) Finally, note that this allows us to make sense of the equation

$$\frac{D\omega}{dt} = 0$$

given $\gamma : (a, b) \to M$, since we now know what $\nabla_{\dot\gamma} \omega$ is.

Covariant derivatives of tensors of arbitrary rank

Finally, we can generalize to (r, s) tensors. This is just a direct application of the results we obtained in the previous two subsections, together with our product rule postulate. Recall that

$$T = T^{\alpha_1 \alpha_2 \cdots \alpha_r}{}_{\beta_1 \beta_2 \cdots \beta_s}\, \partial_{\alpha_1} \otimes \partial_{\alpha_2} \otimes \cdots \otimes \partial_{\alpha_r} \otimes dx^{\beta_1} \otimes \cdots \otimes dx^{\beta_s}.$$

Therefore, using the properties of ∇_X, we obtain (omitting most terms; we get $r+s+1$ terms on the right from the product rule)

$$\begin{aligned}
\nabla_X T = (\nabla_X T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s})\, &\partial_{\alpha_1}\otimes\partial_{\alpha_2}\otimes\cdots\otimes\partial_{\alpha_r}\otimes dx^{\beta_1}\otimes\cdots\otimes dx^{\beta_s} \\
&+ T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s}\,(\nabla_X\partial_{\alpha_1})\otimes\partial_{\alpha_2}\otimes\cdots \\
&\quad\vdots \\
&+ T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s}\,\partial_{\alpha_1}\otimes\partial_{\alpha_2}\otimes\cdots\otimes(\nabla_X dx^{\beta_1})\otimes dx^{\beta_2}\otimes\cdots \\
&\quad\vdots \\
= (X^k T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s,k})\, &\partial_{\alpha_1}\otimes\partial_{\alpha_2}\otimes\cdots\otimes\partial_{\alpha_r}\otimes dx^{\beta_1}\otimes\cdots\otimes dx^{\beta_s} \\
&+ T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s}\,(\Gamma^j_{\alpha_1 k}X^k\partial_j)\otimes\partial_{\alpha_2}\otimes\cdots \\
&\quad\vdots \\
&- T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s}\,\partial_{\alpha_1}\otimes\partial_{\alpha_2}\otimes\cdots\otimes(\Gamma^{\beta_1}_{jk}X^k dx^j)\otimes dx^{\beta_2}\otimes\cdots \\
&\quad\vdots
\end{aligned}$$

In order for the indices in all the tensors to match, we swap j and α_1 in the second term and β_1 and j in the $(r+2)$th term, proceeding in a similar way with the remaining terms. (This is fine, since these are just dummy indices.) We then obtain

$$\boxed{\begin{aligned}
(\nabla_X T)^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s} = (X^k T^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s,k}) &+ \Gamma^{\alpha_1}_{jk}T^{j\alpha_2\cdots\alpha_r}{}_{\beta_1\cdots\beta_s}X^k + \underset{\substack{\uparrow \\ r-1\text{ more terms}}}{\cdots} \\
&- \Gamma^j_{\beta_1 k}T^{\alpha_1\cdots\alpha_r}{}_{j\beta_2\cdots\beta_s}X^k - \underset{\substack{\uparrow \\ s-1\text{ more terms}}}{\cdots}
\end{aligned}}$$

Just like before, this is not just a formula for $\nabla_X T$; it also tells us what ∇T is. For example, for a tensor of rank $(1,1)$ we have

$$T^\alpha{}_{\beta;\gamma} = T^\alpha{}_{\beta,\gamma} + \Gamma^\alpha{}_{\mu\gamma}T^\mu{}_\beta - \Gamma^\mu{}_{\beta\gamma}T^\alpha{}_\mu\,.$$

Generally, there are a lot of terms here, but you can take some comfort in knowing that we rarely work with arbitrary (r,s) tensors. The worst you'll probably see, unless you go to certain subfields, are tensors of type $(0,5)$ or $(1,4)$.

Now that we know how to compute $\nabla_{\dot\gamma}T$ for arbitrary tensors, we can define the parallel transport of an arbitrary tensor. Notice that every expression we've derived so far has depended on our choice of connection coefficients Γ^k_{ij} – in the next lecture, we will pick coefficients so that our parallel transport rule satisfies certain properties.

15 Curvature and torsion; metric compatibility

Recall that we need to put an extra structure on a manifold in order to relate different tangent spaces. This extra structure is the connection, which is a mapping

$$\nabla_X : T^{(r,s)}M \to T^{(r,s)}M.$$

We recall that given a chart, the components for the covariant derivative of a vector and a one-form, respectively, are

$$(\nabla Y)^i{}_j = \partial_j Y^i + \Gamma^i_{jk} Y^k$$

and

$$(\nabla \omega)_{ij} = \partial_j \omega_i - \Gamma^k_{ij} \omega_k.$$

Notice the general pattern here – the components of X in ∇_X get contracted onto a partial derivative of our tensor in the first term, while for the remaining terms we get

$$+\Gamma^{\alpha_1}_{jk} T^{j\alpha_2\cdots\alpha_r}{}_{\beta_1\cdots\beta_s} X^k$$

for the vector slots, where we have a contraction of the first downstairs index of our connection coefficient and the upstairs indices of our tensor. Then for the terms from our one-forms, we have things like

$$-\Gamma^j_{\beta_1 k} T^{\alpha_1\cdots\alpha_r}{}_{j\beta_2\cdots\beta_s} X^k$$

where there is now a minus sign, and we contract over the upstairs index of our connection coefficient and the downstairs indices of our tensor. While computing these components is straightforward in principle, this is a tedious process, and we usually leave this task to a CAS. Also, since we're speaking of a general connection coefficient Γ^k_{ij}, we need to be careful about the ordering of the lower indices.

Uniqueness of parallel transport

Is there a well-defined notion of a parallel-transported tensor? That is, if I have a choice of connection ∇, a curve γ, and some tensor A at a given point p on a manifold, does it make sense to talk about *the* tensor parallel-transported along γ to some other point q? The answer is yes. To see why, suppose we have a chart, so that

$$\dot{\gamma} = \frac{dx^\alpha}{dt}\partial_\alpha.$$

By definition, a parallel-transported tensor A is one that satisfies

$$\frac{DA}{dt} := \nabla_{\dot{\gamma}} A = 0$$

everywhere along γ. Using our expression for the components of $\nabla_{\dot{\gamma}} A$ from the previous lecture, this is equivalent to

$$\frac{dx^\alpha}{dt}\partial_\alpha A^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s} + \dot{x}^m \left(\Gamma^{\alpha_1}_{jm} A^{j\alpha_2\cdots\alpha_r}{}_{\beta_1\cdots\beta_s} + \cdots - \Gamma^j_{\beta_1 m} A^{\alpha_1\cdots\alpha_r}{}_{j\beta_2\cdots\beta_s} - \cdots\right) = 0,$$

which can be rewritten in the form

$$\boxed{\dot{A}^{\alpha_1\cdots\alpha_r}{}_{\beta_1\cdots\beta_s} - S^{\alpha_1\cdots\alpha_r\gamma_1\cdots\gamma_s}{}_{\beta_1\cdots\beta_s\delta_1\cdots\delta_r} A^{\delta_1\cdots\delta_r}{}_{\gamma_1\cdots\gamma_s}(t) = 0,}$$

where S is some tensor. The reason we've introduced S here is to make apparent that this is just $D^{(r+s)}$ first order ordinary differential equations for A; from the theory of ODEs, we know that this has a unique solution. Therefore,

> parallel transport specifies a *unique* tensor field, provided we know the value of our tensor at the initial point.

Finally, we emphasize again – the notions of connection and parallel transport can all be defined without making reference to a metric. The next subsection will have to do with two new geometric objects that can be defined given only a connection; as such, they do not need to have anything to do with a metric.

Curvature and torsion

In what follows, we won't be doing things in a constructive manner – instead, we'll give some definitions, and as we proceed, we will see that these are useful objects.

Definition (Torsion)**.** Given an affine connection ∇ on M, the *torsion* T is a mapping

$$T : TM \times TM \to TM$$

where

$$\boxed{T(X, Y) := \nabla_X Y - \nabla_Y X - [X, Y].}$$

Here, $[X, Y]$ is the so-called commutator of X and Y, defined by

$$[X, Y]f := X(Y f) - Y(X f).$$

You should quickly verify that $[X, Y]$ is a vector to see that our formula for $T(X, Y)$ is well-defined.

Definition (Curvature)**.** Given an affine connection ∇ on M, the *curvature* R is a mapping

$$R : TM \times TM \times TM \to TM$$

where

$$\boxed{R(X, Y)Z := \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]}Z.}$$

Again, you should verify that the object on the right is a vector field. It turns out that T and R are multilinear mappings, which is somewhat remarkable, considering all the derivatives involved. (This fact will be left for you to prove.) The takeaway is that

$$T \in T^{(1,2)}M, \qquad R \in T^{(1,3)}M.$$

You might wonder why the vector Z in the definition of R seems to occupy a privileged spot. The reason is not particularly deep – this is probably just meant to highlight that X and Y appear differently, i.e., play different roles from Z in the defining formula.

Now, one can think of the torsion in a slightly different way, as

$$\begin{aligned} \widetilde{T} : T^*M \times TM \times TM &\to F(M) \\ (\omega, X, Y) &\mapsto \langle \omega, T(X, Y) \rangle =: \widetilde{T}(\omega, X, Y). \end{aligned}$$

In the physics literature, you'll won't see any distinction between $\widetilde{T}$ and T. Similarly for the curvature, one can consider

$$\widetilde{R} : T^*M \times TM \times TM \times TM \to F(M)$$
$$(\omega, Z, X, Y) \mapsto \langle \omega, R(X,Y)Z \rangle =: \widetilde{R}(\omega, X, Y, Z).$$

It's important to note that these definitions only require a connection ∇. So from now on, if you hear the word "curvature", you should remember that there is an underlying connection. Conversely, if we specify a connection, then we automatically get an associated torsion and curvature.

Now let's look for the components of these objects. We start with the torsion, recalling that if we have a chart, we can write

$$T = T_{ij}{}^k \, \partial_k \otimes dx^i \otimes dx^j.$$

Then we have[27]

$$\begin{aligned}
T_{ij}{}^k &= \widetilde{T}(dx^k, \partial_i, \partial_j) \\
&= \langle dx^k, T(\partial_i, \partial_j) \rangle \\
&= \langle dx^k, \nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i - \cancel{[\partial_i, \partial_j]} \rangle \\
&= \langle dx^k, \Gamma^m{}_{ji}\partial_m - \Gamma^m{}_{ij}\partial_m \rangle \\
&= \Gamma^m{}_{ji}\langle dx^k, \partial_m \rangle - \Gamma^m{}_{ij}\langle dx^k, \partial_m \rangle \\
&= \Gamma^m{}_{ji}\delta^k_m - \Gamma^m{}_{ij}\delta^k_m,
\end{aligned}$$

where the commutator in the third line vanishes due to the equality of mixed partials. Thus

$$\boxed{T_{ij}{}^k = \Gamma^k_{ji} - \Gamma^k_{ij}.} \qquad \text{(components of } T\text{)}$$

That is, our connection coefficients are symmetric in their lower indices if and only if $T_\nabla = 0$. Next, we look at the components of the curvature tensor.

$$\begin{aligned}
R^i{}_{jkl} &\overset{1}{=} \widetilde{R}(dx^i, \partial_j, \partial_k, \partial_l) \\
&\overset{2}{=} \langle dx^i, R(\partial_k, \partial_l)\partial_j \rangle \\
&\overset{3}{=} \langle dx^i, \nabla_{\partial_k}\nabla_{\partial_l}\partial_j - \nabla_{\partial_l}\nabla_{\partial_k}\partial_j - \nabla_{\cancel{[\partial_k, \partial_l]}}\partial_k \rangle \\
&\overset{4}{=} \langle dx^i, \nabla_{\partial_k}(\Gamma^s{}_{jl}\partial_s) - \nabla_{\partial_l}(\Gamma^s{}_{jk}\partial_s) \rangle \\
&\overset{5}{=} \langle dx^i, (\nabla_{\partial_k}\Gamma^s{}_{jl})\partial_s + \Gamma^s{}_{jl}\nabla_{\partial_k}\partial_s - (\nabla_{\partial_l}\Gamma^s{}_{jk})\partial_s - \Gamma^s{}_{jk}\nabla_{\partial_l}\partial_s \rangle \\
&\overset{6}{=} \langle dx^i, \Gamma^s{}_{jl,k}\partial_s + \Gamma^s{}_{jl}\Gamma^r{}_{sk}\partial_r - \Gamma^s{}_{jk,l}\partial_s - \Gamma^s{}_{jk}\Gamma^r{}_{sl}\partial_r \rangle \\
&\overset{7}{=} \Gamma^s{}_{jl,k}\langle dx^i, \partial_s \rangle + \Gamma^s{}_{jl}\Gamma^r{}_{sk}\langle dx^i, \partial_r \rangle - \Gamma^s{}_{jk,l}\langle dx^i, \partial_s \rangle - \Gamma^s{}_{jk}\Gamma^r{}_{sl}\langle dx^i, \partial_r \rangle \\
&\overset{8}{=} \Gamma^s{}_{jl,k}\delta^i_s + \Gamma^s{}_{jl}\Gamma^r{}_{sk}\delta^i_r - \Gamma^s{}_{jk,l}\delta^i_s - \Gamma^s{}_{jk}\Gamma^r{}_{sl}\delta^i_r.
\end{aligned}$$

[27] We're offsetting indices on the connection coefficients $\Gamma^i{}_{jk}$ for typographical clarity, not because they represent components of a tensor. (They don't!)

There's a lot to justify here, so we'll go through the whole process. Lines 1, 2, and 3 follow from the definitions, and the last term in line 3 vanishes because $\nabla_{fX}Y = f\nabla_X Y$, thus $f = 0$ implies $\nabla_{fX}Y = 0$. Line 4 uses the definition of connection coefficients, line 5 uses the Leibniz property of ∇_X, and line 6 uses the definition of Γ^k_{ij} again, as well as the comma notation for derivatives. Line 7 uses the linearity of $\langle\,\cdot\,,\,\cdot\,\rangle$, and line 8 uses the identity $\langle dx^i, \partial_j\rangle = \delta^i_j$. Therefore, we have

$$\boxed{R^i{}_{jkl} = \Gamma^i_{jl,k} - \Gamma^i_{jk,l} + \Gamma^s_{jl}\Gamma^i_{sk} - \Gamma^s_{jk}\Gamma^i_{sl}} \qquad \text{(components of } R\text{)}$$

This monstrosity is not something you'll have to memorize, since you can always look this expression up in a book. The important takeaway is that a connection implies a curvature (and a torsion, which we showed earlier). This is clear from the above, since the components of R depend solely on the connection coefficients.

Another thing we can construct given a connection is the so-called Ricci tensor,

$$\mathrm{Ric} = C^1_3 R = \sum_i \widetilde{R}(dx^i, -\,, \partial_i, -\,).$$

Since it is the contraction of a tensor, Ric is a tensor as well, and in particular, $\mathrm{Ric} \in T^{(0,2)}M$. We conclude this part by noting that we've come quite far! A connection tells you how to propagate objects along a manifold, it gives you a notion of "no change", and it specifies a torsion, curvature, and Ricci tensor.

⊛ An alternative characterization of torsion-free

Recall the expression for the covariant derivative of a one-form from the previous lecture:

$$\omega_{\mu;\nu} = \omega_{\mu,\nu} - \Gamma^\rho_{\mu\nu}\omega_\rho.$$

Since $f_{;\alpha} = f_{,\alpha}$, we obtain

$$f_{;\mu\nu} = f_{,\mu\nu} - \Gamma^\rho_{\mu\nu} f_{,\rho}$$

and then we can antisymmetrize to get

$$f_{;[\mu\nu]} = f_{,[\mu\nu]} - \Gamma^\rho_{[\mu\nu]} f_{,\rho}.$$

The first term on the right vanishes in a coordinate basis (partial derivatives commute), thus

$$(f_{;\mu\nu} - f_{;\nu\mu}) = -(\Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu}) f_{,p}.$$

Since f was arbitrary, this shows that

$$\boxed{\text{torsion-free connection} \iff \nabla_a\nabla_b f = \nabla_b\nabla_a f.}$$

Tying a connection to a metric

Geometry is concerned with things like length, but you cannot speak of length without a metric. So far, nothing we've done with connections has made reference to a metric, and for good reason – one can speak of a connection on a manifold without one. However, we *do* have a metric, and back in lecture 13, we discussed some properties that we'd like a parallel transport rule (i.e., a connection) to satisfy. These properties were geometrical – they had to do with the preservation of norms and angles between vectors. In this way, we can tie ∇ to g.

It turns out that the norm-preserving and angle-preserving properties for parallel transport are equivalent! Let's start with the forward implication. Suppose

$$\boxed{\nabla_W V = 0 \implies \nabla_W\big(g(V,V)\big) = 0 \quad \text{for all vectors } V, W.}$$

It's straightforward to see how this relates to our $\mathbb{R}^3$ property – if V is parallelly transported, then its norm $g(V,V)$ should be preserved, which is expressed by the vanishing on the right. Now let's see how angle-preservation follows. Suppose U and V are vectors such that that $\nabla_W U = \nabla_W V = 0$. By the linearity of ∇_W, it follows that $\nabla_W(U+V) = 0$. Then

$$\nabla_W\big(g(U+V,U+V)\big) = \nabla_W\big(g(U,U)\big) + \nabla_W\big(g(V,V)\big) + 2\nabla_W\big(g(U,V)\big),$$

where we've used the symmetry of g. By assumption, the left-hand side and the first two terms on the right are zero, which shows that

$$\boxed{\nabla_W\big(g(U,V)\big) = 0 \quad \text{for all vectors } U, V, W \text{ such that } \nabla_W U = \nabla_W V = 0,}$$

and this is what we wanted. The reverse implication is obvious – we just let $U = V$. Thus

$$\boxed{\text{The two statements above are equivalent.}}$$

Metric compatibility

Now we'll show how this property can be more concisely stated; first we establish an identity.

Lemma. Given any two vectors U, V and metric g, we have

$$g(V,U) = CC(g \otimes V \otimes U),$$

where C denotes a contraction.

Proof. Expand the left-hand side out in a coordinate basis:

$$\begin{aligned}
(g_{ij}\, dx^i \otimes dx^j)(V^l\partial_l, U^k\partial_k) &= g_{ij}\, dx^i(V^l\partial_l)\, dx^j(U^k\partial_k) \\
&= g_{ij} V^l U^k \delta^i{}_l \delta^j{}_k \\
&= g_{ij} V^i U^j.
\end{aligned}$$

Then do the same for the right-hand side:

$$\begin{aligned}
CC(g \otimes V \otimes U)(\,-\,,\,-\,,\,-\,,\,-\,) &= C\sum_i (g \otimes V \otimes U)(\,-\,,\partial_i,\,-\,,dx^i) \\
&= \sum_j\sum_i (g \otimes V \otimes U)(\partial_j,\partial_i,dx^j,dx^i) \\
&= \sum_j\sum_i g(\partial_j,\partial_i)V(dx^j)U(dx^i) \\
&= \sum_j\sum_i g_{ji}V^jU^i \equiv g_{ji}V^jU^i.
\end{aligned}$$

While we chose specific indices to contract over (namely C^2_4) in the first line for the purposes of writing a concrete expression, the particular choice does not matter! This is because g is symmetric, so $g_{ij} = g_{ji}$. If we had chosen, say, C^2_3, then we would get $g_{ji}V^iU^j = g_{ij}V^iU^j$. □

Next we note that $\nabla g \in T^{(0,3)}M$, which is true because we can feed in a vector to get $\nabla_X g$. We know that this object is a $(0,2)$ tensor, because g is. Thus, we may write

$$(\nabla g)(V,U,W) := (\nabla_W g)(V,U).$$

Now we proceed as follows. Assume that U and V are vectors parallelly transported along some curve with tangent vector W. Then

$$\begin{aligned}
\nabla_W\big(g(V,U)\big) &= \nabla_W\big(CC(g\otimes V\otimes U)\big) && \text{our lemma}\\
&= CC\big(\nabla_W(g\otimes V\otimes U)\big) && C\circ\nabla_W = \nabla_W\circ C\\
&= CC\big[\nabla_W g\otimes V\otimes U + g\otimes\nabla_W V\otimes U + g\otimes V\otimes\nabla_W U\big] && \text{product rule for } \nabla_W\\
&= (\nabla_W g)(V,U) + g(\cancel{\nabla_W V},U) + g(V,\cancel{\nabla_W U}) && \text{lemma, assumption}\\
&= (\nabla_W g)(V,U)\\
&=: (\nabla g)(V,U,W).
\end{aligned}$$

The inner product between U and V is preserved, meaning that the left-hand side is zero, and so $(\nabla g)(V,U,W) = 0$. But since W was arbitrary, we conclude that $\nabla g = 0$.

$$\boxed{\begin{array}{c}\text{our parallel transport rule}\\ \text{preserves inner products/norms}\end{array} \iff \nabla g = 0.}$$

This is called *metric compatibility*, and is to be thought of as a condition on ∇ rather than on g. Imposing this allows us to retain the $\mathbb{R}^3$ notion of preserved lengths and angles.

⊛ The same proof, different notation

This is a bit cleaner in abstract index notation, because the contractions are hidden in the indices. Suppose that v and w are vectors parallelly transported along some curve, so that $t^a\nabla_a v^b = 0$ and $t^a\nabla_a w^b = 0$. Then since ∇_a commutes with contractions and obeys a product rule, we have

$$\begin{aligned}
t^a\nabla_a(g_{bc}v^b w^c) &= t^a v^b w^c\nabla_a g_{bc} + t^a g_{bc}w^c\nabla_a v^b + t^a g_{bc}v^b\nabla_a w^c\\
&= t^a v^b w^c\nabla_a g_{bc}.
\end{aligned}$$

So if we require that parallel transport preserves inner products, it follows that

$$t^a v^b w^c\nabla_a g_{bc} = 0,$$

which holds for all parallelly transported vectors along arbitrary curves if and only if $\nabla_a g_{bc} = 0$. Conversely, $\nabla_a g_{bc} = 0$ implies that parallel transport preserves inner products. □

16 The Levi-Civita connection; abstract index notation

We saw in the previous lecture that metric compatibility, or $\nabla g = 0$, is equivalent to requiring that parallel transport preserve inner products. Let's see how to express this in components. Since ∇g is a $(0,3)$ tensor, we have

$$\begin{aligned}
(\nabla g)_{ijk} &= (\nabla g)(\partial_i, \partial_j, \partial_k) && \\
&= (\nabla_{\partial_k} g)(\partial_i, \partial_j) && \text{def. of } \nabla g \\
&= \nabla_{\partial_k}\big(g(\partial_i, \partial_j)\big) - g(\nabla_{\partial_k}\partial_i, \partial_j) - g(\partial_i, \nabla_{\partial_k}\partial_j) && \text{product rule and } \nabla_W \circ C = C \circ \nabla_W \\
&= g_{ij,k} - g(\Gamma^l{}_{ik}\partial_l, \partial_j) - g(\partial_i, \Gamma^l{}_{jk}\partial_l) && \text{comma notation, def. of } \Gamma^k_{ij} \\
&= g_{ij,k} - \Gamma^l{}_{ik} g_{lj} - \Gamma^l{}_{jk} g_{il}. && g \text{ is a tensor}
\end{aligned}$$

Recalling the semicolon notation, we see that metric compatibility amounts to

$$\boxed{g_{ij;k} = 0.}$$

Specifying a unique connection

We've spoken at length about properties that a connection ought to satisfy. Does metric compatibility suffice to get a unique connection? Let us do some counting. First we recall that a connection is specified by a choice of D^3 functions. Then

$$g_{ij;k} = 0$$

D independent equations

$D(D+1)/2$ independent equations

To see what we're doing here, note that each permutation of the indices amounts to one equation for the connection coefficients. However, not all of these are independent – exchanging the first two indices gives the same equation, since g is symmetric. Therefore, imposing metric compatibility leaves us with

$$D^3 - \frac{D^2(D+1)}{2} = \frac{D^2(D-1)}{2}$$

free functions. Our naive counting argument suggests that we have this many connection coefficients left to specify. We need more conditions! What we need to notice now is

$$\widetilde{T} \in T^{(1,2)}M$$

and

$$\begin{aligned}
\widetilde{T}(\omega, U, V) &:= \langle \omega, T(U,V) \rangle \\
&= \langle \omega, \nabla_U V - \nabla_V U - [U,V] \rangle,
\end{aligned}$$

which reveals that $\widetilde{T}$ is antisymmetric:

$$\widetilde{T}(\omega, U, V) = -\widetilde{T}(\omega, V, U).$$

Recall also the expression for the components of the torsion tensor:

$$\widetilde{T}^i{}_{jk} = \Gamma^i{}_{kj} - \Gamma^i{}_{jk}$$

Therefore, if we impose the extra condition that $\widetilde{T} = 0$, then the antisymmetry in the lower indices gives $D(D-1)/2$ independent equations, while the upstairs index contributes D equations. That is, our torsion-free condition specifies $D^2(D-1)/2$ functions, which is precisely what we needed. To summarize, then,

$$\text{naive counting} \quad \Longrightarrow \quad \begin{array}{c}\text{there exists a } \textit{unique}, \\ \text{metric-compatible,} \\ \text{torsion-free connection } \nabla.\end{array}$$

This is not yet a proof, but we will see one shortly.

The metric-compatible, torsion-free connection on $\mathbb{R}^3$

Note that $\mathbb{R}^3$ is a special kind of manifold. All of its connection coefficients are zero – parallel-transporting a vector along any curve leaves that vector untouched. (General manifolds have more 'moving parts'.) We can prove this by a direct computation – if we know how the connection acts on all our basis vectors, then we can determine the coefficients from the definition

$$\nabla_k \partial_j = \Gamma^i_{jk} \partial_i.$$

This is a bit cumbersome, but let's do this as an exercise. We are working in $\mathbb{R}^3$, so we have $g_{ij} = \operatorname{diag}(1,1,1)$. First let us determine what $\nabla_{\partial_x} \partial_x$ is – we will get some combination of our basis vectors, which will specify the connection coefficients Γ^i_{xx}.

We take a sideways approach to this problem. First note that

$$0 = \nabla_{\partial_x}\big(g(\partial_x, \partial_x)\big) = \cancel{(\nabla_{\partial_x} g)}(\partial_x, \partial_x) + g(\nabla_{\partial_x}\partial_x, \partial_x) + g(\partial_x, \nabla_{\partial_x}\partial_x),$$

where the first term vanishes by metric compatibility, so the symmetry of g implies

$$\boxed{g(\nabla_{\partial_x}\partial_x, \partial_x) = 0.}$$

A similar computation gives

$$g(\nabla_{\partial_x}\partial_x, \partial_y) = \nabla_{\partial_x}\big(\cancel{g(\partial_x, \partial_y)}\big) - g(\partial_x, \nabla_{\partial_x}\partial_y),$$

where we've already discarded the $(\nabla_{\partial_x} g)$ term, and where the first term on the right vanishes since $g_{ij} = \operatorname{diag}(1,1,1)$. To treat the remaining term on the right, we use our torsion-free condition to write

$$\begin{aligned} g(\nabla_{\partial_x}\partial_x, \partial_y) &= -g(\partial_x, \cancel{[\partial_x, \partial_y]} + \nabla_{\partial_y}\partial_x) \\ &= -g(\partial_x, \nabla_{\partial_y}\partial_x) \\ &= -\frac{1}{2}\nabla_{\partial_y}\big(g(\partial_x, \partial_x)\big) \\ &= 0, \end{aligned}$$

where we used the CC-lemma[28] from the previous lecture in the third line, together with the symmetry of g. Therefore, we have

$$\boxed{g(\nabla_{\partial_x}\partial_x, \partial_y) = 0,}$$

[28] This is a made-up name, no one calls it this

and the same argument shows that

$$\boxed{g(\nabla_{\partial_x}\partial_x, \partial_z) = 0.}$$

Since any vector V can be expressed as a linear combination of our basis vectors, this implies

$$g(\nabla_{\partial_x}\partial_x, V) = 0$$

for all V. By nondegeneracy, this implies that

$$\boxed{\nabla_{\partial_x}\partial_x = 0.}$$

In other words, the connection coefficients Γ^i_{xx} are all zero, and a similar computation shows that *all* the connection coefficients vanish, as we claimed.

The Koszul formula

The following formula, which applies for a metric-compatible, torsion-free connection, will be useful in the next subsection. We have

$$\begin{aligned} g(\nabla_X Y, Z) &= \nabla_X\big(g(Y,Z)\big) - g(Y, \nabla_X Z) \\ &= \nabla_X\big(g(Y,Z)\big) - g(Y, [X,Z] + \nabla_Z X) \\ &= \nabla_X\big(g(Y,Z)\big) - g(Y, [X,Z]) - g(\nabla_Z X, Y), \end{aligned}$$

where we used $\nabla g = 0$ in the first line and $T = 0$ in the second line. By permuting X, Y, and Z to get two more equations, we can eliminate all the terms like $g(\nabla_Z X, Y)$ to get

$$\begin{aligned} g(\nabla_X Y, Z) = \frac{1}{2}\Big[&\nabla_X\big(g(Y,Z)\big) + \nabla_Y\big(g(Z,X)\big) - \nabla_Z\big(g(X,Y)\big) \\ &- g(Y,[X,Z]) + g(X,[Z,Y]) - g(Z,[Y,X])\Big]. \end{aligned}$$

This is the so-called *Koszul formula.* This was a generalization of the computation in the previous subsection – this time, we do not assume anything special about g or our vectors X, Y, Z.

The Levi-Civita connection

Say we have a local chart $\{x^\mu\}$, so that we can put $X = \partial_\alpha$, $Y = \partial_\beta$, and $Z = \partial_\mu$. Then the commutators in the Koszul formula vanish, and we obtain

$$\begin{aligned} & g(\nabla_\alpha\partial_\beta, \partial_\mu) = \frac{1}{2}\Big[\nabla_\alpha\big(g(\partial_\beta,\partial_\mu)\big) + \nabla_\beta\big(g(\partial_\mu,\partial_\alpha)\big) - \nabla_\mu\big(g(\partial_\alpha,\partial_\beta)\big)\Big] \\ \iff \quad & \Gamma^\gamma_{\alpha\beta}\, g_{\gamma\mu} = \frac{1}{2}(g_{\beta\mu,\alpha} + g_{\mu\alpha,\beta} - g_{\alpha\beta,\mu}) \\ \iff \quad & \Gamma^\gamma_{\alpha\beta} = \frac{1}{2}g^{\gamma\mu}(g_{\beta\mu,\alpha} + g_{\mu\alpha,\beta} - g_{\alpha\beta,\mu}). \end{aligned}$$

Let's recap – if our connection is metric-compatible and torsion-free, we get the Koszul formula. Then if we have a local chart, we can use this formula to obtain the connection coefficients from the definition. We see that we have no choice – any metric-compatible and torsion-free connection must have coefficients given by the above. We call this the *Levi-Civita connection*, and write

$$\boxed{\Gamma^{\gamma}_{\alpha\beta} = \frac{1}{2}g^{\gamma\mu}(g_{\beta\mu,\alpha} + g_{\mu\alpha,\beta} - g_{\alpha\beta,\mu}).} \qquad \text{(Levi-Civita)}$$

For example, if we put $g_{ij} = \operatorname{diag}(1,1,1)$, then we get a flat connection, as we found earlier.

You may be confused by the notation. Up until this point, we've worked with general *connection coefficients* Γ^i_{jk}, which are defined without reference to a metric. On the other hand, the *Christoffel symbols* are a very particular combination of the metric and its derivatives. These tend to be denoted Γ^i_{jk} as well, because in many cases the two are the same thing! But generally these are distinct objects – see the next subsection.

A few more comments. In gauge theory, you don't generally have a metric, so you don't use the Levi-Civita connection. And there are occasions in GR where you are looking at two spacetimes – a background spacetime, and a perturbed spacetime. Then you will have two (Levi-Civita) connections based on your two spacetimes.

⊛ What does this tell us about geodesics?

To quickly recap, we introduced geodesics as length-extremizing curves, where the notion of length was provided to us by $g_{\mu\nu}$. We found that these satisfied the equation

$$\frac{d^2x^\sigma}{du^2} + \left\{ \begin{matrix} \sigma \\ \alpha\beta \end{matrix} \right\} \frac{dx^\alpha}{du}\frac{dx^\beta}{du} = 0,$$

where the expressions in braces are the Christoffel symbols,

$$\left\{ \begin{matrix} \sigma \\ \alpha\beta \end{matrix} \right\} := \frac{1}{2}g^{\sigma\lambda}(g_{\alpha\lambda,\beta} + g_{\beta\lambda,\alpha} - g_{\alpha\beta,\lambda}).$$

This is one privileged class of curves. But there's another class of curves that we can make sense of even *without a metric*; one can ask about curves that are as straight as possible. As we discussed earlier, the notion of straightness requires a parallel transport rule (connection).

To be more precise, suppose we have a curve $x^\sigma = x^\sigma(u)$, and define a vector field X^σ along the curve by

$$\frac{dx^\sigma}{du} = X^\sigma.$$

We will say that a curve is autoparallel (as straight as possible) if parallel transport along the curve leaves the direction of the tangent vector unchanged:

$$\nabla_X X^\sigma = \lambda X^\sigma.$$

It can be shown[29] that it's possible to reparametrize the curve in such a way that $\lambda = \lambda(u)$ vanishes; in this case we say that we have an *affine parameter*. Thus, without loss of generality, we can say that autoparallel curves satisfy

$$\nabla_X X^\sigma = 0. \qquad \text{(affine parameter)}$$

Now here's the punchline – in components, this expression reads

$$\frac{d^2x^\sigma}{du^2} + \Gamma^\sigma{}_{\alpha\beta}\frac{dx^\alpha}{du}\frac{dx^\beta}{du} = 0.$$

[29]See, for instance, Section 1.3 and Problem 1.13.2 of *A Relativist's Toolkit* by Poisson.

A priori, the connection coefficients ${\Gamma^\sigma}_{\alpha\beta}$ can be anything. However, what we did in this lecture was show that for a torsion-free, metric-compatible connection, we end up with

$$ {\Gamma^\sigma}_{\alpha\beta} = \frac{1}{2} g^{\sigma\lambda}(g_{\alpha\lambda,\beta} + g_{\beta\lambda,\alpha} - g_{\alpha\beta,\lambda}) = \left\{ \begin{matrix} \sigma \\ \alpha\beta \end{matrix} \right\}. $$

So by comparing the defining equations for the two classes of curves, we see that they coincide. This gives us two ways of thinking about geodesics:

(1) Geodesics are length-extremizing curves.

(2) Geodesics are curves that 'curve' as little as possible.

Some texts introduce geodesics as curves that are as straight as possible; we've shown that this is an equivalent definition, provided that you work with a Levi-Civita connection. Otherwise, you need to be careful!

⦿ An alternative approach to covariant derivatives

Here we'll do things a bit differently, reproducing a few definitions and results from section 3.1 in Wald and section 1.7 of Malament. This is very clean and efficient; we will cover a lot of ground in the next couple of pages.

Definition. A *derivative operator* ∇ (also *covariant derivative*) on a manifold is a map which takes each smooth tensor field of type (k,l) to a smooth tensor field of type $(k,l+1)$ and satisfies the properties listed below. We will denote the tensor field resulting from the action of ∇ on $T \in \mathscr{T}(k,l)$ by

$$ \nabla_c {T^{a_1\cdots a_k}}_{b_1\cdots b_l}, $$

and we'll often refer to the derivative operator itself by ∇_c, although this is in some sense an abuse of notation, because ∇_c is not a dual vector.

(1) Linearity: For all $A, B \in \mathscr{T}(k,l)$ and $\alpha, \beta \in \mathbb{R}$,

$$ \nabla_c(\alpha {A^{a_1\cdots a_k}}_{b_1\cdots b_l} + \beta {B^{a_1\cdots a_k}}_{b_1\cdots b_l}) = \alpha\nabla_c {A^{a_1\cdots a_k}}_{b_1\cdots b_l} + \beta\nabla_c {B^{a_1\cdots a_k}}_{b_1\cdots b_l} $$

(2) Leibniz rule: For all $A \in \mathscr{T}(k,l)$, $B \in \mathscr{T}(k',l')$,

$$ \begin{aligned} &\nabla_e[{A^{a_1\cdots a_k}}_{b_1\cdots b_l} {B^{c_1\cdots c_{k'}}}_{d_1\cdots d_{l'}}] \\ &\qquad = [\nabla_e {A^{a_1\cdots a_k}}_{b_1\cdots b_l}] {B^{c_1\cdots c_{k'}}}_{d_1\cdots d_{l'}} + {A^{a_1\cdots a_k}}_{b_1\cdots b_l} [\nabla_e {B^{c_1\cdots c_{k'}}}_{d_1\cdots d_{l'}}] \end{aligned} $$

(3) Commutativity with contraction: For all $A \in \mathscr{T}(k,l)$,

$$ \nabla_d({A^{a_1\cdots c\cdots a_k}}_{b_1\cdots c\cdots b_l}) = \nabla_d {A^{a_1\cdots c\cdots a_k}}_{b_1\cdots c\cdots b_l} $$

(4) Consistency with the notion of tangent vectors as directional derivatives on scalar fields: For all $f \in \mathscr{F}$ and all $t^a \in V_p$

$$ t(f) = t^a \nabla_a f. $$

(5) Torsion-free: For all $f \in \mathscr{F}$,

$$ \nabla_a \nabla_b f = \nabla_b \nabla_a f. $$

Remark. Apparently, proving that a manifold admits even one derivative operator is fairly involved, but we will ignore these subtleties, as you'll see now. △

Example. Given a coordinate system ψ with associated coordinate bases $\{\partial_\mu\}$ and $\{dx^\mu\}$, there exists a corresponding derivative operator ∂_a, called the *ordinary derivative*, defined by

$$T^{a_1\cdots a_k}{}_{b_1\cdots b_l} \mapsto \partial_c T^{a_1\cdots a_k}{}_{b_1\cdots b_l} \equiv \frac{\partial(T^{\mu_1\cdots\mu_k}{}_{\nu_1\cdots\nu_l})}{\partial x^\sigma}\, dx^\sigma \otimes \partial_{\mu_1} \otimes \cdots \otimes \partial_{\mu_k} \otimes dx^{\nu_1} \otimes \cdots \otimes dx^{\nu_l}.$$

Indeed, it is clear (is it? be careful) that the new object is a $(k, l+1)$ tensor field, and one can show that ∂_a obeys properties (1)–(5). But this is not a natural way to create tensor fields from other tensor fields on a manifold, because it privileges the ψ coordinate system; in any other chart, the components of this tensor field will not generally look like partial derivatives of $T^{\mu_{1'}\cdots\mu_{k'}}{}_{\nu_{1'}\cdots\nu_{l'}}$ with respect to the new coordinates. △

It turns out that the difference of two derivative operators, $\tilde\nabla_a - \nabla_a$, defines a tensor field of type $(1,2)$. But before we describe how this is done, we will show that if ω'_b and ω_b are dual vector fields that agree at p, then at p, the operator $\tilde\nabla_a - \nabla_a$ acts on them in the same way. To see this, first let γ_b be a dual vector field such that $\gamma_b|_p = 0$, and let v^b be a vector field. Now consider the scalar field $\gamma_b v^b$. We have

$$0 = (\tilde\nabla_a - \nabla_a)(\gamma_b v^b) = \gamma_b(\tilde\nabla_a - \nabla_a)v^b + v^b(\tilde\nabla_a - \nabla_a)\gamma_b$$

everywhere; in particular, at p we have $v^b(\tilde\nabla_a - \nabla_a)\gamma_b = 0$. Since v^b was arbitrary, this implies $(\tilde\nabla_a - \nabla_a)\gamma_b = 0$. Then our desired result follows, for if ω'_b and ω_b are smooth dual vector fields such that $(\omega'_b - \omega_b)|_p = 0$, then we can put $\gamma_b = \omega'_b - \omega_b$ in the above and conclude that at p,

$$(\tilde\nabla_a - \nabla_a)\omega'_b = (\tilde\nabla_a - \nabla_a)\omega_b.$$

Now we can describe the construction of our $(1,2)$ tensor field, which will be denoted $C^c{}_{ab}$. If we assign to every $p \in M$ a dual vector $\overset{0}{\omega}_c$, we may define $C^c{}_{ab}$ at p by

$$C^c{}_{ab}\overset{0}{\omega}_c = (\tilde\nabla_a - \nabla_a)\omega_b,$$

where ω_b is any smooth dual vector field such that $\omega_b|_p = \overset{0}{\omega}_c$. (Our previous work shows that ω_b can be totally arbitrary at points away from p.) In particular, we can let the covector field make the assignments for us, so for any smooth covector field ω_b, we have

$$\nabla_a\omega_b = \tilde\nabla_a\omega_b - C^c{}_{ab}\omega_c. \tag{16.1}$$

With a bit more work, one can use the torsion-free property of the derivative operator to show that $C^c{}_{ab} = C^c{}_{ba}$. The remaining properties can be used to extend Eq. (16.1) to an analogous expression for arbitrary tensor fields:

$$\nabla_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} = \tilde\nabla_a T^{b_1\cdots b_k}{}_{c_1\cdots c_l} + \sum_i C^{b_i}{}_{ad} T^{b_1\cdots d\cdots b_k}{}_{c_1\cdots c_l} - \sum_j C^d{}_{ac_j} T^{b_1\cdots b_k}{}_{c_1\cdots d\cdots c_l}$$

This shows that the tensor field $C^c{}_{ab}$ completely describes how the two derivative operators $\tilde\nabla_a$ and ∇_a differ. Conversely, (this needs to be verified) one can start from a derivative operator $\tilde\nabla_a$ and a tensor field $C^c{}_{ab}$, and then *define* a new derivative operator ∇_a by the equation above.

Remark. If the last assertion sounds familiar, it's because starting from the derivative operator $\tilde\nabla_a = \partial_a$ corresponds to our notion (in the lectures) of a connection. Then we write $\Gamma^c{}_{ab}$ for the tensor field[30] and, e.g.,

$$\nabla_a t^b = \partial_a t^b + \Gamma^b{}_{ac} t^c.$$

[30]Wald calls these "Christoffel symbols", but unless you want to confuse yourself, you should think of these as connection coefficients (associated with a torsion-free connection), because they have nothing to do with a metric.

The choice of a derivative operator ∇_a then amounts to the choice of a tensor field $\Gamma^c{}_{ab}$. The same comments are in order – while $\Gamma^c{}_{ab}$ is in fact a tensor field, we need to note that it involves the ordinary derivative ∂_a, so we don't expect its components to transform in a tensorial way. △

To round out our discussion, we'll rederive the main result of this lecture.

Proposition. Given a metric g_{ab} and derivative operator $\tilde{\nabla}_a$, there exists a unique derivative operator satisfying $\nabla_a g_{bc} = 0$.

Proof. Let $\tilde{\nabla}_a$ be a derivative operator, and let ∇_a be a derivative operator satisfying $\nabla_a g_{bc} = 0$. Then we have

$$0 = \nabla_a g_{bc} = \tilde{\nabla}_a g_{bc} - C^d{}_{ab} g_{dc} - C^d{}_{ac} g_{bd},$$

which can be rewritten

$$C_{cab} + C_{bac} = \tilde{\nabla}_a g_{bc}.$$

By cycling thru indices, we can get two more equations, and by using the symmetry in the last two indices of C_{abc}, we arrive at

$$C^c{}_{ab} = \frac{1}{2} g^{cd} (\tilde{\nabla}_a g_{bd} + \tilde{\nabla}_b g_{ad} - \tilde{\nabla}_d g_{ab}).$$

This tensor field is manifestly unique, and it completely specifies ∇_a, so we're done. □

Remark. If we put $\tilde{\nabla}_a = \partial_a$, then we recover the familiar expression for the Christoffel symbols!

Abstract index notation

I've misplaced my notes here, but you'll find a nice discussion of this in Section 4.9, *Epilogue: Indices? Indices!* of Matthias Blau's lecture notes.

17 Locally Cartesian coordinates; symmetries of the Riemann tensor

Is it possible to find a chart $\{x'^{\mu}\}$ such that the line element is Cartesian?

$$\begin{aligned} ds^2 &= (dx'^1)^2 + (dx'^2)^2 + (dx'^3)^2 + \cdots \\ &= \delta_{ab}\, dx'^a dx'^b \end{aligned}$$

For a Lorentzian manifold, we replace δ_{ab} with η_{ab}, but the same question can be asked. This is a very physics-y way of thinking about things, but it will help to highlight what curvature (as embodied by the Riemann tensor) really means. We've written $\{x'^{\mu}\}$ with the primes to anticipate the following question: say we have the line element

$$ds^2 = g_{ij}(x)\, dx^i\, dx^j$$

in some chart $\{x^{\alpha}\}$. If we move to some other chart $\{x'^{\mu}\}$, then we have D coordinate functions

$$x'^{\mu} = x'^{\mu}(x^{\alpha}).$$

Then we can give a more precise statement of our question – does there exist a chart satisfying

$$\boxed{\begin{aligned} &(1)\quad g'_{\mu\nu}(p) = \delta_{\mu\nu} \qquad (\text{or } \eta_{\mu\nu}) \\ &(2)\quad \left.\frac{\partial g'_{\mu\nu}}{\partial x'^{\mu}}\right|_p = 0 \end{aligned}}$$

The first condition by itself is somewhat trivial – you can think of this as a diagonalization problem. It is condition (2) that makes things interesting – this says that deviations from the Euclidean (Minkowski) metric occur at second order. We will see that the existence of such a chart amounts to a mathematical statement of the Equivalence Principle.

Some counting

This will not be a proof, but we will give a counting argument to make plausible the statement that a chart satisfying (1) and (2) exists. First note that

$$g'_{\mu\nu}(x') = \frac{\partial x^{\alpha}}{\partial x'^{\mu}} \frac{\partial x^{\beta}}{\partial x'^{\nu}} g_{\alpha\beta}\big(x(x')\big).$$

The conditions we require apply to a neighborhood of p, so we expand in a power series:

$$\begin{aligned} x^{\alpha}(x') = x'^{\alpha}_p &+ \left.\frac{\partial x^{\alpha}}{\partial x'^{\beta}}\right|_p (x'^{\beta} - x'^{\beta}_p) + \frac{1}{2} \left.\frac{\partial^2 x^{\alpha}}{\partial x'^{\beta} \partial x'^{\gamma}}\right|_p (x'^{\beta} - x'^{\beta}_p)(x'^{\gamma} - x'^{\gamma}_p) \\ &+ \frac{1}{6} \left.\frac{\partial^3 x^{\alpha}}{\partial x'^{\beta} \partial x'^{\gamma} \partial x'^{\delta}}\right|_p (x'^{\beta} - x'^{\beta}_p)(x'^{\gamma} - x'^{\gamma}_p)(x'^{\delta} - x'^{\delta}_p) + \mathcal{O}\big[(x' - x'_p)^4\big] \end{aligned}$$

There are some assumptions being made here, such as the existence of such a power series, but we will say that our manifold is nice enough to allow this. (We've gone up to third order, and we could go higher, but this will suffice for our purposes.)

First note that the constant in front (x'^{α}_p) is irrelevant – it amounts to a choice of origin, and doesn't enter into either (1) or (2), since it gets killed by the derivatives. So we start counting from first order:

$$\frac{\partial x^\alpha}{\partial x'^\beta} \quad \longrightarrow \quad D^2 \text{ individual values to choose at } p$$

$$\frac{\partial^2 x^\alpha}{\partial x'^\beta \partial x'^\gamma} \quad \longrightarrow \quad D^2(D+1)/2 \text{ individual values to choose at } p$$

$$\frac{\partial^3 x^\alpha}{\partial x'^\beta \partial x'^\gamma \partial x'^\delta} \quad \longrightarrow \quad D^2(D+1)(D+2)/6 \text{ indiv. values to choose at } p$$

To see why the second assertion is true, note that the index in the numerator contributes D independent values, while the two indices below contribute $D(D+1)/2$ values due to the equality of mixed partials.

The third-order term requires a bit more thought. Since the order in which we differentiate does not matter (partial derivatives commute), this is equivalent to the following combinatorial problem – we have $N = 3$ indistinguishable balls that we toss into D bins. You can use a short 'stars-and-bars' argument to show that the number of ways we can do this is

$$\# = \frac{(N+D-1)!}{(D-1)!N!}.$$

In particular, for the third-order term, we have

$$\# = \frac{(D+2)!}{(D-1)!\,3!} = \frac{1}{6}(D+2)(D+1)D,$$

which we multiply to the D values contributed by the α index to get the result above.

Can we find such a chart?

Now we impose condition (1), which says that $g'_{\alpha\beta}(p) = \eta_{\alpha\beta}$. Since g is symmetric, this amounts to $D(D+1)/2$ conditions. Then

$$D^2 - \frac{D(D+1)}{2} = \frac{D(D-1)}{2},$$

and since we are in $D = 4$ dimensions, this number is 6. In other words, at a given point, we have six degrees of freedom with which to choose a chart in which the metric (evaluated at that point) looks like $\eta_{\mu\nu}$. And this is not unexpected! The Lorentz group has 6 parameters, which correspond to three boosts and three rotations.

Let's look at (2) next. In view of our expression for $g'_{\alpha\beta}$, we see that only the second-order terms in the power series for $x^\alpha(x')$ can enter into this condition, since all the higher-order contributions become zero when evaluated at p. Now, the symmetry of the metric implies

$$\left.\frac{\partial g'_{\alpha\beta}}{\partial x'^\gamma}\right|_p = 0 \quad \longrightarrow \quad D^2(D+1)/2 \text{ values to choose at } p,$$

which happens to be exactly the number of values you need to specify to determine the chart to second order! So although we haven't given a proof, it should seem plausible that it is possible to pick a chart satisfying (1) and (2). In fact, our counting argument in (1) suggests that there are infinitely many such charts.

Can we go further?

Now suppose we want to strengthen (2) by demanding that the second derivatives of $g'_{\alpha\beta}$ vanish at p as well; the deviation from Minkowski would then occur at third order. Is this possible? Let's count how many conditions this would imply.

$$\left.\frac{\partial^2 g'_{\alpha\beta}}{\partial x'^\gamma \partial x'^\delta}\right|_p = 0$$

The symmetry of g implies that we get $D(D+1)/2$ conditions from the indices on top, and since partial derivatives commute, the indices in the bottom give $D(D+1)/2$ conditions. And by the same argument as earlier, only the third-order term in the power series for $x^\alpha(x')$ can contribute. We counted earlier how many conditions we get at third order, so we subtract:

$$\frac{D^2(D+1)^2}{4} - \frac{D^2(D+1)(D+2)}{6} = \frac{D^2(D^2-1)}{12}.$$

This turns out to be more than we can allow for – for $D = 4$, this number equals 20. But this number will turn out to be interesting. In fact, this is the number of independent components of the Riemann tensor (as we will show in a bit). So the best we can do to satisfy (1) and (2) is find a chart such that

$$g_{\alpha\beta} = \eta_{\alpha\beta} + \text{“}\mathrm{Riem}_{\alpha\beta\mu\nu}\Delta x^\mu \Delta x^\nu\text{”}.$$

The Riemann curvature will turn out to be the first measure of deviation from flat space. As a final note, we call coordinates satisfying (1) and (2) *Riemann normal coordinates.*

Ricci identity; algebraic symmetries of the Riemann tensor

Now let's revisit the Riemann tensor – from now on we'll be working with abstract index notation. By the definition of $R^a{}_{bcd}$ and properties of ∇_a, we have

$$\begin{aligned}
R^a{}_{bcd}Z^bX^cY^d &= X^e\nabla_e(Y^f\nabla_f Z^a) - Y^e\nabla_e(X^f\nabla_f Z^a) - [X,Y]^e\nabla_e Z^a \\
&= (X^e\nabla_e Y^f)\nabla_f Z^a + X^eY^f\nabla_e\nabla_f Z^a - (Y^e\nabla_e X^f)\nabla_f Z^a \\
&\qquad\qquad\qquad - Y^eX^f\nabla_e\nabla_f Z^a - [X,Y]^e\nabla_e Z^a \\
&= \Big(X^e\nabla_e Y^f - Y^e\nabla_e X^f - [X,Y]^f\Big)\nabla_f Z^a \\
&\qquad\qquad\qquad + X^eY^f\nabla_e\nabla_f Z^a - X^eY^f\nabla_f\nabla_e Z^a \\
&= T_{cd}{}^f X^cY^d\nabla_f Z^a + X^eY^f[\nabla_e,\nabla_f]Z^a \\
&= X^cY^d[\nabla_c,\nabla_d]Z^a,
\end{aligned}$$

where in the last line we use the fact that our connection is torsion-free. Since X^c and Y^d were arbitrary, we conclude that

$$\boxed{[\nabla_c,\nabla_d]Z^a = R^a{}_{bcd}Z^b.} \qquad \text{(Ricci identity)}$$

Put another way, the curvature measures the lack of commutativity between the ∇_c's. This also tells us about how a commutator acts on a vector. From here, we can derive a similar identity for one-forms:

$$\begin{aligned}[\nabla_a, \nabla_b]Z_c &= [\nabla_a, \nabla_b](g_{cd}Z^d)\\ &= g_{cd}R^d{}_{fab}Z^f\\ &= R_{cfab}Z^f,\end{aligned}$$

where we used metric compatibility and the Ricci identity in the second line. Then by raising and lowering the bound indices, we get

$$\boxed{[\nabla_a, \nabla_b]Z_c = R_c{}^f{}_{ab}Z_f.}$$

We can change some labels to make this look more like our previous result:

$$[\nabla_c, \nabla_d]Z_a = R_a{}^b{}_{cd}\, Z_b.$$

To establish another important index symmetry of the Riemann tensor, we'll first show that

$$[\nabla_a, \nabla_b]f = 0.$$

It suffices to show that $X^aY^b(\nabla_a\nabla_b - \nabla_b\nabla_a)f = 0$ for all $X, Y \in TM$. (A warning: this is true if and only if our connection is torsion-free!) We have

$$\begin{aligned}X^aY^b(\nabla_a\nabla_b - \nabla_b\nabla_a)f &= (\nabla_X\nabla_Y - \nabla_Y\nabla_X - \nabla_{[X,Y]})f\\ &= \nabla_X(Yf) - \nabla_Y(Xf) - [X,Y]f\\ &= X(Yf) - Y(Xf) - [X,Y]f\\ &= 0.\end{aligned}$$

In particular, then, $[\nabla_a, \nabla_b](t^cZ_c) = 0$. Let's examine both terms we get from the commutator:

$$\begin{aligned}\nabla_a\nabla_b(t^cZ_c) &= \nabla_a\big((\nabla_bt^c)Z_c + t^c\nabla_bZ_c\big)\\ &= (\nabla_a\nabla_bt^c)Z_c + \nabla_bt^c\nabla_aZ_c + \nabla_at^c\nabla_bZ_c + t^c\nabla_a\nabla_bZ_c.\end{aligned}$$

we also have

$$\nabla_b\nabla_a(t^cZ_c) = (\nabla_b\nabla_at^c)Z_c + \nabla_at^c\nabla_bZ_c + \nabla_bt^c\nabla_aZ_c + t^c\nabla_b\nabla_aZ_c.$$

Putting these together, we get

$$\begin{aligned}[\nabla_a, \nabla_b](t^cZ_c) &= Z_c[\nabla_a, \nabla_b]t^c + t^c[\nabla_a, \nabla_b]Z_c\\ &= Z_cR^c{}_{dab}t^d + t^c[\nabla_a, \nabla_b]Z_c\end{aligned}$$

But we showed that the left-hand side equals zero, hence

$$\begin{aligned}t^c[\nabla_a, \nabla_b]Z_c &= -t^dR^c{}_{dab}Z_c\\ &= -t^cR^d{}_{cab}Z_d.\end{aligned}$$

Then since this holds for all t^c, we have

$$\boxed{[\nabla_a, \nabla_b]Z_c = -R^d{}_{cab}Z_d.}$$

In view of our previous expression, we obtain

$$-R^d{}_{cab} = R_c{}^d{}_{ab},$$

which is one of the many algebraic symmetries of the Riemann tensor. We can get a downstairs version by acting the lowering operator on both sides:

$$-R_{fcab} = R_{cfab}.$$

By the definition of the Riemann tensor, it is antisymmetric in the last two indices:

$$R^d{}_{cab} = -R^d{}_{cba}.$$

There is a third symmetry that will be left for you to prove as an exercise, called the *first Bianchi identity*. To collect everything, then, we have

$$\boxed{\begin{aligned} &(1)\ \ R^d{}_{cab} = -R^d{}_{cba} \\ &(2)\ \ R_{cdab} = -R_{dcab} \\ &(3)\ \ R_{a[bcd]} = 0 \end{aligned}}$$

If you want to rewrite (3) without the antisymmetrization brackets, this becomes (exercise)

$$\boxed{(3)\ \ R_{abcd} + R_{adbc} + R_{acdb} = 0.}$$

These three symmetries can be used to derive a fourth one (another exercise):

$$\boxed{(4)\ \ R_{abcd} = R_{cdab}.}$$

We care about which properties are independent of one another, since we will count the number of independent components of the Riemann tensor. Let us begin with (1) and (2):

$$\underbrace{R_{\alpha\beta\gamma\delta}}_{R_{AB}}$$

$B = \gamma\delta$ contributes $D(D-1)/2$ by property (1)

$A = \alpha\beta$ contributes $D(D-1)/2$ by property (2)

By symmetry (4), the Riemann tensor is symmetric in the 'bundled' indices A and B, giving

$$\frac{1}{2}\left[\frac{D(D-1)}{2}\right]\left[\frac{D(D-1)}{2}+1\right] = \frac{D^4 - 2D^3 + 3D^2 - 2D}{8} \quad \text{components.}$$

Then you can show (exercise) that symmetry (4) implies

$$\boxed{(5)\ \ R_{[\alpha\beta\gamma\delta]} = 0.}$$

As a final exercise, you can show that symmetries (1), (2), (4) and (5) are equivalent to (1)–(3),[31] so we can use these to do the counting.

[31] If you've worked the previous exercises, then it suffices to show that (3) follows from these four properties. And while this may seem like a lot of work, it's important that you familiarize yourself with the Riemann tensor – it's essentially the left-hand side of the Einstein equation!

Now note that we can write (why?)

$$R_{\alpha\beta\gamma\delta} = X_{\alpha\beta\gamma\delta} + R_{[\alpha\beta\gamma\delta]}.$$

Conditions (1), (2), and (4) apply only to $X_{\alpha\beta\gamma\delta}$; the second term satisfies these automatically by virtue of being a completely antisymmetric tensor. Then condition (5) tells us that we've overcounted by $\binom{D}{4}$ (sketchy, justify), so we subtract to get

$$\frac{D^4 - 2D^3 + 3D^2 - 2D}{8} - \frac{D(D-1)(D-2)(D-3)}{24} = \frac{D^2(D^2-1)}{12} \text{ \textit{independent} components.}$$

This is our final result for the number of independent components of the Riemann tensor. You may recognize this from earlier – it's also the number of components of the metric that cannot be made zero by a suitable choice of chart. In the case of $D = 4$ dimensions, this number is 20.

18 Path-dependence of parallel transport; 'warped time' spacetime

Unlike the familiar case of Euclidean $\mathbb{R}^n$, where you can slide vectors around with impunity, parallel transport on a curved manifold is path-dependent: if you take some vector at a point p and parallel-transport it along two different curves to some other point q, you will generally get two different vectors.[32]

Recall that we can write the condition

$$\frac{DV^\alpha}{d\lambda} = 0$$

in components as

$$\frac{dV^\alpha}{d\lambda} + \Gamma^\alpha_{\beta\gamma} V^\beta \frac{dx^\gamma}{d\lambda} = 0.$$

If we transport a vector along these two curves in succession, we get the same vector. (show this. it is supposed to follow from the components expression)

(figure here)

Now consider a curve C on M that lies in some local chart. We can fill this curve with many small curves C_N that share boundaries:

(figure here)

Then the parallelly transported vector ΔV^α around C equals the sum

$$\Delta V^\alpha = \sum_N (\Delta V^\alpha)_N,$$

since all the interior contributions cancel out by the previous argument. Now let's consider the parallelly transported vector around one of the small curves C_N.

$$\frac{dV^\alpha}{d\lambda} = -\Gamma^\alpha_{\beta\gamma} V^\beta \frac{dx^\gamma}{d\lambda}$$

$$\implies V^\alpha(\lambda) = V^\alpha_p - \int_{\lambda_p}^{\lambda} \Gamma^\alpha_{\beta\gamma} V^\beta \frac{dx^\gamma}{d\lambda}\, d\lambda$$

Next we look at a couple of Taylor expansions. We have

$$\Gamma^\alpha_{\beta\gamma}(\lambda) = (\Gamma^\alpha_{\beta\gamma})_p + (\partial_\delta \Gamma^\alpha_{\beta\gamma})_p \big(x^\delta(\lambda) - x^\delta_p\big) + \mathcal{O}(\Delta x^2),$$

and also

$$V^\alpha(\lambda) = V^\alpha_p - (\Gamma^\alpha_{\beta\gamma})_p V^\beta_p \big(x^\gamma(\lambda) - x^\gamma_p\big) + \mathcal{O}(\Delta x^2).$$

Substituting these into our integral expression (this is an exercise in minding indices), we obtain

$$\begin{aligned} V^\alpha(\lambda) = V^\alpha_p - (\Gamma^\alpha_{\beta\gamma})_p V^\beta_p \int_{\lambda_p}^{\lambda} \frac{dx^\gamma}{d\lambda}\, d\lambda - (\partial_\delta \Gamma^\alpha_{\beta\gamma})_p V^\beta_p \int_{\lambda_p}^{\lambda} \big(x^\delta(\lambda) - x^\delta_p\big) \frac{dx^\gamma}{d\lambda}\, d\lambda \\ - (\Gamma^\alpha_{\beta\gamma})_p (\Gamma^\beta_{\mu\nu})_p V^\mu_p \int_{\lambda_p}^{\lambda} \big(x^\nu(\lambda) - x^\nu_p\big) \frac{dx^\gamma}{d\lambda}\, d\lambda + \mathcal{O}(\Delta x^2). \end{aligned}$$

Since C_N is closed, at some later $\lambda = \lambda_f$ we end back up at p, and then $\Delta V^\alpha = V^\alpha(\lambda_f) - V^\alpha(\lambda_p)$. A bunch of integrals on the right become zero because $x^\gamma(\lambda_f) = x^\gamma(\lambda_p)$, and we get

$$\Delta V^\alpha = -(\partial_\delta \Gamma^\alpha_{\mu\gamma} - \Gamma^\alpha_{\beta\gamma}\Gamma^\beta_{\mu\delta})_p V^\mu_p \oint x^\delta\, dx^\gamma,$$

[32] The canonical example is of parallel transport of a vector around a closed curve on a sphere; if you look at a few books, you're guaranteed to find at least one picture of this.

where we've shuffled a few dummy indices around. Let's tidy up this expression to make some symmetries apparent. First we interchange δ and γ to get

$$\Delta V^\alpha = -(\partial_\gamma \Gamma^\alpha_{\mu\delta} - \Gamma^\alpha_{\beta\delta}\Gamma^\beta_{\mu\gamma})_p V^\mu_p \oint x^\gamma\, dx^\delta,$$

and then we observe that

$$0 = \oint d(x^\gamma x^\delta) = \oint x^\gamma\, dx^\delta + \oint x^\delta\, dx^\gamma,$$

which implies

$$\oint x^\gamma\, dx^\delta = -\oint x^\delta\, dx^\gamma.$$

Thus (using our expression with δ and γ swapped)

$$\Delta V^\alpha = (\partial_\gamma \Gamma^\alpha_{\mu\delta} - \Gamma^\alpha_{\beta\delta}\Gamma^\beta_{\mu\gamma})_p V^\mu_p \oint x^\delta\, dx^\gamma,$$

so we can add our two expressions to obtain

$$\Delta V^\alpha = -\frac{1}{2}\left(\partial_\delta \Gamma^\alpha_{\mu\gamma} - \partial_\gamma \Gamma^\alpha_{\mu\delta} + \Gamma^\alpha_{\beta\delta}\Gamma^\beta_{\mu\gamma} - \Gamma^\alpha_{\beta\gamma}\Gamma^\beta_{\mu\gamma}\right)_p V^\mu_p \oint x^\delta\, dx^\gamma.$$

But the expression in brackets is familiar! We see that

$$\boxed{\Delta V^\alpha = -\frac{1}{2}(R^\alpha{}_{\mu\delta\gamma})_p V^\mu_p \oint x^\delta\, dx^\gamma + \text{higher order terms.}}$$

In other words, the Riemann tensor is a measure of the noncommutativity (path dependence?) of parallel transport.

(why did we introduce the big loop C?)

Newtonian gravity as 'warped time' spacetime

We move on to the most 'physics' description of the Riemann tensor. We start with a thought experiment. Suppose we have a mass m at rest at $\vec{x}_0$, so that $E = mc^2$. Now suppose it falls to $\vec{x}$; in doing so it acquires kinetic energy

$$\frac{1}{2}mv^2 = m\big[U(\vec{x}) - U(\vec{x}_0)\big].$$

Note the difference in convention here – our potential U is defined so that $\vec{F} = \nabla U$, whereas one typically has $\vec{F} = -\nabla\phi$. There's nothing particularly deep behind this choice, it just turns out that omitting the minus sign makes some calculations easier. By conservation of energy, then, we have

$$E = mc^2\left[1 + \frac{U(\vec{x})}{c^2} - \frac{U(\vec{x}_0)}{c^2}\right].$$

Now here is where the 'thought experiment' aspect comes in – suppose that at this point, all this energy is converted into a photon. Then

$$\hbar\omega(\vec{x}) = mc^2\left[1 + \frac{U(\vec{x})}{c^2} - \frac{U(\vec{x}_0)}{c^2}\right].$$

Finally, suppose the photon climbs back up to $\vec{x}_0$. (a figure would be helpful) Energy conservation requires $\hbar\omega(\vec{x}_0) = mc^2$, hence

$$\boxed{\frac{\omega(\vec{x})}{\omega(\vec{x}_0)} = 1 + \frac{U(\vec{x})}{c^2} - \frac{U(\vec{x}_0)}{c^2}.}$$

Now since U is a physical potential, we have $\lim_{|\vec{x}_0|\to\infty} U(\vec{x}_0) = 0$, so we may write

$$\frac{\omega(\vec{x})}{\omega_\infty} = 1 + \frac{U(\vec{x})}{c^2}.$$

Finally, assuming that $|U(\vec{x})| \ll c^2$ (which is reasonable), we obtain

$$\boxed{\frac{\omega_\infty}{\omega(\vec{x})} \approx 1 - \frac{U(\vec{x})}{c^2}.}$$

So as a photon climbs back out to infinity, its frequency decreases, becoming 'redder'. (double check)

Perhaps we can come up with a spacetime that accommodates this redshift effect. It turns out that we can do this by working with the assumption that time flows differently in different points in a gravitational field. So let us start by introducing

$dt(\vec{x})$ — time interval between ticks at $\vec{x}$, as measured by a stationary observer there.

Since $dt \propto 1/\omega$, we get

$$\frac{\omega_\infty}{\omega(\vec{x})} = \frac{dt(\vec{x})}{dt_\infty}.$$

What we would like to find is some spacetime such that the right-hand side equals $1 - U(\vec{x})/c^2$. This cannot be true for flat spacetime, which is described by the line element

$$ds^2 = -d(ct)^2 + d\vec{x}\cdot d\vec{x}.$$

To see why, consider some static observer, for which $d\vec{x} = 0$. Then we have

$$d\tau = \frac{\sqrt{-ds^2}}{c},$$

which is independent of $\vec{x}$. Hence $dt(\vec{x})/dt_\infty = 1$ for all $\vec{x}$, which is not what we want. This means that we need to change our metric! So let us write

$$ds^2 = g_{00}\, d(ct)^2 + d\vec{x}\cdot d\vec{x},$$

where $g_{00} \to -1$ as $|\vec{x}| \to \infty$. Now since $d\tau = \sqrt{-g_{00}}\, dt$ (we just computed this), we have

$$\frac{dt(\vec{x})}{dt_\infty} = \frac{\sqrt{-g_{00}}\, dt}{dt} = \sqrt{-g_{00}}.$$

We can make this equal to $1 - U(\vec{x})/c^2$ if we pick

$$g_{00} \approx -\left(1 - \frac{2U(\vec{x})}{c^2}\right),$$

where the approximation arises from our assumption $|U(\vec{x})| \ll c^2$. Then we get

$$ds^2 = -\left(1 - \frac{2U(\vec{x})}{c^2}\right) d(ct)^2 + d\vec{x} \cdot d\vec{x}.$$

So by construction, this 'warped time' spacetime is one that accommodates the redshift effect. (But note that this has been a heuristic derivation – for example, we have not paid any attention to whether this satisfies the Einstein field equations.)

Now here is the key point. What does freefall look like? We consider the geodesic equation

$$\frac{d^2 x^\alpha}{d\tau^2} + \Gamma^\alpha{}_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

This is now nontrivial due to the component g_{00}. If we still had $g_{ij} = \mathrm{diag}(-1, 1, 1, 1)$, then all the Christoffel symbols would be zero; geodesics would be 'straight lines'. We could compute $\Gamma^\alpha{}_{\mu\nu}$ by brute force from the components of the metric, or we could start from the Lagrangian, which is what we'll do here.

$$\begin{aligned} L &= g_{\alpha\beta} \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} \\ \iff \quad L &= -\left(1 - \frac{2U(\vec{x})}{c^2}\right) c^2 \left(\frac{dt}{d\tau}\right)^2 + \delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} \end{aligned}$$

Now the x^j equations are

$$\begin{aligned} \frac{d}{d\tau}\left(\frac{\partial L}{\partial \dot{x}^j}\right) &= \frac{\partial L}{\partial x^j} \\ \iff \quad \frac{d}{d\tau}(2\dot{x}^j) &= -\frac{\partial}{\partial x^j}\left(1 - \frac{2U}{c^2}\right) c^2 \left(\frac{dt}{d\tau}\right)^2 \\ \iff \quad \frac{d^2 x^j}{d\tau^2} &= \frac{\partial U}{\partial x^j} \left(\frac{dt}{d\tau}\right)^2 \end{aligned}$$

To make this look familiar, we can combine these three equations into

$$\frac{d^2 \vec{r}}{d\tau^2} = (\nabla U) \left(\frac{dt}{d\tau}\right)^2. \tag{18.1}$$

Now if we compare this against the geodesic equation, we see that (recall that $x^0 = ct$)

$$\Gamma^i{}_{tt} = -\frac{\nabla U}{c^2} \quad \text{and} \quad \Gamma^i{}_{jk} = 0.$$

This method for computing Christoffel symbols is often more efficient than the brute-force method. (It's also useful during exams.) Now, it will be convenient to rewrite Eq. (18.1) as

$$\frac{d^2 t}{d\tau^2} \frac{d\vec{r}}{dt} + \left(\frac{dt}{d\tau}\right)^2 \frac{d^2 \vec{r}}{dt^2} = (\nabla U) \left(\frac{dt}{d\tau}\right)^2, \tag{18.2}$$

for reasons that will be clear in a moment. Consider the x^0 equation next:

$$\begin{aligned}
& & \frac{d}{d\tau}\left(\frac{\partial L}{\partial \dot{t}}\right) &= \cancel{\frac{\partial L}{\partial t}} \\
&\iff & \frac{d}{d\tau}\left[-2\left(1-\frac{2U}{c^2}\right)c^2\dot{t}\right] &= 0 \\
&\iff & -2c^2\left(-\frac{2\dot{U}}{c^2}\right)\dot{t} - 2c^2\left(1-\frac{2U}{c^2}\right)\ddot{t} &= 0 \\
&\iff & \left(1-\frac{2U}{c^2}\right)\frac{d^2t}{d\tau^2} &= \frac{2}{c^2}\left((\partial_j U)\frac{dx^j}{d\tau}\right)\frac{dt}{d\tau}
\end{aligned}$$

Rearranging and keeping only terms leading order in $1/c$, we have

$$\frac{d^2t}{d\tau^2} \approx \frac{2}{c^2}\,(\partial_j U)\frac{dx^j}{d\tau}\frac{dt}{d\tau}, \tag{18.3}$$

and as an aside we note that (by comparison with the geodesic equation) this implies

$$\Gamma^0{}_{0j} \approx -\partial_j\left(\frac{2U}{c^2}\right).$$

To put Eq. (18.3) in more Newtonian language, we may rewrite it as

$$\frac{d^2t}{d\tau^2} \approx \frac{2}{c^2}(\vec{v}\cdot\nabla U)\left(\frac{dt}{d\tau}\right)^2.$$

Then upon substitution into Eq. (18.2), the $dt/d\tau$ terms cancel out, and we get

$$\boxed{\vec{a} \approx \nabla U - \frac{2}{c^2}\,\vec{v}(\vec{v}\cdot\nabla U).} \tag{18.4}$$

This is the geodesic equation in 'warped time' spacetime. The only difference from the Newtonian equation of motion is the second term, which is $\mathcal{O}(c^{-2})$. In particular, in the $c\to\infty$ limit we get

$$\vec{a} \approx \nabla U.$$

As another aside, the second term in Eq. (18.4) is called a 1 PN correction, with 'PN' short for 'post-Newtonian'. More generally, terms that are $\mathcal{O}(v^{2n}/c^{2n})$ are called n PN. This series converges really slowly; it's not impossible to see a paper with a fifty-page-long equation, listing out correction terms up to some PN order.

Note the change in perspective. What was 'accelerated motion' in Newtonian is 'unaccelerated motion' in GR, in the sense that it comes from the geodesic equation.

19 The geodesic deviation equation; Ricci and Weyl tensors

We know that we can make $g \to \eta$ and $\partial g \to 0 \implies \Gamma \to 0$ at a point, but is this the whole story? The important point is that this principle is *local*. The one exception is where $\nabla\Phi = \text{const}$; a uniform gravitational field is globally indistinguishable from a fictitious force. But this is a special case. What happens in general?

Consider a one-parameter family of geodesics $\gamma(s, \lambda)$, where s is a parameter along the geodesics, and λ serves as a label.

(figure here)

Adopting s and λ as coordinates, we obtain the vector fields $u = \partial/\partial s$ and $z = \partial/\partial\lambda$. Then a bit of work is needed to show that we may choose our parameter s on each curve to get $g(u, z) = 0$ everywhere, but we will not do this here.[33] Now since u is the tangent to a geodesic, it satisfies

$$\nabla_u u = 0,$$

whereas z is to be interpreted as a geodesic deviation vector, representing the displacement to an infinitesimally nearby geodesic. Coordinates can make this clearer:

$$\gamma^\alpha(s, \lambda) = \gamma^\alpha(s, \lambda_0) + z^\alpha(\lambda - \lambda_0) + \mathcal{O}\big[(\lambda - \lambda_0)^2\big]$$

The question now is, what are the dynamics of z? For example, do geodesics move closer to each other? To answer this, we want to know

$$\frac{D^2 z}{ds^2} = \,?$$

To build some intuition, in flat spacetime all geodesics are straight lines, so their separation may change with s, but this change must be linear. Thus, flat spacetime necessitates a zero right-hand side. Equivalently, a nonzero right-hand side reveals the presence of curvature.

First we establish a general identity. By working in a coordinate basis, we have

$$\begin{aligned}
[v, w]^a &= v^b \partial_b w^a - w^b \partial_b v^a \\
&= v^b \partial_b w^a + v^b \Gamma^a_{cb} w^c - \big(w^b \partial_b v^a + w^b \Gamma^a_{cb} v^c\big) \\
&= v^b \nabla_b w^a - w^b \nabla_b v^a,
\end{aligned}$$

where we've used the fact that $\Gamma^a_{cb} = \Gamma^a_{bc}$.[34] Now since u and z are coordinate basis vectors, their

[33] See, for instance, page 46 of Wald.

[34] Alternatively, let v^a and w^b be vector fields, let $f \in \mathscr{F}$, and let ∇_a be a derivative operator, so that

$$\begin{aligned}
[v, w](f) &= v\{w(f)\} - w\{v(f)\} \\
&= v^a \nabla_a (w^b \nabla_b f) - w^a \nabla_a (v^b \nabla_b f) \\
&= v^a w^b \nabla_a \nabla_b f + v^a \nabla_b f \nabla_a w^b - w^a v^b \nabla_a \nabla_b f - w^a \nabla_a v^b \nabla_b f \\
&= \{v^a \nabla_a w^b - w^a \nabla_a v^b\} \nabla_b f,
\end{aligned}$$

where (among other things) we've used the torsion-free property of derivative operators. Since f was arbitrary, the desired identity follows.

commutator vanishes, so it follows from the identity that $u^b\nabla_b z^a = z^b\nabla_b u^a$. Then

$$\left(\frac{Dz}{ds}\right)^a \equiv u^b\nabla_b z^a = z^b\nabla_b u^a,$$

$$\begin{aligned}\left(\frac{D^2z}{ds^2}\right)^a \equiv u^c\nabla_c(u^b\nabla_b z^a) &= u^c\nabla_c(z^b\nabla_b u^a)\\ &= (u^c\nabla_c z^b)\nabla_b u^a + z^b u^c\nabla_c\nabla_b u^a\\ &= (z^c\nabla_c u^b)\nabla_b u^a + z^b u^c(\nabla_b\nabla_c u^a + R^a{}_{dcb}u^d)\\ &= z^c\nabla_c\cancel{(u^b\nabla_b u^a)} + R^a{}_{dcb}u^c z^b u^d,\end{aligned}$$

where we use the Ricci identity in the third line. To summarize, then, we have

$$\boxed{\begin{aligned}\frac{D^2z^a}{ds^2} &\equiv u^c\nabla_c(u^b\nabla_b z^a)\\ &= R^a{}_{dcb}u^c z^b u^d.\end{aligned}} \qquad \text{(geodesic deviation eq.)}$$

In mathy circles, this is known as the *Jacobi equation.* This tells us that curvature is what drives the growth of deviation! Therefore, one can detect gravity by examining the behavior of nearby geodesics. Or, put another way, the Riemann curvature measures the relative acceleration of nearby geodesics.

Now, in comoving coordinates, where

$$u^\alpha \stackrel{*}{=} (1,0,0,0),$$

this expression simplifies to

$$\frac{D^2z^i}{ds^2} = -R^i{}_{0j0}z^j. \tag{19.1}$$

Now consider Newtonian physics, with two nearby freely falling objects $\overline{x}^i(t)$ and $x^i(t)$, and let

$$\overline{x}^i(t) = x^i(t) + z^i(t).$$

Then

$$\frac{d^2x^i}{dt^2} = \delta^{ij}\partial_j U(x),$$

$$\frac{d^2\overline{x}^i}{dt^2} = \delta^{ij}\partial_j U(\overline{x}),$$

and by our definition, the second equation becomes (using a Taylor expansion around x)

$$\frac{d^2x^i}{dt^2} + \frac{d^2z^i}{dt^2} = \delta^{ij}\partial_j\Big[U(x) + z^k\partial_k U + \cdots\Big].$$

Subtracting yields

$$\begin{aligned}\frac{d^2z^i}{dt^2} &= \delta^{ij}(\partial_j\partial_k U)z^k\\ &= (\partial^i\partial_j U)z^j.\end{aligned}$$

Compare to Eq. (19.1); this new equation also expresses relative acceleration between nearby free-falling bodies. This motivates the idea of the so-called *Newtonian tidal tensor*,

$$\boxed{S^i{}_j = \delta^{ik}\partial_k\partial_j U.}$$

Squinting a bit, one can generalize this (in the sense that we involve all spacetime indices) to the *tidal stress tensor*, (check ordering of the indices)

$$\boxed{S^\mu{}_\gamma = R^\mu{}_{\beta\gamma\delta}u^\beta u^\delta.}$$

Earlier, we discussed how the Riemann tensor is the part of the metric that cannot be set to zero. Therefore, this equation tells us that the best you can do is remove gravity at a point; you cannot eliminate its tidal effects. If we identify $R^i{}_{0j0}$ with $S^i{}_j$, we get (recalling that $U = -\Phi$)

$$R^i{}_{0j0} = \partial^i\partial_j\Phi.$$

As a final comment, one can think of the tidal stress tensor as a projection of the Riemann tensor onto the two legs u^a and u^b.

An index-free derivation

We can give another derivation of the Jacobi equation without using indices. Recall that

$$R(X,Y)Z := [\nabla_X, \nabla_Y]Z - \nabla_{[X,Y]}Z,$$

and that the torsion is defined by

$$T(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y].$$

Now, we chose u and z such that $[u,z] = 0$ and $\nabla_u u = 0$. So since our connection is torsion-free (i.e. $T = 0$), we have $\nabla_u z = \nabla_z u$. Thus

$$\begin{aligned} R(u,z)u &= \nabla_u\nabla_z u - \nabla_z\cancel{\nabla_u u} - \nabla_{\cancel{[u,z]}}u \\ &= \nabla_u\nabla_u z, \end{aligned}$$

which establishes that

$$\boxed{\nabla_u\nabla_u z = R(u,z)u.} \qquad \text{(Jacobi equation)}$$

It's worth noting that none of the subtleties regarding the parametrization of our geodesics entered into this derivation. Presumably, this is only relevant when it comes to interpreting z as a geodesic deviation vector.

A first attempt at relating matter and curvature

Recall that (we're identifying the Newtonian tidal tensor with the tidal stress tensor)

$$R^i{}_{ajb}u^a u^b = \partial^i\partial_j\Phi,$$

and we will promote i and j to spacetime indices:

$$R^c{}_{adb}u^a u^b = \partial^i\partial_j\Phi.$$

Then we contract to get

$$\begin{aligned} R^c{}_{acb} u^a u^b &= \partial^c \partial_c \Phi \\ &= -\frac{1}{c^2}\partial_t^2 \Phi + \nabla^2 \Phi. \end{aligned}$$

In the $c \to \infty$ limit, then (with units $G_N = 1$)

$$R^c{}_{acb} u^a u^b = \nabla^2 \Phi = 4\pi\rho.$$

This is a scalar equation, but can we promote it to a tensorial one? We need to find a tensor whose 00-th component is ρ. (Remember that u^a gives our time direction in Newtonian physics.) And this exists – we call this T_{ab}, the stress-energy tensor. So the equation above reads

$$R^c{}_{acb} u^a u^b = 4\pi T_{ab} u^a u^b.$$

We might then guess that the more general relation

$$R_{ab} = 4\pi T_{ab}$$

holds. This will turn out to be incorrect, but it is a good first candidate for relating matter content and curvature.

Ricci and Weyl tensors

Recall that the Ricci tensor was defined to be the contraction of the first and third indices of the Riemann tensor,

$$\boxed{R_{ab} := R^c{}_{acb} = g^{cd} R_{cadb}.} \qquad \text{(Ricci tensor)}$$

You should show that the symmetries of the Riemann tensor imply this is the only (independent) trace you can take. If you then take the trace of the Ricci tensor, you get the *scalar curvature* or *Ricci tensor*, defined by

$$\boxed{R := g^{ab} R_{ab}.} \qquad \text{(Ricci scalar)}$$

Finally, there is the *Weyl tensor*, which is the first term in the decomposition

$$\boxed{R_{abcd} = C_{abcd} + \frac{2}{D-2}\Big(g_{a[c}R_{d]b} - g_{b[c}R_{d]a}\Big) - \frac{2}{(D-1)(D-2)} R g_{a[c} g_{d]b},}$$

where $D \geq 3$ is the dimension of the manifold. This tensor has the property

$$g^{bd} C_{abcd} = g^{ac} C_{abcd} = 0,$$

and one can show that it shares the algebraic symmetries of the Riemann tensor. Together with the property above, this implies that C_{abcd} is trace-free on all pairs of its indices; therefore, you can think of it as the trace-free part of the Riemann tensor.

The Weyl tensor is very conceptually important; it says that even if $R_{ab} = 0$, one can still have curvature. So this is not just a mathematical nicety! For example, gravitational waves are nothing but a nonzero Weyl tensor.

20 The Einstein equation and a crash course in classical field theory

In the previous lecture, we encountered a first candidate for relating matter and curvature:

$$R_{ab} = \kappa T_{ab}, \tag{20.1}$$

where κ is some constant. To understand why we might reject Eq. (20.1), at least on theoretical grounds, we need to acquaint ourselves with the *principle of minimal coupling*. If you have a four-vector equation in SR, you make the replacements $\eta \to g$, $\partial \to \nabla$ to get the corresponding equation in GR. So if we argue for conservation of the stress-energy tensor in flat space,[35] then

$$\partial^\alpha T_{\alpha\beta} = 0 \implies \nabla^a T_{ab} = 0.$$

The converse is true, given the right basis – the statement $\nabla_a T^{ab} = 0$ can be written

$$\partial_\alpha T^{\alpha\beta} + \Gamma^\alpha_{\alpha\mu} T^{\mu\beta} + \Gamma^\beta_{\alpha\mu} T^{\alpha\mu} = 0,$$

and we can make the Christoffel symbols vanish by working in an inertial frame.

A heuristic derivation of the Einstein equation

It turns out that this divergence-free condition on T_{ab} is inconsistent with Eq. (20.1), because $\nabla^a R_{ab} \neq 0$. Instead of adjusting the right-hand side, we will (following Einstein) assume that the stress-energy tensor is proportional to traces of the Riemann tensor, and then we'll look for a combination that's divergence-free. So let's assemble a symmetric $(0,2)$ tensor E_{ab} of this form. In general, we can write

$$E_{ab} = \alpha R_{ab} + \beta g_{ab} R + \text{'derivatives of traces'}.$$

We'll ignore any terms after the first two, although these do show up in alternative theories of gravity. To justify this, note that any subsequent terms are multiplied to constants with dimensions of length... but the fundamental length scale involved is

$$\ell_p^2 = \frac{\hbar G}{c^2} \sim 10^{-35}\,\mathrm{m},$$

which is very small, so the higher-order terms will be practically negligible from an observational standpoint. In this class, then, we'll ignore these terms, but don't take this to mean that they are irrelevant! This leaves us with a tensor

$$E_{ab} = \alpha R_{ab} + \beta g_{ab} R, \tag{20.2}$$

together with the condition that this be divergence-free ($\nabla^a E_{ab} = 0$). Equivalently,

$$\alpha \nabla^a R_{ab} + \beta \nabla_b R = 0.$$

To determine what α and β should be, we consider the so-called second Bianchi identity. This is slightly different in character from the first; it is a differential identity, not an algebraic one. First we collect a couple of identities.

$$(\nabla_a \nabla_b - \nabla_b \nabla_a)\nabla_c \omega_d = R_{abc}{}^e \nabla_e \omega_d + R_{abd}{}^e \nabla_c \omega_e, \tag{20.3}$$

$$\nabla_a(\nabla_b \nabla_c \omega_d - \nabla_c \nabla_b \omega_d) = (\nabla_a R_{bcd}{}^e)\omega_e + R_{bcd}{}^e \nabla_a \omega_e. \tag{20.4}$$

Informally, the first identity follows from treating each index as though it were a one-form index and applying the identity for the Riemann tensor and one-forms.

[35] In lecture 22, we'll offer some justification for why we expect $\partial^\alpha T_{\alpha\beta} = 0$.

Proposition (Second Bianchi identity)**.**

$$\nabla_{[a} R_{bc]de} = 0.$$

Proof. Antisymmetrizing over $\{a, b, c\}$ on the left-hand sides of Eqs. (20.3) and (20.4) shows that the two are equal, hence the antisymmetrized right-hand sides are equal. That is,[36]

$$R_{[abc]}{}^{e} \nabla_e \omega_d + R_{[ab|d|}{}^{e} \nabla_{c]} \omega_e = \omega_e \nabla_{[a} R_{bc]d}{}^{e} + R_{[bc|d|}{}^{e} \nabla_{a]} \omega_e .$$

The first term on the left vanishes due to the symmetries of the Riemann tensor, while the second terms on both sides cancel out, since one is obtained from the other by exchanging two pairs of indices. Thus

$$\nabla_{[a} R_{bc]d}{}^{e} = 0,$$

and the result follows from lowering the last index. □

Using the algebraic symmetries of the Riemann tensor, the second Bianchi identity becomes

$$\nabla_a R_{bcde} + \nabla_b R_{cade} + \nabla_c R_{abde} = 0,$$

and then we act g^{bd} to get (by metric compatibility and the definition of the Ricci tensor)

$$\begin{aligned} & \nabla_a R_{ce} + \nabla_b R_{ca}{}^{b}{}_{e} + \nabla_c R_{a}{}^{d}{}_{de} = 0 \\ \Longrightarrow \qquad & \nabla_a R_{ce} + \nabla_b R^{b}{}_{eca} - \nabla_c R_{ae} = 0, \end{aligned}$$

and finally act g^{ae} to get

$$\begin{aligned} & \nabla_a R^{a}{}_{c} + \nabla_b R^{b}{}_{c} - \nabla_c R = 0 \\ \Longrightarrow \qquad & 2\nabla_a \Big(R^{a}{}_{c} - \frac{1}{2} \delta^{a}{}_{c} R \Big) = 0 \\ \Longrightarrow \qquad & \nabla^a \Big(R_{ac} - \frac{1}{2} g_{ac} R \Big) = 0. \end{aligned}$$

Evidently, the combination in parentheses is a tensor satisfying the conditions from Eq. (20.2). This object is of great historical importance; it is the *Einstein tensor*, written G_{ab}. One can think of this as the part of the Ricci tensor that is divergence-free.

Now, notice that we can always write

$$\begin{aligned} E_{ab} = \alpha R_{ab} + \beta g_{ab} R &= \alpha \Big(R_{ab} - \frac{1}{2} g_{ab} R \Big) + \Big(\beta + \frac{1}{2} \alpha \Big) g_{ab} R \\ &= \alpha G_{ab} + \gamma g_{ab} R. \end{aligned}$$

Using what we just found about G_{ab}, the condition $\nabla^a E_{ab} = 0$ implies that $\gamma \nabla_b R = 0$, thus either $\gamma = 0$ or $\nabla_b R = 0$. Would the second conclusion be justified? To determine this, recall that our proposed field equation was

$$E_{ab} = \kappa T_{ab}.$$

Taking the trace of both sides gives

$$\alpha \Big(R - \frac{1}{2} (4) R \Big) + \gamma (4) = \kappa T,$$

[36] Vertical bars around a free index indicate that it should be excluded from symmetrization or antisymmetrization. We don't bother with this notation for bound indices.

thus $R \propto T$, and then $\nabla_b R = 0 \implies \nabla_b T = 0$, which turns out to be unreasonable. (This is okay in E&M, but not in GR.) (explain) So we must take $\gamma = 0$, and we obtain

$$\boxed{G_{ab} = \kappa T_{ab},}$$

where the proportionality constant κ needs to be fixed by experiment. It also follows that

$$\nabla^a T_{ab} = 0.$$

We can already see how things couple here – the first equation can be thought of as matter telling spacetime how to curve, while the second is spacetime telling matter how to move. (Remember that the covariant derivative involves Christoffel symbols, which in turn depend on the metric.)

Since we are practically done with understanding the left-hand side of Einstein's equation, let us now look at the right-hand side. You can think of T_{ab} as what "sources" the curavture – this is where the physics comes in.

A crash course in classical field theory

The canonical example of a classical field theory is electromagnetism, which is described by Maxwell's equations; we discuss these here.

Some preliminaries

(A better way of understanding integration on manifolds is to use the language of differential forms, but we don't have the time for this. Instead we will use the following approach – not as satisfying, but it works.) First we observe that $\sqrt{-g}\, d^4x$ is an invariant volume element. To see this, recall that

$$g'_{\alpha\beta} = \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial x^\nu}{\partial x'^\beta}\, g_{\mu\nu},$$

hence $g' = J^2 g$, with J the Jacobian of the transformation and g the determinant of the metric. This implies that

$$J = \sqrt{\frac{g'}{g}}.$$

But under the same coordinate transformation we have

$$\begin{aligned} d^4x &= J\, d^4x' \\ &= \sqrt{\frac{g'}{g}}\, d^4x', \end{aligned}$$

so we get $\sqrt{-g}\, d^4x = \sqrt{-g'}\, d^4x'$, as was to be shown. The other thing we'll need here is a covariant Dirac delta function. First recall that

$$\int d^4x\, \underbrace{\delta(x^0 - x'^0)\delta(x^1 - x'^1)\cdots}_{\equiv \delta^{(4)}(x^\mu - x'^\mu)} = 1,$$

but this is not invariant, due to the d^4x. We therefore define

$$\delta_4(x, x') = \frac{\delta^{(4)}(x^\mu - x'^\mu)}{\sqrt{-g}},$$

so that (note the invariant volume element)

$$\int \sqrt{-g}\, d^4x\, \delta_4(x,x') = 1.$$

Then we have

$$\int f(x)\delta_4(x,x')\sqrt{-g}\, d^4x = f(x')$$

and similarly

$$\int f(x')\delta_4(x,x')\sqrt{-g'}\, d^4x' = f(x),$$

as we expect from a delta function.

Maxwell's equations

We now want to recast Maxwell's equations in a way that is manifestly covariant.[37] We start with the Faraday 2-form

$$F = F_{\mu\nu}\, dx^\mu \wedge dx^\nu.$$

E&M is actually embodied by this tensor, and this is how you make the theory appear manifestly special-relativistic. The components are[38]

$$\begin{aligned} F^{ij} &= F_{ij} = \epsilon^{ijk} B^k, \\ F^{0i} &= -F^{i0} = E^i/c, \end{aligned}$$

and you can write this all out in a 4×4 matrix, if you like. But notice that electric and magnetic fields are not physical, in the sense that they depend on the observer. What is really physical is F, and electric and magnetic fields are just components.

We also have the Lorentz force law,

$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}u_\nu.$$

For structureless, spinless, massive particles, we have $p^\mu = mu^\mu$, and then we get

$$\frac{du^\mu}{d\tau} = \frac{q}{m}F^{\mu\nu}u_\nu$$

and also

$$\begin{aligned} p^\mu = (p^0, p^i) &= (E/c, p^i) \\ &= m\gamma(c, v^i). \end{aligned}$$

Then

$$E = m\gamma c^2 \approx mc^2 + \frac{1}{2}mv^2 + \mathcal{O}(v^4).$$

Now let's examine the Lorentz force law equations corresponding to the spatial parts of p^μ.

$$\frac{dp^i}{d\tau} = q\Big(u_0 F^{i0} + u_j F^{ij}\Big).$$

[37] Oftentimes, this part doesn't make it into the the 131/132 sequence. We didn't when I took it!

[38] Raising or lowering indices on the spatial components is free, since our metric is $\eta_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$.

To express this in Newtonian language, we bring out a $dt/d\tau$.

$$\begin{aligned}\frac{dt}{d\tau}\frac{dp^i}{dt} &= q\Big(-u_0 E^i/c + u_j\epsilon^{ijk}B^k\Big)\\ &= q\Big(u^0 E^i/c + \epsilon^{ijk}u^j B^k\Big)\\ &= q\Big((\gamma c)E^i/c + \epsilon^{ijk}\gamma v^j B^k\Big).\end{aligned}$$

Then we cancel out the common γ to get

$$\frac{dp^i}{dt} = q(E^i + \epsilon^{ijk}v^j B^k),$$

or just

$$\boxed{\frac{d\vec{p}}{dt} = q\big(\vec{E} + \vec{v}\times\vec{B}\big).}$$

A caveat – this is not quite the Lorentz force law you're used to, since $p^i = m\gamma v^i \neq p^i_{(3)}$. But in the low-velocity limit it is, and either way, this is exact if you understand $\vec{p}$ to be the spatial components of the four-momentum. Meanwhile, the zeroth component gives

$$\begin{aligned}\frac{dp^0}{d\tau} &= qF^{0i}u_i\\ \iff \quad \gamma\frac{d}{d\tau}\frac{E}{c} &= q\left(\frac{E^i}{c}\right)(\gamma v^i),\end{aligned}$$

which is

$$\boxed{\frac{dE}{dt} = q\vec{E}\cdot\vec{v}.}$$

This equation says that the work done on a charged particle is related only to the electric field – magnetic fields do no work!

21 Maxwell's equations in covariant form; the Maxwell SET

We continue our discussion on how Maxwell's equations can be expressed in covariant (that is, manifestly special-relativistic) form. We had the Lorentz equations of motion,

$$\frac{dp^\mu}{d\tau} = qF^{\mu\nu}_{\text{ext}} u_\nu.$$

(The 'ext' subscript is a subtlety that we glossed over earlier.) Now, the covariant form of Maxwell's equations is

$$\boxed{\begin{aligned} &\partial_\nu F^{\mu\nu} = \mu_0 J^\mu, && \text{(inhomogeneous)} \\ &\partial_{[\alpha} F_{\mu\nu]} = 0. && \text{(homogeneous)} \end{aligned}}$$

However, you're probably more familiar with

$$\begin{aligned} \nabla \cdot \vec{E} &= \rho/\varepsilon_0, \\ \nabla \times \vec{B} - \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} &= \mu_0 \vec{J}, \\ \nabla \cdot \vec{B} &= 0, \\ \nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} &= 0. \end{aligned}$$

In a moment, we'll see how these fall out from the four-vector equations. First we note that J^μ, which is known as the *four-current*, is defined by

$$J^\mu = (c\rho, J^i),$$

where the three-vector $J^i = \rho v^i$ is what you're used to calling the current. Next we note that the Maxwell equations must satisfy an integrability condition to be solvable:

$$\begin{aligned} && \partial_\mu \partial_\gamma F^{\mu\gamma} &= \mu_0 \partial_\mu J^\mu \\ \Longrightarrow && 0 &= \partial_\mu J^\mu. \end{aligned}$$

That is, the four-current must be divergence-free. This condition has a physical interpretation – we may expand this out as

$$\begin{aligned} && \frac{1}{c}\frac{\partial}{\partial t}(c\rho) + \nabla \cdot \vec{J} &= 0 \\ \Longleftrightarrow && \frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} &= 0 \\ \Longleftrightarrow && \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) &= 0. \end{aligned}$$

This is just the continuity equation! This is also a conservation law. To see this, we can integrate over a $t =$ const. surface of Minkowski and apply Stokes' theorem to get

$$\begin{aligned} && \int_\Sigma d^3x\, \partial_t \rho &= -\int_\Sigma \nabla \cdot (\rho \vec{v})\, d^3x \\ \Longrightarrow && \frac{d}{dt}\int_\Sigma d^3x\, \rho &= -\oint_{\partial\Sigma} \vec{J} \cdot d\vec{A} \\ \Longrightarrow && \frac{dQ}{dt} &= -\oint_{\partial\Sigma} \vec{J} \cdot d\vec{A}. \end{aligned}$$

We can imagine pushing the boundary $\partial\Sigma$ out to infinity; for any physical configuration, the integrand will be zero there. Therefore, the right-hand side will vanish, and then $Q = \text{const}$. Now let's look at the individual components of the inhomogeneous Maxwell equations. For the zeroth component, we have

$$\partial_\nu F^{0\nu} = \mu_0 J^0 = \mu_0 \rho c,$$

and since $F^{\mu\nu}$ is antisymmetric, we have $F^{00} = 0$, and this becomes

$$\begin{aligned}
& & \partial_i F^{0i} &= \mu_0 \rho c \\
&\iff & \partial_i \frac{E^i}{c} &= \mu_0 \rho c \\
&\iff & \partial_i E^i &= \mu_0 c^2 \rho \\
&\iff & \nabla \cdot \vec{E} &= \rho/\varepsilon_0,
\end{aligned}$$

so we have recovered Gauss' law. Meanwhile, the spatial components ($\mu = i$) give

$$\begin{aligned}
& & \partial_\nu F^{i\nu} &= \mu_0 J^i \\
&\iff & \partial_0 F^{i0} + \partial_j F^{ij} &= \mu_0 J^i \\
&\iff & -\frac{1}{c}\frac{\partial}{\partial t} F^{0i} + \partial_j F^{ij} &= \mu_0 J^i \\
&\iff & -\frac{1}{c^2}\,\partial_t E^i + \epsilon^{ijk}\partial_j B^k &= \mu_0 J^i \\
&\iff & -\frac{1}{c^2}\frac{\partial \vec{E}}{\partial t} + \nabla \times \vec{B} &= \mu_0 \vec{J},
\end{aligned}$$

which is Ampere's law. It's left to you to show that the homogeneous equations give the remaining Maxwell equations.

⊛ The homogeneous equations

(If you haven't done the exercise above, then please do – it was a problem set question! This is just here for completeness.) Due to the antisymmetry of the Faraday tensor, the homogeneous equations are equivalent to

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0.$$

If we take $(\alpha, \beta, \gamma) = (1, 2, 3)$, then this becomes

$$\begin{aligned}
& & \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} &= 0 \\
&\iff & \frac{\partial B^x}{\partial x} + \frac{\partial B^y}{\partial y} + \frac{\partial B^z}{\partial z} &= 0,
\end{aligned}$$

which in vector notation is $\nabla \cdot \vec{B} = 0$. Now take one index to be zero, and take the remaining two to be spatial indices to get

$$\begin{aligned}
& & \partial_0 F_{ij} + \partial_i F_{j0} + \partial_j F_{0i} &= 0 \\
&\iff & \partial_0(\epsilon^{ijk} B^k) + \partial_i(E^j/c) - \partial_j(E^i/c) &= 0 \\
&\iff & \epsilon^{ijk}\partial_t B^k + \partial_i E^j - \partial_j E^i &= 0.
\end{aligned}$$

Then act $\epsilon_{ij\ell}$ on this equation to get

$$\begin{aligned} & 2\delta^k_\ell \partial_t B^k + \epsilon_{ij\ell}\big(\partial_i E^j - \partial_j E^i\big) = 0 \\ \iff \qquad & \partial_t B^\ell + \epsilon_{ij\ell}\partial_i E^j = 0, \end{aligned}$$

where we've used $\epsilon_{ij\ell}\epsilon^{ijk} = 2\delta^k_l$ and the fact that $\epsilon_{ij\ell}$ is totally antisymmetric. This last line is is equivalent to Faraday's law,

$$\frac{\partial \vec{B}}{\partial t} + \nabla \times \vec{E} = 0.$$

Maxwell equations as a conservation law

We will manipulate Maxwell's equations to make something called the *Maxwell stress-energy tensor* appear. This is similar to what's done in Griffiths, but there he distinguishes between contributions from the Poynting vector and from the 'stress tensor'. Here we deal with all four spacetime indices. We start with the inhomogeneous equations:

$$\begin{aligned} & \partial_\nu F^{\mu\nu} = \mu_0 J^\mu \\ \implies \qquad & F_{\mu\lambda}\partial_v F^{\mu\nu} = \mu_0 F_{\mu\lambda} J^\mu \\ \implies \qquad & \partial_\nu(F_{\mu\lambda}F^{\mu\nu}) - F^{\mu\nu}\partial_\nu F_{\mu\lambda} = \mu_0 F_{\mu\lambda} J^\mu \end{aligned}$$

Now we use the homogeneous equations to simplify the second term on the left. Recalling how antisymmetrization brackets work, we get

$$\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} - \partial_\mu F_{\lambda\nu} - \partial_\nu F_{\mu\lambda} - \partial_\lambda F_{\nu\mu} = 0.$$

(In general, you bring in a combinatorial factor when you antisymmetrize, but we've ignored it here because the right-hand side is zero.) Then since F is antisymmetric, you can swap all the indices on the last three terms to get

$$\partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} = 0.$$

We mention this here because the same reasoning can be applied to the first Bianchi identity. Anyway, this implies that

$$\begin{aligned} F^{\mu\nu}\partial_\nu F_{\mu\nu} &= -F^{\mu\nu}\partial_\mu F_{\nu\lambda} - F^{\mu\nu}\partial_\nu F_{\lambda\mu} \\ &= -F^{\nu\mu}\partial_\nu F_{\mu\lambda} + F^{\mu\nu}\partial_\nu F_{\mu\lambda} \\ &= 2F^{\mu\nu}\partial\nu F_{\mu\lambda}, \end{aligned}$$

and this last term is what we had in our equation, so we substitute to get

$$\begin{aligned} \mu_0 F_{\mu\lambda} J^\mu &= \partial_\nu(F_{\mu\lambda}F^{\mu\nu}) - \frac{1}{2}F^{\mu\nu}\partial_\lambda F_{\mu\nu} \\ &= \partial_\nu(F_{\mu\lambda}F^{\mu\nu}) - \frac{1}{4}\partial_\lambda(F^{\mu\nu}F_{\mu\nu}) \\ &= \partial_\nu\left[F^{\mu\nu}F_{\mu\lambda} - \frac{1}{4}\delta^\nu{}_\lambda F^{\alpha\beta}F_{\alpha\beta}\right]. \end{aligned}$$

We define the term in brackets to be the *electromagnetic stress-energy tensor*,

$$\boxed{T^{\mu\nu}_{(\text{em})} = \frac{1}{\mu_0}\Big(F^{\lambda\mu}F_\lambda{}^\nu - \frac{1}{4}\delta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}\Big).}$$

(If you look closely, you'll notice that we've changed some labels and raised an index. To make things clearer, the tensor we started with is written $T^{\nu}{}_{\lambda}$.) It will be left for you to show that

$$T^{00}_{(\text{em})} \propto \frac{1}{2}\left(\varepsilon_0|\vec{E}|^2 + \frac{1}{\mu_0}|\vec{B}|^2\right),$$

and you're also invited to look at the remaining components to see what you get. At this point we're told that the whole point of electromagnetism is to build intuition for fields[39] – in particular, E&M is not all boundary value problems.

Notice that we can now write

$$\boxed{\partial_\nu T^{\mu\nu}_{(\text{em})} = F_{\alpha}{}^{\mu} J^{\alpha}.}$$

If there were no J^α (our source term) and we had only the Faraday tensor, then what this tells us is that there are 4 things conserved. Otherwise, if there is charge to interact with, then this isn't conserved. (Put another way, the energy in an electromagnetic field is not conserved!)

Interpretation of the components of a SET

(look at a fluid dynamics textbook)

A distributional identity

Does non-conservation of $T_{(\text{em})}$ spell non-conservation of energy in general? The answer is no – we need to account for the mechanical energy of the particle itself. We will use the following distributional identities later: recall that

$$\begin{aligned}
\int d\tau\, \delta_4\big(x, z(\tau)\big) &= \int d\tau\, \frac{\delta^{(4)}\big(x^\mu - z^\mu(\tau)\big)}{\sqrt{-g}} \\
&= \int \frac{d\tau}{\sqrt{-g}} \delta\big(x^0 - z^0(\tau)\big)\delta^{(3)}\big(x^i - z^i(\tau)\big),
\end{aligned}$$

then perform a change of variables,

$$\begin{aligned}
&= \int \frac{1}{\sqrt{-g}} dx^0 \left(\frac{d\tau}{dx^0}\right)\delta\big(x^0 - z^0(\tau)\big)\delta^{(3)}\big(x^i - z^i(\tau)\big) \\
&= \frac{1}{\sqrt{-g}}\frac{1}{c}\frac{d\tau}{dt}\delta^{(3)}\big(x^i - z^i(t)\big)\Big|_{x^0 = ct = z^0(t)}.
\end{aligned}$$

So this is our first distributional identity:

$$\boxed{\int d\tau\, \delta_4\big(x, z(\tau)\big) = \frac{1}{\sqrt{-g}}\frac{1}{u^0}\delta^{(3)}\big(x^i - z^i(t)\big)\Big|_{x^0 = ct = z^0(\tau)}}$$

[39] Of course, there is a theorist's bias here, but you don't have to take Dr. Vega's word for it! You can see this in the first few chapters of Traveling at the Speed of Thought, which discuss how early practitioners made extensive use of analogies with electromagnetism to investigate the existence of gravitational radiation.

22 Point particles in field theory; conservation of SET; perfect fluids

There was not much GR in the previous lecture – we were sorting out what conservation of the stress-energy tensor had to do with the equations of motion. We showed that Maxwell's equations implied

$$\partial_\nu T^{\mu\nu}_{(\mathrm{em})} = -F^{\mu\nu} J_\nu,$$

and that the absence of a four-current implies conservation of the stress-energy tensor. Now, it's important for spacetime modeling to have an interpretation for the components of T.

(insert intepretations here, or just put these in lec. 21)

SET of a particle

We come back to the distributional identity derived in the previous lecture. Let's get some practice with converting from 4-language to 3-language. Consider a massive particle of rest mass m. Then

$$\boxed{T^{\mu 0} = m u^\mu \delta^{(3)}\big(x^i - z^i(t)\big).}$$

(The Dirac delta is what makes this a density.) Now what about the components $T^{\mu i}$? Consider the following picture.

(figure here)

We want the integral over a (spatial) hypersurface to have the right amount of four-momentum crossing it. Thus

$$\begin{aligned} T^{\mu x} c\Delta t\, \Delta y\, \Delta z &= (m u^\mu) N \\ &= m u^\mu \rho\, \Delta x\, \Delta y\, \Delta z \\ &= m u^\mu \rho v^x\, \Delta t\, \Delta y\, \Delta z. \end{aligned}$$

For a single particle ρ will just be the density from earlier, so we have

$$\boxed{T^{\mu x} = m u^\mu \frac{v^x}{c} \delta^{(3)}\big(x^i - z^i(t)\big).}$$

Then for the components involving y and z, we just let $v^x \to v^y, v^z$. To make these consistent with the general covariance of our spacetime, we convert these Dirac deltas into covariant ones:

$$\begin{aligned} T^{\mu 0}_{(\mathrm{p})} &= m u^\mu \delta^{(3)}\big(x^i - z^i(t)\big) \\ &= m \int d\tau\, u^\mu u^0 \delta_4(x, z) \\ &= m u^\mu \int d\tau\, u^0 \delta_4(x, z). \end{aligned}$$

Here we've used (a variant of) our distributional identity from earlier, and the subscript is meant to indicate that this is the SET of a point particle. Similarly, we get

$$
\begin{aligned}
T^{\mu i}_{(\mathrm{p})} &= mu^\mu \frac{1}{c}\frac{dz^i}{dt}\delta^{(3)}\big(x^i - z^i(t)\big) \\
&= mu^\mu \frac{1}{c}\frac{dz^i}{d\tau}\frac{d\tau}{dt}\delta^{(3)}\big(x^i - z^i(t)\big) \\
&= mu^\mu \frac{dz^i}{d\tau}\left[\frac{1}{u^0}\delta^{(3)}\big(x^i - z^i(t)\big)\right] \\
&= m\int d\tau\, u^\mu u^i \delta_4(x,z).
\end{aligned}
$$

Putting everything together, we have

$$
\boxed{T^{\mu\nu}_{(\mathrm{p})}(x) = \int d\tau\, u^\mu u^\nu \delta_4(x,z).}
$$

Most of the time, you don't have this neat point-particle picture for computing stress-energy. But we're doing this to illustrate an important point, and in the next lecture we'll discuss a systematic way to obtain SETs.

Now recall that $J^\mu = (c\rho, \vec{J}) = (c\rho, \rho\vec{v})$, so for a point particle, $J^0 = cq\delta^{(3)}\big(x^i - z^i(t)\big)$, which can be written

$$
J^0 = cq\int d\tau\, u^0 \delta_4(x,z).
$$

Next we consider the spatial components:

$$
\begin{aligned}
J^i &= qv^i\delta^{(3)}\big(x^i - z^i(t)\big) \\
&= q\frac{dz^i}{d\tau}\frac{d\tau}{dt}\delta^{(3)}\big(x^i - z^i(t)\big) \\
&= q\frac{dz^i}{d\tau}c\left(\frac{1}{c}\frac{d\tau}{dt}\delta^{(3)}(x^i - z^i)\right) \\
&= cq\int d\tau\, u^i \frac{d\tau}{dt}\delta_4(x,z).
\end{aligned}
$$

Putting everything together, we get

$$
\boxed{J^\mu(x) = cq\int d\tau\, u^\mu \delta_4(x,z).}
$$

From here it's worth checking that this satisfies the integrability condition $\partial_\mu J^\mu = 0$. We have

$$
\begin{aligned}
\partial_0 J^0 + \partial_i J^i &= \frac{1}{c}\frac{\partial}{\partial t}(cq\delta^{(3)}) + \frac{\partial}{\partial x^i}(qv^i\delta^{(3)}) \\
&= q\left(-\frac{dz^i}{dt}\right)\partial_i\delta^{(3)} + qv^i\partial_i\delta^{(3)} \\
&= 0,
\end{aligned}
$$

where we applied the chain rule in the second line. This is what we were looking to find, but it's worth noting that – strictly speaking – you cannot take derivatives of distributions. But we are physicists, and the final conclusion is correct, anyway.

Conservation of SET

Going back to our expressions for the components of the particle SET, we find that

$$\begin{aligned}
\partial_\nu T^{\mu\nu}_{(\mathrm{p})} &= \partial_0 T^{\mu 0} + \partial_i T^{\mu i} \\
&= \frac{1}{c}\frac{\partial}{\partial t}(m u^\mu \delta^{(3)}) + \partial_i\big((m u^\mu/c) v^i \delta^{(3)}\big) \\
&= \frac{m}{c}\frac{du^\mu}{dt}\delta^{(3)} - \cancel{\frac{m u^\mu}{c}\frac{dz^i}{dt}\partial_i\delta^{(3)}} + \cancel{\frac{m u^\mu}{c} v^i \partial_i \delta^{(3)}} \\
&= \frac{m}{c}\frac{du^\mu}{dt}\delta^{(3)}.
\end{aligned}$$

Again, we've taken the derivative of a distribution, so if you like, you can think of this as a heuristic derivation. Also, the partial derivative ∂_i can move inside the parentheses because it is with respect to field points, but u^μ is a function of time.

The conclusion is that we do not in general have a conservation law, since

$$\boxed{\partial_\nu T^{\mu\nu}_{(\mathrm{p})} = \int d\tau\, m\frac{du^\mu}{d\tau}\delta_4.}$$

Of course, we do get conservation if this integral is zero. And this comes with a nice physical interpretation – the integrand can be thought of as a four-force density, so that

$$\partial_\nu T^{\mu\nu}_{(\mathrm{p})} = \int d\tau\, f^\mu =: F^\mu.$$

This means that

$$\partial_\nu T^{\mu\nu}_{(\mathrm{p})} = 0 \iff \frac{du^\mu}{d\tau} = 0.$$

Therefore, this stress-energy tensor is divergence-free (gives a conservation law) if and only if the particle moves along a geodesic. Now suppose that the particle is accelerated by an external EM field. Then by the Lorentz force law,

$$\frac{du^\mu}{d\tau} = \frac{q}{m}F^{\mu\nu}u_\nu.$$

Taken together with our previous work, this implies that

$$\begin{aligned}
\partial_\nu T^{\mu\nu}_{(\mathrm{p})} &= \int d\tau\, m\frac{du^\mu}{d\tau}\delta_4 \\
&= \int d\tau\, qF^{\mu\nu}u_\nu\delta_4 \\
&= F^{\mu\nu}\big(z(\tau)\big)\int d\tau\, q u_\nu \delta_4 \\
&= F^{\mu\nu}J_\nu.
\end{aligned}$$

That is,

$$\boxed{\partial_\nu T^{\mu\nu}_{(\mathrm{p})} = F^{\mu\nu}J_\nu.}$$

If we now recall from the previous lecture that

$$\partial_\nu T^{\mu\nu}_{(\mathrm{em})} = -F^{\mu\nu}J_\nu,$$

we see that this implies

$$\boxed{\partial_\nu\left(T^{\mu\nu}_{(\mathrm{em})} + T^{\mu\nu}_{(\mathrm{p})}\right) = 0.}$$

Now, *this* is a conservation law – all the energy and momentum must slosh back and forth between these two tensors. This illustrates the point we were trying to make! Taking everything into account, one finds that $\partial_\nu T^{\mu\nu} = 0$. We discussed all this because physics books tend to demand that $\nabla_\nu T^{\mu\nu} = 0$ without offering any motivation. What we've done is show that this follows from the principle of minimal coupling.[40]

SET of a perfect fluid

A fluid is a regime where we can approximate things by coarse-grained fluid elements. This is an idealization. A *perfect fluid* is isotropic, meaning that if you move to the rest frame of any element, all directions are equivalent. There is also no viscosity and/or shear (meaning that the T^{xy} components, etc. are zero). Finally, there is no heat transfer or friction in a perfect fluid.

Translated into symbols, this means

$$\begin{aligned} T^{0i} &= 0, \\ T^{ij} &\propto \delta^{ij}, \end{aligned}$$

in a locally inertial comoving frame of a given fluid element. In particular, let ρ and P denote the energy density and pressure, respectively, in the fluid rest frame. Then

$$\boxed{\begin{aligned} T^{00} &\overset{*}{=} \rho, \\ T^{ij} &\overset{*}{=} P\delta^{ij}, \end{aligned}}$$

with stars to denote that these are components in the comoving frame. How do we express this in a covariant way? Recall that

$$u^\alpha \overset{*}{=} (1, 0, 0, 0),$$

and notice that

$$\begin{aligned} \eta^{\mu\nu} + u^\mu u^\nu &\overset{*}{=} \mathrm{diag}(0, 1, 1, 1) \\ &= \delta^{ij}, \end{aligned}$$

so that we may write

$$T^{\alpha\beta} = \rho u^\alpha u^\beta + P(\eta^{\alpha\beta} + u^\alpha u^\beta).$$

So this is the four-dimensional generalization of the above two equations. This is often written (equivalently) as

$$\boxed{T^{\alpha\beta} = (\rho + P)u^\alpha u^\beta + P\eta^{\alpha\beta}.} \qquad \text{(perfect fluid SET)}$$

To wrap up, SETs are important because they model matter fields in GR. So if we have ϕ, what's important is finding the corresponding SET, because our governing equation is

$$G_{\alpha\beta} = 8\pi T_{\alpha\beta},$$

[40] One can show that $\partial_\nu T^{\mu\nu} = 0$ in flat space; this is derivable from first principles, but we won't do this here.

with the integrability condition
$$\nabla_\alpha T^{\alpha\beta} = 0.$$
You might ask how to construct a SET in general, because the process we took to get it for a point particle was rather cumbersome. There is a more systematic way based on the Lagrangian formulation, and this will be the topic of next lecture.

23 SETs via the Lagrangian formulation

Our "derivation" of the Einstein equation in lecture 20 was very heuristic, relying on comparisons with Newtonian gravity. But this does not reflect how modern physical theories are constructed. Instead, an *action principle* is what defines a theory, and then your equations of motion are derived via the principle of stationary action.

A priori, the variational principle is not more correct than other approaches, but it turns out to work, and yields laws that agree with experiment.

Let us begin with the Euler-Lagrange equations in flat spacetime. Our action will be

$$S = \int d^4x\, \mathcal{L}(\phi, \partial_\mu \phi),$$

where we could consider Lagrangian densities with different functional dependence, but we won't do so here. Then

$$\begin{aligned} S = \int dt\, L &= \int dt \int d^3x\, \mathcal{L} \\ &= \int d^4x\, \mathcal{L}. \end{aligned}$$

Then we take the variation of this,

$$\begin{aligned} \delta S &= S[\phi + \delta\phi] - S[\phi] \\ &= \int d^4x \left(\frac{\delta S}{\delta \phi}\right) \delta\phi. \end{aligned} \tag{23.1}$$

Note that ϕ pertains to an arbitrary field; here it is a scalar field, but we could just as well have ϕ^ν, for example. Now, we have

$$S[\phi + \delta\phi] - S[\phi] = \int d^4x\, \mathcal{L}\big(\phi + \delta\phi, \partial_\mu\phi + \delta(\partial_\mu\phi)\big) - \int d^4x\, \mathcal{L}(\phi, \partial_\mu\phi),$$

and since

$$\mathcal{L}\big(\phi + \delta\phi, \partial_\mu\phi + \delta(\partial_\mu\phi)\big) = \mathcal{L}(\phi, \partial_\mu\phi) + \frac{\partial \mathcal{L}}{\partial \phi}\,\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\delta(\partial_\mu\phi) + \mathcal{O}\big[(\delta\phi)^2, \big(\delta(\partial_\mu\phi)\big)^2\big],$$

the second integral cancels out with the first term on the right. Then

$$\begin{aligned} \delta S &= S[\phi + \delta\phi] - S[\phi] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\,\delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta(\partial_\mu\phi)\right] \\ &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi}\,\delta\phi - \partial_\mu\!\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right)\delta\phi + \partial_\mu\!\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi\right)\right]. \end{aligned}$$

Finally, by Stokes' theorem,

$$\delta S = \int_\Omega d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\right]\delta\phi + \oint_{\partial\Omega} \frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\delta\phi\, ds_\mu,$$

where $ds_\mu = \sqrt{-\gamma}\, n_\mu\, d^3\sigma$. But the point is that this last integral is a boundary term, so it is going to be discarded. In this case we'll say that on $\partial\Omega$, we have $\phi(\partial\Omega) = f(\partial\Omega)$. This condition means that the boundary term vanishes – therefore, by comparing against Eq. (23.1), we see that we get

$$\boxed{\frac{\delta S}{\delta \phi} := \frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)},} \qquad \text{(flat spacetime)}$$

and the Euler-Lagrange equation for ϕ is

$$\boxed{\frac{\delta S}{\delta \phi} = 0.}$$

When we move to curved spacetime, we make the replacements $\partial_\mu \to \nabla_\mu$ and $d^4x \to \sqrt{-g}\, d^4x$. Then by similar considerations,

$$\begin{aligned} S &= \int d^4x \sqrt{-g}\, \mathcal{L}(\phi, \nabla_\mu \phi; g_{\alpha\beta}), \\ \delta S &= \int d^4x \sqrt{-g} \left(\frac{\partial \mathcal{L}}{\partial \phi} - \nabla_\mu \frac{\partial \mathcal{L}}{\partial(\nabla_\mu \phi)} \right) \delta\phi. \end{aligned}$$

Since $\nabla_\mu \phi = \partial_\mu \phi$ for a scalar function ϕ, you'll sometimes see the E-L equations written as

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} = \nabla_\mu \frac{\partial \mathcal{L}}{\partial \phi_{,\mu}}.} \qquad \text{(curved spacetime)}$$

The Einstein equation from a variational principle

This isn't usually discussed in an introductory GR course, but we do it here because it is the easiest way to systematically account for matter fields. We introduce the Einstein-Hilbert action

$$S^{EH} = \frac{c^3}{16\pi G} \int d^4x \sqrt{-g}\, R. \tag{23.2}$$

Then we look at the variation,

$$\delta S^{EH} = \frac{c^3}{16\pi G} \int d^4x\, \frac{\delta(\sqrt{-g}\, R)}{\delta g^{\mu\nu}}\, \delta g^{\mu\nu}. \tag{23.3}$$

We're varying with respect to the inverse metric because it is more convenient (not because we have to); the resulting term will then have indices like $G_{\alpha\beta}$, which is what we're used to seeing in the Einstein equation. Now

$$\delta(\sqrt{-g}\, R) = (\delta\sqrt{-g}\,)R + \sqrt{-g}\, \delta R, \tag{23.4}$$

and we'll show at the end of this lecture that (since the calculation takes a bit of work)

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\, \delta g^{\mu\nu},$$

which is an important identity in field theory. The variation of the remaining term is

$$\begin{aligned} \delta R &= \delta(R_{\mu\nu} g^{\mu\nu}) \\ &= R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}, \end{aligned}$$

so we plug both of these results into Eq. (23.4) and then back into Eq. (23.3) to get

$$\delta S^{EH} = \frac{c^3}{16\pi G}\left[\int d^4x \left(R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R\right)\sqrt{-g}\,\delta g^{\mu\nu} + \int d^4x\, g^{\mu\nu}\delta R_{\mu\nu}\sqrt{-g}\right].$$

One can already see the Einstein equation here, but there's an extra integral that complicates things. We'll show that it is possible to eliminate its contribution, but we'll relegate the proof to the end of the lecture. So if we take this on faith for now, it follows that

$$\begin{aligned}\frac{\delta S^{EH}}{\delta g^{\mu\nu}} &:= R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \\ &= G_{\mu\nu}.\end{aligned}$$

Now, in general we have

$$S = S^{EH} + S^{\phi}, \qquad \delta S = 0,$$

for all $\delta g^{\mu\nu}$ that vanish on $\partial\Omega$. Then since (similar to the calculation for δS^{EH}, but with a general Lagrangian density $\mathcal{L}$)

$$\begin{aligned}\delta S^{\phi} &= \int d^4x\,\delta\big[\sqrt{-g}\,\mathcal{L}\big] \\ &= \int d^4x\,\big[(\delta\sqrt{-g})\mathcal{L} + \sqrt{-g}\,\delta\mathcal{L}\big] \\ &= \int d^4x \left(\frac{\partial\mathcal{L}}{\partial g^{\mu\nu}} - \frac{1}{2}g_{\mu\nu}\mathcal{L}\right)\sqrt{-g}\,\delta g^{\mu\nu},\end{aligned}$$

we can combine this with the previous result to get

$$\delta S = \frac{c^3}{16\pi G}\int d^4x\,\sqrt{-g}\left[G_{\mu\nu} + \frac{16\pi G}{c^3}\left(\frac{\partial\mathcal{L}}{\partial g^{\mu\nu}} - \frac{1}{2}g_{\mu\nu}\mathcal{L}\right)\right]\delta g^{\mu\nu}.$$

Now compare this to the Einstein equation (with dimensions restored to c, G)

$$G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}.$$

For $\delta S = 0$ to be consistent with this, we must set

$$\boxed{T^{(\phi)}_{\mu\nu} = -2c\left(\frac{\partial\mathcal{L}}{\partial g^{\mu\nu}} - \frac{1}{2}g_{\mu\nu}\mathcal{L}\right).}$$

This is the systematic approach to obtaining SETs that we mentioned in the previous lecture.

Tying up loose ends

Here we'll justify the steps that we skipped over. First, we establish the identity

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}\,g_{\mu\nu}\delta g^{\mu\nu}.$$

Recalling that $g = g(g_{\mu\nu})$, we have that

$$\delta g = \frac{\partial g}{\partial g_{\mu\nu}}\delta g_{\mu\nu}.$$

Now, we can obtain the determinant g using an expansion by minors:

$$g = \sum_{\nu} (-1)^{\mu+\nu} g_{\mu\nu} M_{\mu\nu}.$$

And one can also compute inverses using determinants:

$$g^{\mu\nu} = \frac{1}{g} M_{\mu\nu} (-1)^{\mu+\nu}.$$

Comparing these two, we find

$$\frac{\partial g}{\partial g_{\mu\nu}} = (-1)^{\mu+\nu} M_{\mu\nu} = g g^{\mu\nu},$$

and finally

$$\delta g = g g^{\mu\nu} \delta g_{\mu\nu}.$$

This is not exactly what we want (the indices are in the wrong place), but by noting that $0 = \delta(g_{\mu\nu} g^{\mu\nu}) = g^{\mu\nu}\delta g_{\mu\nu} + g_{\mu\nu}\delta g^{\mu\nu}$, we get

$$\delta g = -g g_{\mu\nu} \delta g^{\mu\nu}.$$

From here, it just takes the chain rule to compute

$$\begin{aligned}
\delta\sqrt{-g} &= -\frac{1}{2}\frac{1}{\sqrt{-g}}\delta g \\
&= -\frac{1}{2}\frac{-g}{\sqrt{-g}} g_{\mu\nu}\delta g^{\mu\nu} \\
&= -\frac{1}{2}\sqrt{-g}\, g_{\mu\nu}\delta g^{\mu\nu},
\end{aligned}$$

as we wanted to establish. Next, we'll show that

$$\int_{\Omega} g^{\mu\nu}\delta R_{\mu\nu}\sqrt{-g}\, d^4x = \oint_{\partial\Omega}\left(g^{\mu\nu}\delta\Gamma^{\alpha}_{\mu\nu} - g^{\mu\alpha}\delta\Gamma^{\lambda}_{\mu\lambda}\right) ds_{\alpha}. \tag{23.5}$$

Before we continue, a comment – this is rather peculiar, because it involves variations in the Christoffel symbols, rather than $\delta g^{\mu\nu}$. But all we can do is set the latter to zero! This means that this term does not generally vanish. The normal fix is to include a counter-term to the total action to kill this.

We won't actually show how the counter-term results in the necessary cancellation, but we'll prove Eq. (23.5) here anyway.

(incomplete)

24 The Schwarzschild solution

We showed in the previous lecture that the Einstein equations follow from a variational principle, where the Einstein-Hilbert action is

$$S_{EH} = \int d^4x \sqrt{-g}\, R + \oint (\quad)\, ds_\alpha$$

and we demand that

$$\frac{\delta S_{EH}}{\delta g^{\mu\nu}} = 0.$$

We went over this in order to get a recipe for stress-energy tensors. In particular, we found

$$T_{ab} \propto \frac{\delta S_\phi}{\delta g^{ab}}.$$

There's also the so-called Palatini formulation, where Γ and g are treated independently. This comes up in theories where your connection is not tied to your metric (i.e. you don't demand metric compatibility).

Today we'll ask the following question – assuming that

$$G_{ab} = 8\pi T_{ab}$$

is the correct description, then how do we solve it?

Solving the Einstein equation is difficult!

Here a schematic picture of the Einstein equation (with integrability condition) is drawn; the point is that we get nonlinear terms, and these complicate things greatly. This is a notoriously difficult system of nonlinear, coupled PDEs.

One can compare this against electromagnetism – schematically, you just have

$$\partial F = J,$$
$$\partial J = 0,$$

and since these are linear, you get nice things like superposition. This is not the case in GR. And it's not just the form of the equations – for example, the problem of specifying initial data is conceptually more difficult.

Then Dr. Vega tells the story of how when he was a grad student, two groups solved the notoriously difficult problem in numerical relativity of two black holes evolving around each other. In fact there was a singular effort by Frans Pretorius a month before either of these two (massive) groups published. His work was based on a completely different idea (something about excision). Before this, he was not so well known – now he is a big name in GR.

Spherically symmetric solutions

One might call this the "poor man's approach", but it is essentially how Schwarzschild himself did things. First, a general principle – in order to find *exact* solutions, you need to assume some symmetries of your spacetime. Einstein actually believed at first that *no* exact solutions existed! But as we'll see, Schwarzschild managed to find one, even if he did not understand the true meaning of his result. This took something like 30 years, and – seeing as how Schwarzschild died a few months after the publication of his work – the interpretation did not come from him.[41]

[41] For an entertaining paper that both describes the original 1916 derivation and illustrates the importance of choosing appropriate coordinates, see The Schwarzschild metric: It's the coordinates, stupid! by Fromholz et al.

Anyway, let us proceed with the calculation. We will say that spherical symmetry implies no preferred spatial direction, and so

$$ds^2 = g_{\alpha\beta}\, dx^\alpha\, dx^\beta$$

must involve only rotationally invariant differentials. Leveraging our intuition from $\mathbb{R}^3$, these are

$$dt, \quad d\vec{x}\cdot d\vec{x}, \quad \vec{x}\cdot d\vec{x}.$$

Now

$$\begin{aligned} d\vec{x}^2 &= dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\phi^2) \\ &= dr^2 + r^2 d\Omega^2, \end{aligned}$$

$$\vec{x}\cdot d\vec{x} = r\, dr,$$

$$r^2 = \vec{x}\cdot\vec{x}.$$

So with x to denote $\{t, r, \theta, \phi\}$, we have

$$ds^2 = -U(x)\, dt^2 - 2V(x)\, dt\, dr + G(x)\, d\vec{x}^2 + H(x)(\vec{x}\cdot d\vec{x})^2,$$

and using the expressions above, this becomes

$$ds^2 = -U(x)\, dt^2 - 2V(x)\, dt\, dr + W(x)\, dr^2 + X(x)\, d\Omega^2.$$

We now argue that (for the same reasons) the functions U, V, W, X cannot depend on θ and ϕ, so we eliminate this dependence from our free functions. Next, we introduce

$$\begin{aligned} \tilde{r} &= X(t, r), \\ d\tilde{r} &= \partial_t X\, dt + \partial_r X\, dr, \end{aligned}$$

so that our line element becomes (inverting the above to eliminate dr)

$$ds^2 = -U(t,\tilde{r})\, dt^2 - 2V(t,\tilde{r})\, dt \left(\frac{d\tilde{r} - \partial_t X\, dt}{\partial_r X}\right) + W(t,\tilde{r})\left(\frac{d\tilde{r} - \partial_t X\, dt}{\partial_r X}\right)^2 + \tilde{r}^2\, d\Omega^2.$$

Finally

$$ds^2 = -\tilde{U}(t,\tilde{r})\, dt^2 - 2\tilde{V}(t,\tilde{r})\, dt\, d\tilde{r} + \tilde{W}(t,\tilde{r})\, d\tilde{r}^2 + \tilde{r}^2\, d\Omega^2.$$

The tildes indicate that we've absorbed some terms into our functions. But these are still arbitrary, so we'll drop them. So now we have

$$ds^2 = -U(t,r)\, dt^2 - 2V(t,r)\, dt\, dr + W(t,r)\, dr^2 + r^2\, d\Omega^2. \tag{24.1}$$

So far, all we've really done has been to adopt one of our free functions as our coordinate. We now want to eliminate the cross term. To do this, we will define a new coordinate

$$\begin{aligned} \tilde{t} &= \phi(t, r), \\ d\tilde{t} &= \xi(t, r)(U\, dt + V\, dr). \end{aligned}$$

We appeal to a result from the theory of ODEs to say that such a ξ exists. (but you should verify this yourself) Thus we can write

$$dt = U^{-1}\xi^{-1}\, d\tilde{t} - U^{-1}V\, dr.$$

From here it's straightforward to show by direct substitution that

$$U(t,r)\,dt^2 + 2V(t,r)\,dt\,dr = \frac{1}{U\xi^2}\,d\tilde{t}^2 + \left(\frac{V^2}{U} - \frac{2V^2}{U}\right)dr^2,$$

so this choice kills the cross terms in Eq. (24.1), we can absorb what's left into our free functions, and we can relabel $\tilde{t} \to t$. So we've shown that spherical symmetry implies

$$\boxed{ds^2 = -A(t,r)\,dt^2 + B(t,r)\,dr^2 + r^2\,d\Omega^2.} \qquad \text{(spherically symm.)}$$

This is a significant reduction from the 10 free functions that a general line element carries. Now, this calculation was naive because it didn't pay much attention to geometry – we've borrowed heavily from our $\mathbb{R}^3$ intuition. One should look for a deeper understanding, and one that is better suited to arbitrary symmetries. (For example, what about cylindrical or planar symmetries?)

The Schwarzschild solution

Now we add the additional assumption that our spacetime is *static*, and this comes with a precise mathematical meaning:

$$\boxed{\begin{aligned} &(1)\ \ \exists \text{ timelike Killing vector field } \xi^\alpha, \\ &(2)\ \ \exists \text{ spacelike hypersurface orthogonal to } \xi^\alpha. \end{aligned}} \qquad \text{(static)}$$

Condition (1) alone defines a *stationary* spacetime. One can get to the following result in a more rigorous fashion, via differential geometry, but for now we'll say that a static spacetime allows us to eliminate the t dependence from A and B. Thus

$$\boxed{ds^2 = -e^{2\alpha(r)}\,dt^2 + e^{2\beta(r)}\,dr^2 + r^2\,d\Omega^2.} \qquad \text{(static, spher.)}$$

There appears to be some loss of generality here – A and B were originally arbitrary, but these new functions never change sign. But it turns out that we need to assume $A, B > 0$ anyway in order to protect the meanings of our coordinates. (More on this, presumably, in the future.)

Now we will look for a *vacuum* solution – this is one where the SET is zero. Notice that

$$\begin{aligned} G_{\alpha\beta} = 0 \quad &\implies \quad R_{\alpha\beta} - \frac{1}{2}g_{\alpha\beta}R = 0 \\ &\implies \quad R - \frac{1}{2}(4)R = 0 \\ &\implies \quad R = 0. \end{aligned}$$

Substituting this back into the definition of $G_{\alpha\beta}$, we find $R_{\alpha\beta} = 0$; conversely, it's not hard to see that $R_{\alpha\beta} = 0$ implies a zero Einstein tensor. This is what's called a *Ricci flat* solution. This is relatively easy to impose, since you just need to set all the components of the Ricci tensor to zero.[42] So we compute the Ricci tensor from our line element[43] and find that

$$\begin{aligned} R_{tt} &= e^{2(\alpha-\beta)}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - (\partial_r\alpha)(\partial_r\beta) + \frac{2}{r}\partial_r\alpha\right], \\ R_{rr} &= -\partial_r^2\alpha - (\partial_r\alpha)^2 + (\partial_r\alpha)(\partial_r\beta) + \frac{2}{r}\partial_r\beta, \end{aligned}$$

[42] Generally, you need to compute the Einstein tensor and match up its components with the stress-energy.

[43] This was done using Mathematica, but you should do this by hand at least once! Afterwards, you will be much more appreciative of computers.

$$
\begin{aligned}
R_{\theta\theta} &= e^{-2\beta}\big[r(\partial_r\beta) - r(\partial_r\alpha) - 1\big] + 1,\\
R_{\phi\phi} &= \sin^2\theta R_{\theta\theta}.
\end{aligned}
$$

Now we set all of these to zero. Notice that if $R_{rr} = R_{tt} = 0$, then

$$
\begin{aligned}
0 &= e^{2(\beta-\alpha)}R_{tt} + R_{rr}\\
&= \frac{2}{r}\frac{d}{dr}(\alpha+\beta).
\end{aligned}
$$

This gives $\alpha(r) + \beta(r) = C$ for some constant C, so in particular $\alpha(r) = -\beta(r) + C$. Putting this back into our line element, we get

$$
ds^2 = -e^{-2\beta+2C}dt^2 + e^{2\beta}dr^2 + r^2\,d\Omega^2,
$$

and we can get rid of the constant by rescaling $t \to e^{-C}t$. Then

$$
ds^2 = -e^{-2\beta}dt^2 + e^{2\beta}dr^2 + r^2\,d\Omega^2. \tag{24.2}
$$

This leaves us with the $R_{\theta\theta} = 0$ condition, since the one on $R_{\phi\phi}$ follows immediately from this.

$$
\begin{aligned}
& e^{-2\beta}(2r\partial_r\beta - 1) = -1\\
\iff\quad & e^{-2\beta}\left(1 - 2r\frac{d\beta}{dr}\right) = 1\\
\iff\quad & \frac{d}{dr}\left(re^{-2\beta(r)}\right) = 1.
\end{aligned}
$$

From here, we get

$$
e^{-2\beta(r)} = 1 - \frac{R_s}{r},
$$

where R_s is a constant of integration. Putting this back into Eq. (24.2), we have

$$
\boxed{ds^2 = -\left(1 - \frac{R_s}{r}\right)dt^2 + \left(1 - \frac{R_s}{r}\right)^{-1}dr^2 + r^2\,d\Omega^2.} \qquad \text{(Schwarzschild)}
$$

This is a very simple solution (supposedly)! It is a one-parameter family of solutions parametrized by R_s, which is called the *Schwarzschild radius.* So we've solved the Einstein equation, but we're still missing an interpretation. It turns out that this is just a local description – our coordinates cannot describe everything. We call $r = R_s$ the *Schwarzschild singularity*, and there's another singularity at $r = 0$. Physical interpretations will follow in the next lecture.

Finally, a point that textbooks tend to overlook: we did not actually solve the system

$$
\begin{aligned}
R_{tt} &= 0,\\
R_{rr} &= 0,
\end{aligned}
$$

but instead we worked with the reduced set

$$
e^{2(\beta-\alpha)}R_{tt} + R_{rr} = 0.
$$

Obviously the first two equations imply the third, but this doesn't necessarily mean that the converse is true, so the last thing you need to do is to go back and verify that we do indeed get $R_{tt} = R_{rr} = 0$, and then you're done.

25 Interpreting Schwarzschild coordinates; the TOV equation

A priori, coordinates have no intrinsic meaning. The metric is what provides our meanings for t, r, etc. In, say, Newtonian physics, we can interpret equations like $\vec{F} = m\vec{a}$ because we know what our coordinates mean. But this is a much subtler matter in GR.

Coordinates for spherical symmetry

Consider flat spacetime, which is clearly spherically symmetric. One choice for a metric is

$$ds^2 = -dt^2 + dr^2 + r^2\, d\Omega^2.$$

Now, a question: what does r mean? We typically understand r to be a distance from the origin, defined like $r = \sqrt{x^2 + y^2 + z^2}$. But another way of thinking, and one that generalizes to our present situation, is to think of r as a label for the 2-sphere the given point belongs to. So consider a surface of constant t and r, so that the restricted metric on that surface is

$$ds_\Omega^2 = d\ell^2 = r^2\, d\Omega^2.$$

We know that each 2-sphere has an area of $4\pi r^2$. So we can say that

$$\boxed{r = \sqrt{\frac{\text{Area}}{4\pi}}.}$$

From this point of view, r is what's called an *areal radius.*

One definition of spherical symmetry is to say that every point of spacetime belongs to a 2-sphere. Now recall that the most general line element for a spherically symmetric spacetime is

$$ds^2 = g(t, r')\, dt^2 + h(t, r')\, dr'^2 + f(t, r')\, d\Omega^2.$$

For constant t and r' surfaces, then, we have

$$\text{Area} = 4\pi f(t, r').$$

This interpretation is given to us by our metric. Notice that r' is not an areal radius, but we can define $r = \sqrt{f(t, r')}$ to get this interpretation. Generally, there is no *a priori* relation between r and the (proper) distance "from the center". A center may not even exist!

A naive comparison, and then a proper one

We showed in the previous lecture that the Schwarzschild solution was given by

$$ds^2 = -\left(1 - \frac{R}{r}\right) dt^2 + \left(1 - \frac{R}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2.$$

Now, sometimes people make the following (incorrect) comparison to imbue R with a physical meaning. The final answer is correct, but the method is illegal, so we'll point out the mistake in the reasoning and then show the correct way.

For weakly-curved spacetime, we have

$$ds^2 = -\left(1 - \frac{2U}{c^2}\right) c^2\, dt^2 + \left(1 + \frac{2U}{c^2}\right) d\vec{x} \cdot d\vec{x},$$

where $\nabla^2 U = -4\pi G\rho$. For a point mass $\rho = M\delta(\vec{r})$, it follows that $U = GM/r$, and the line element becomes

$$ds^2 = -\left(1 - \frac{2GM}{r}\right)dt^2 + \left(1 + \frac{2GM}{r}\right)d\vec{x}\cdot d\vec{x}.$$

A naive comparison suggests that $R = 2M$ (in geometric units). But this is careless – it mixes up variables and geometry. When you write $U = GM/r$, you invite confusion by using the same label r, but this isn't the same thing as the one in the Schwarzschild solution. To see this, look at constant t and r surfaces in weakly curved spacetime.

$$\begin{aligned} d\ell^2 &= \left(1 + \frac{2M}{r}\right)d\vec{x}\cdot d\vec{x} \\ &= \left(1 + \frac{2M}{r}\right)\left(\cancel{dr^2} + r^2\,d\theta^2 + r^2\sin^2\theta\,d\phi^2\right) \\ &= \left(1 + \frac{2M}{r}\right)r^2\,d\Omega^2. \end{aligned}$$

So the areal radius in this case is $\overline{r} = r\sqrt{1 + 2M/r}$, and this is what we ought to be comparing against the r in the Schwarzschild solution.

But we'll do something slightly different, and we'll change coordinates in the Schwarzschild solution so that we can compare directly against the weakly-curved solution.[44] We define *isotropic coordinates*

$$r = r_{\text{iso}}\left(1 + \frac{R}{4r_{\text{iso}}}\right)^2,$$

so that our line element becomes

$$ds^2 = -\left(\frac{1 - R/4r_{\text{iso}}}{1 + R/4r_{\text{iso}}}\right)^2 dt^2 + \left(1 + \frac{R}{4r_{\text{iso}}}\right)^4 (dr_{\text{iso}}^2 + r_{\text{iso}}^2\,d\Omega^2).$$

Evidently r_{iso} is not an areal radius. But now we can compare to the weakly-curved solution:

$$\begin{aligned} g_{00} &= -\left(\frac{1 - R/4r_{\text{iso}}}{1 + R/4r_{\text{iso}}}\right)^2, \\ g_{ij} &= \delta_{ij}\left(1 + \frac{R}{4r_{\text{iso}}}\right)^4. \end{aligned}$$

If $r_{\text{iso}} \gg R$, we obtain

$$\begin{aligned} g_{00} &= -1 + \frac{R}{r_{\text{iso}}} - \frac{1}{2}\left(\frac{R}{r_{\text{iso}}}\right)^2 + \cdots \\ g_{ij} &= \delta_{ij}\left(1 + \frac{R}{r_{\text{iso}}} + \frac{3}{8}\left(\frac{R}{r_{\text{iso}}}\right)^2 + \cdots\right) \end{aligned}$$

and by comparison against the weakly-curved metric elements,

$$\begin{aligned} g_{00}^{\text{wc}} &= -1 + \frac{2M}{r_{\text{iso}}}, \\ g_{ij}^{\text{wc}} &= \delta_{ij}\left(1 + \frac{2M}{r_{\text{iso}}}\right), \end{aligned}$$

we see that we can make the identification $R = 2M$. So this is the same answer, but this is the proper way to arrive at it. Notice that this argument only really applies to cases where $r_{\text{iso}} \gg R$, but this identification can be made more generally. (Something about geodesics mentioned here.)

[44] You're invited to go from $r \to \overline{r}$ in the weakly-curved solution and make the comparison from here, though.

The Tolman-Oppenheimer-Volkoff equation

The next simplest thing to look at is a static perfect fluid, which has stress-energy tensor

$$T_{\alpha\beta} = (\rho + P)u_\alpha u_\beta + P g_{\alpha\beta}.$$

(Actually, this is just the SET of a perfect fluid – we haven't imposed the static condition yet.) Now recall from the previous lecture that the most general line element for a static spacetime is

$$ds^2 = -e^{2\Phi}dt^2 + e^{2\Lambda}dr^2 + r^2\,d\Omega^2.$$

There are no cross terms with dt – at best, all the fluid elements move in the $\partial/\partial t$ direction:

$$u^\alpha = A\delta^\alpha_0.$$

Then for this to be a four-velocity, it must be normalized, hence

$$g_{\alpha\beta}u^\alpha u^\beta = -1 \implies u^0 = e^{-\Phi(r)}.$$

Lowering the index then gives $u_0 = -e^{+\Phi}$. Substituting all this back into the SET, we obtain

$$\begin{aligned} T_{00} &= \rho e^{2\Phi}, \\ T_{rr} &= P e^{2\Lambda}, \\ T_{\theta\theta} &= r^2 P, \\ T_{\phi\phi} &= \sin^2\theta\, T_{\phi\phi}. \end{aligned}$$

Everything is going to be substituted into

$$G_{\alpha\beta} = 8\pi T_{\alpha\beta},$$

which comes automatically with the integrability condition

$$\nabla_\beta T^{\alpha\beta} = 0.$$

There's an argument to show that the θ, ϕ components must be zero (the hairy ball theorem is mentioned here), while the r component becomes

$$(P+\rho)\frac{d\Phi}{dr} = -\frac{dP}{dr},$$

which is the equation for hydrostatic equilibrium. (The P doesn't show up on the left in the Newtonian case, because it comes with c's in non-geometric units, and therefore goes to zero in the $c \to \infty$ limit.)

Now, we no longer have a Ricci flat solution (there is stuff, i.e. a nonzero SET), so we actually need to compute the components of the Einstein tensor. You can show that

$$\begin{aligned} G_{tt} &= \frac{1}{r^2}e^{2\Phi}\frac{d}{dr}\Big[r(1-e^{-2\Lambda})\Big], \\ G_{rr} &= -\frac{1}{r^2}e^{2\Lambda}(1-e^{-2\Lambda}) + \frac{2}{r}\frac{d\Phi}{dr}, \\ G_{\theta\theta} &= r^2 e^{-2\Lambda}\left[\frac{d^2\phi}{dr^2} + \left(\frac{d\Phi}{dr}\right)^2 + \frac{1}{r}\frac{d\Phi}{dr} - \frac{d\Phi}{dr}\frac{d\Lambda}{dr} - \frac{1}{r}\frac{d\Lambda}{dr}\right], \\ G_{\phi\phi} &= \sin^2\theta\, G_{\theta\theta}. \end{aligned}$$

Writing out the tt component of the Einstein equation, a factor of $e^{2\Phi}$ cancels out, and we obtain

$$\frac{1}{r^2}\frac{d}{dr}\left[r(1-e^{-2\Lambda})\right] = 8\pi\rho.$$

Here it will be useful (and meaningful!) to define

$$\boxed{m(r) := \frac{1}{2}r(1-e^{-2\Lambda}).}$$

We now observe that

$$e^{-2\Lambda} = 1 - \frac{2m(r)}{r},$$

which allows us to write the tt equation as

$$\frac{dm}{dr} = 4\pi\rho(r)\,r^2$$

It follows that

$$m(\bar{r}) = \int_0^{\bar{r}} 4\pi\rho(r)\,r^2\,dr.$$

In Newtonian physics, the quantity on the left takes the interpretation of the mass contained inside a sphere of radius $\bar{r}$. For this reason, we say that

$$m(r) \quad \text{— mass function.}$$

A bit of manipulation shows that the rr equation can be written

$$\boxed{\frac{d\Phi}{dr} = \frac{m(r)+4\pi r^3 P(r)}{r\big(r-2m(r)\big)}.} \tag{25.1}$$

and for convenience, we'll collect the other two equations we've found. The next one followed from the integrability condition,

$$\boxed{(\rho+P)\frac{d\Phi}{dr} = -\frac{dP}{dr},} \tag{25.2}$$

and the remaining one is just the tt equation in terms of m.

$$\boxed{\frac{dm}{dr} = 4\pi\rho(r)r^2.} \tag{25.3}$$

You might wonder about the $\theta\theta$ and $\phi\phi$ equations, but the latter depend on the former, and it turns out (as you should show) that the $\theta\theta$ equation follows from the three boxed equations above. So we have three equations and four unknowns – we're still missing an equation. In particular, what we need is an *equation of state* for the fluid:

$$P = P(\rho, S, T, \ldots).$$

We'll restrict ourselves to cases where

$$\boxed{P = P(\rho).} \tag{25.4}$$

(Otherwise we'd need to put in more equations to account for the new unknowns.) So this is what we have to work with! Plugging Eq. (25.1) into (25.2), we get

$$\boxed{\frac{dP}{dr} = -\frac{(\rho + P)\big(m(r) + 4\pi r^3 P(r)\big)}{r\big(r - 2m(r)\big)}.} \qquad \text{(TOV equation)}$$

Some work is needed to turn this into an autonomous system of ODEs (one where r does not appear explicitly on the right). Then we need to supply initial conditions in order to solve this. In particular, we need

$$\Phi, \ m, \ P \quad \text{at } r = 0.$$

Choosing $m(r = 0)$ is easy; this will be zero for a reasonable ρ, in view of Eq. (25.3). Meanwhile, $\Phi(r = 0)$ is arbitrary (why?), so we only really specify the central pressure $P_c \equiv P(r = 0)$.

If your fluid satisfies what are known as energy conditions (these are imposed on ρ and P), then dP/dr will always be negative, and the P–r diagram looks something like this:

(figure here)

In practice, we integrate outward until some $r = R$ where $P(R) = 0$; this marks the boundary of your star.[45] In this sense, R and $m(R) \equiv M$ are outputs, rather than something you can set from the beginning. Here we wonder whether it's possible to reformulate things to work with

$$\frac{dm}{dr} \quad \text{and} \quad \frac{dR}{dr},$$

but it's not clear whether this can be done.

[45] Mathematically, $dP/dr < 0$ does not ensure that the pressure eventually becomes zero – it could asymptote to some nonzero value. But this situation doesn't correspond to a physical star.

26 Solving the TOV equation

In the previous lecture, we started from a static, spherically symmetric line element,

$$ds^2 = -e^{2\Phi(r)}dt^2 + e^{2\Lambda(r)}dr^2 + r^2\, d\Omega^2,$$

and we assumed that our matter was described by the SET for a perfect fluid. We then defined the mass function

$$m(r) = \frac{1}{2}\, r(1 - e^{-2\Lambda}),$$

and obtained the following from the Einstein equation:

$$\frac{dm}{dr} = 4\pi r^2 \rho, \tag{26.1}$$

$$\frac{d\Phi}{dr} = \frac{m + 4\pi r^3 P}{r(r-2m)}, \tag{26.2}$$

$$\frac{dP}{dr} = -(\rho + P)\frac{m + 4\pi r^3 P}{r(r-2m)}. \tag{26.3}$$

This gives us the functions Φ, m, ρ, and P to determine – the first two can be thought of as metric functions, while the remaining two are material functions (they have to do with the fluid). Finally, we assumed an equation of state of the form $P = P(\rho)$. So how do we solve these? We start from (26.1) and (26.3), which do not involve Φ, and then we use these to solve (26.2).

TOV in vacuum

Let's first look at what happens when $\rho \equiv 0$ and $P \equiv 0$. Then Eq. (26.1) says that $dm/dr = 0$, so $m = M$ for some constant of integration M. There's nothing to do with Eq. (26.3), and finally Eq. (26.2) tells us that

$$\frac{d\Phi}{dr} = \frac{M}{r(r-2M)} \implies \Phi(r) = K + \int \frac{M}{r^2(1-2M/r)}\, dr.$$

Doing the integral gives

$$e^{2\Phi} = e^{2K}\left(1 - \frac{2M}{r}\right).$$

As before, we can get rid of the integration constant by a rescaling $t \to e^{-K}t$. Putting everything back into our line element, we obtain

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dr^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2,$$

which is the Schwarzschild solution. This is a case of *Birkhoff's theorem*, which says that all spherically symmetric spacetimes with $R_{ab} = 0$ are static. And this is quite a powerful statement – for example, it can be interpreted as meaning that spherically symmetric gravitational waves cannot exist.

Constant-density stars

Here we fix ρ to be some constant, but P can still vary. This is far less trivial to deal with, but it still isn't a very realistic model; if we recall that the speed of sound v is given by

$$v = \sqrt{\frac{dP}{d\rho}},$$

then what this says is that v must be infinite... which is forbidden. But these are still used to gain insight when studying, for example, very stiff stars. Anyway, now Eq. (26.1) reads

$$\frac{dm}{dr} = 4\pi r^2 \rho \quad \text{for } r \leq R,$$

which means that

$$m(r) = \frac{4}{3}\pi r^3 \rho$$

inside the star. Our boundary condition is

$$m(r) = M, \quad r \geq R,$$

so in particular $m(R) = M$ and

$$\boxed{\frac{4}{3}\pi \rho R^3 = M.}$$

Therefore, M and R aren't independent, and we just need to determine one. A short calculation shows that Eq. (26.3) leads to

$$\ln\left(\frac{\rho + 3P}{\rho + P}\right) = \frac{1}{2}\ln u + \ln A, \quad \text{where } u = 1 - \frac{8\pi\rho}{3}r^2.$$

To fix the constant of integration A, we put $r = 0$, so that $u = 1$ and $P = P_c$. Then we find

$$\boxed{\frac{\rho + 3P_c}{\rho + P_c} = A,}$$

so this is where the central pressure enters. With this, we can now write

$$\frac{\rho + 3P}{\rho + P} = \left(\frac{\rho + 3P_c}{\rho + P_c}\right) u^{1/2}.$$

Finally, since

$$\frac{m(r)}{r} = \frac{4\pi}{3}\rho r^2,$$

we may write

$$u(r) = 1 - \frac{2m(r)}{r},$$

and finally we obtain

$$\boxed{\frac{\rho + 3P(r)}{\rho + P(r)} = \left(\frac{\rho + 3P_c}{\rho + P_c}\right)\left(1 - \frac{2m(r)}{r}\right)^{1/2}.}$$

One can invert and get only $P(r)$ on the left, but there's no need to do this. From here we can also get an explicit solution for R. Since $P(R) = 0$ and $m(R) = M$, the above becomes

$$1 = \frac{\rho + 3P_c}{\rho + P_c}\left(1 - \frac{2M}{R}\right)^{1/2},$$

and a bit of rearranging yields

$$R^2 = \frac{3}{8\pi\rho}\left[1 - \left(\frac{\rho + P_c}{\rho + 3P_c}\right)^2\right].$$

This gives us the radius of the star as a function of the central pressure P_c. But recall that M and R are not independent, so we can write this as

$$\frac{2M}{R} = 1 - \left(\frac{\rho + P_c}{\rho + 3P_c}\right)^2.$$

Note that the plot has an asymptote at $2M/R = 8/9$:

(figure here)

In reality, one needs huge central pressures to get near the asymptote, because P is subdominant to ρ in non-geometric units. But the point is that $2M/R$ is bounded, and in particular can never attain a value of 1. This is an intimation of *Buchdahl's limit*, which is a more general result (applicable to arbitrary (perfect?) fluid equations of state). Briefly, a body cannot exceed a particular compactness.

Of course, your result is only as good as your assumptions. We've assumed that our central pressure is well-behaved, but if it is divergent, you can get something that exceeds Buchdahl. Maybe not too surprising – if you work with infinity, very weird things can happen.

Newtonian equations of stellar structure

Here we have the familiar equation

$$\frac{dm}{dr} = 4\pi r^2 \rho(r),$$

and each fluid element has mass given by

$$dm = \rho\, dA\, dr.$$

In hydrostatic equilibrium, the pressure balances out the pull of gravity, so we get

$$0 = -[P(r + dr) - P(r)]\, dA - \frac{Gm(r)}{r^2}\, dm.$$

Expanding to linear order in dr, we obtain

$$\boxed{\frac{dP}{dr} = -\frac{Gm\rho}{r^2}.}$$

Since we're studying GR, we should be looking to compare this against our work from earlier. To translate back into non-geometric units, we note that our ρ from earlier is actually an energy density, so we write

$$\epsilon(r) = \rho(r)c^2.$$

(A useful mnemonic for remembering the right dimensions is $E = mc^2$.) Then we get

$$\begin{aligned}
\frac{dm}{dr} &= 4\pi r^2 \frac{\epsilon(r)}{c^2}, \\
\frac{dP}{dr} &= -\frac{G\epsilon(r)m(r)}{c^2 r^2}\left(1 + \frac{P}{\epsilon}\right)\left(1 + \frac{4\pi r^3 P}{mc^2}\right)\left(1 - \frac{2Gm}{c^2 r}\right)^{-1},
\end{aligned}$$

while the Newtonian equation is

$$\frac{dP}{dr} = -\frac{G\epsilon(r)m(r)}{c^2 r^2}.$$

One can think of the terms in parentheses (in the GR equation) as correction terms. But where do these come from? First we note that

$$P \sim \frac{k_F^2}{2m} \sim \frac{1}{2} m \langle v^2 \rangle,$$

where k_F is the Fermi momentum of the fluid (check). So the first two correction terms are special-relativistic, because they scale as $(v/c)^2$. On the other hand, the last term is all geometry.

Now we notice that all the correction terms are greater than 1 (excepting trivial cases), which explains why people say that GR is stronger than Newtonian gravity, at least in this case. The absence of these in NG means that one can construct very, very stiff stars. But in GR, one cannot find a strong enough EOS!

Polytropic equations of state

It's common to deal with so-called *polytropic equations of state*, where

$$P = K\epsilon^\gamma,$$

or in geometric units,

$$P = K\rho^\gamma,$$

where K, γ are constants. These are convenient to deal with, because these correspond to things in statistical mechanics. For instance,

$$\begin{aligned} \gamma &= \tfrac{4}{3} && \text{— relativistic electron gas,} \\ \gamma &= \tfrac{5}{3} && \text{— nonrelativistic electron gas,} \\ \gamma &= 1 && \text{— relativistic neutron gas,} \\ \gamma &= \tfrac{5}{3} && \text{— nonrelativistic neutron gas.} \end{aligned}$$

The number 5/3 is typically understood to be the boundary between stiff and soft equations of state. (This particular choice is mostly convention – one needs to draw a line somewhere.)

Finally, it's common to draw M–R diagrams – a typical one is below.

(figure here)

While this looks a bit odd, this curve is parametrized by P_c and is traversed from right to left. Different branches correspond to different star regimes. For example, the upward-left moving branch on the far right is known as the white dwarf branch; it is stable. This sort of picture can give a heuristic understanding of the qualitative behavior of the star, but I get the impression that there is a lot of work underlying this.

(For example, if I understand correctly, this is obtained by matching two polytropic equations of state together. A single such EOS gives a curve that spirals into itself – this is unphysical, which tells us that our assumptions are no longer accurate.)